VMware Vsphere Cluster Project

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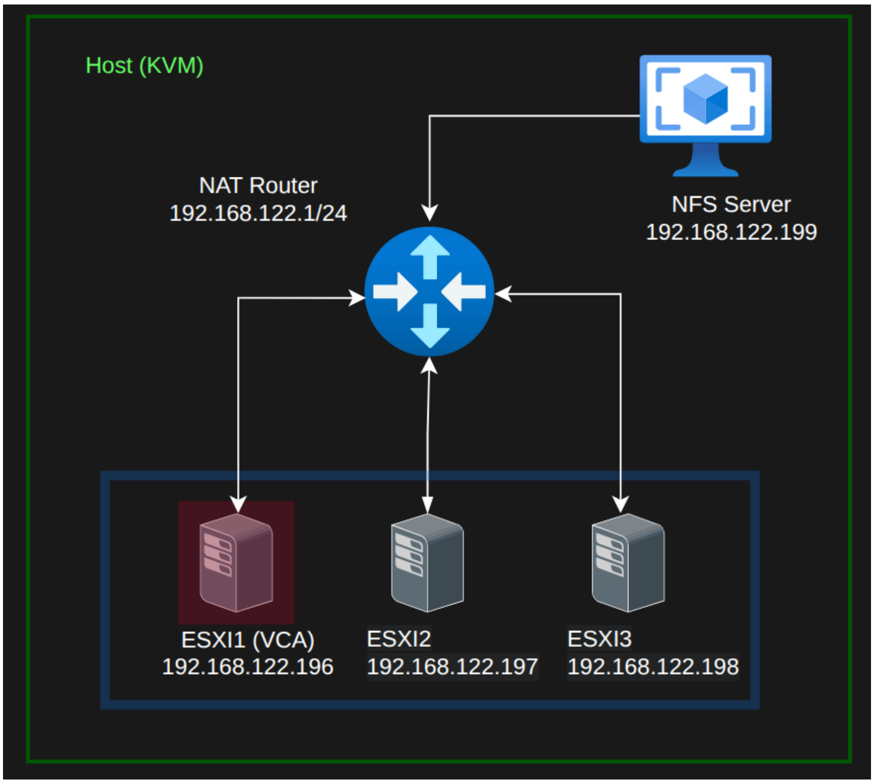
Introduction:

This project aims to design and implement a VMware vSphere environment by deploying multiple ESXi hosts and a centralized vCenter Server. The setup demonstrates core virtualization concepts such as clustering, shared storage, networking, and advanced features like vMotion, High Availability (HA), Distributed Resource Scheduler (DRS), and Fault Tolerance (FT). The environment provides a practical lab for managing virtual machines and validating enterprise level virtualization capabilities.

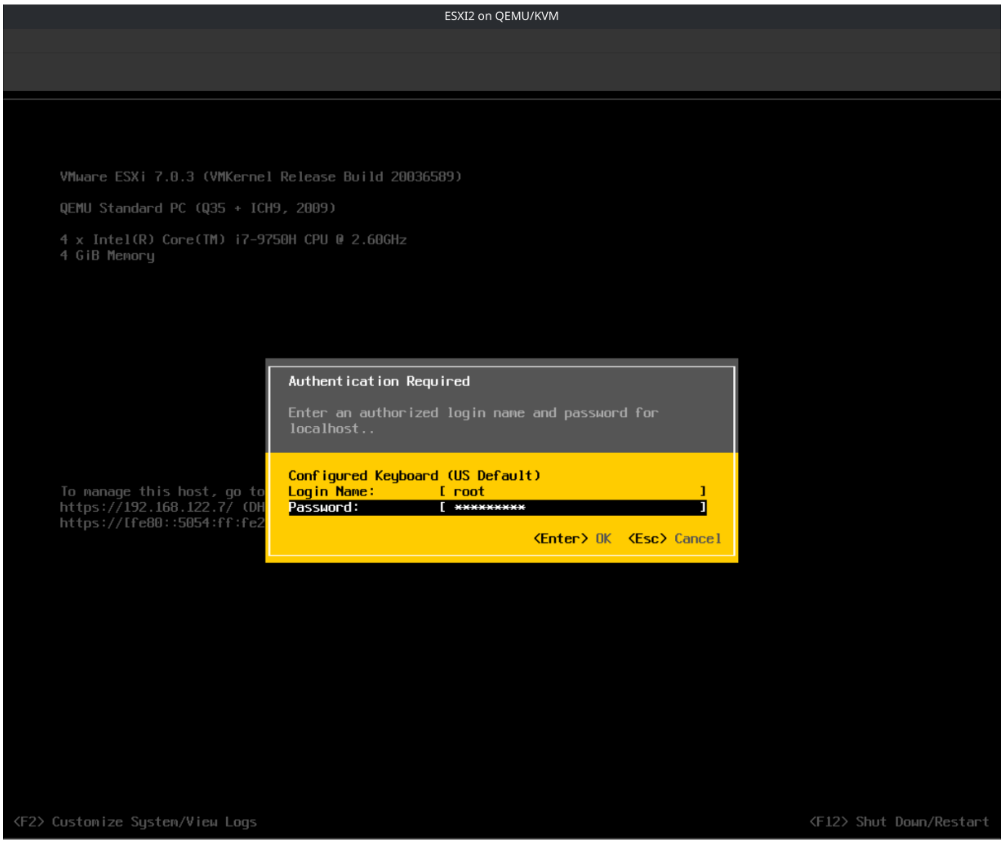
Architecture Overview:

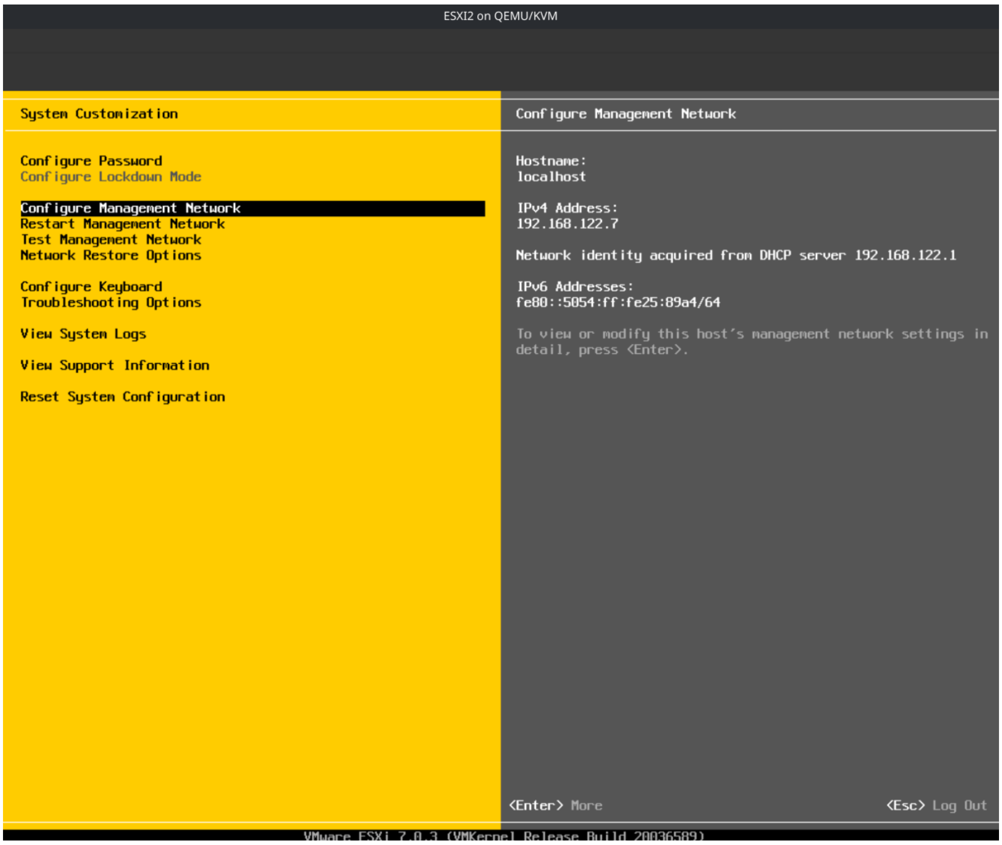


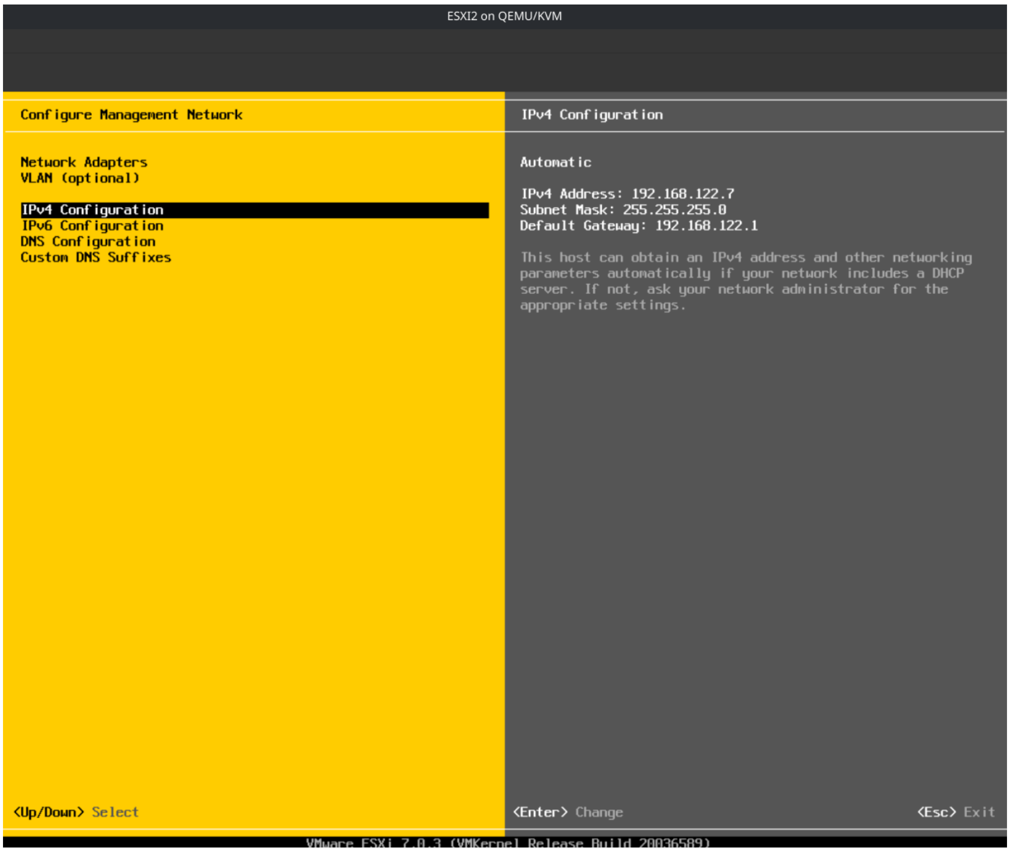
- **ESXI Hosts:**
  - ESXI1 hosts the vCenter Server Appliance and has 4 vCpus, 20 GB of RAM and 500GB of storage on 2 drives (50 GB for the hypervisor and 500 GB for additional storage).
  - ESXI2, ESXI3 each server has 4 vCpus, 4 GB of RAM and 100GB of storage on 2 drives (50 GB for the hypervisor and 50 GB for additional storage).
- **NFS Server:** RHEL 9 Server to act as an NFS shared Datastore for the ESXI cluster and it is configured with 2 vcpus and 2 GB of RAM.
- All of these servers have 2 NICs and are virutalized on a Linux system (Fedora 43) using KVM and are connected to each other using a NAT network setup.

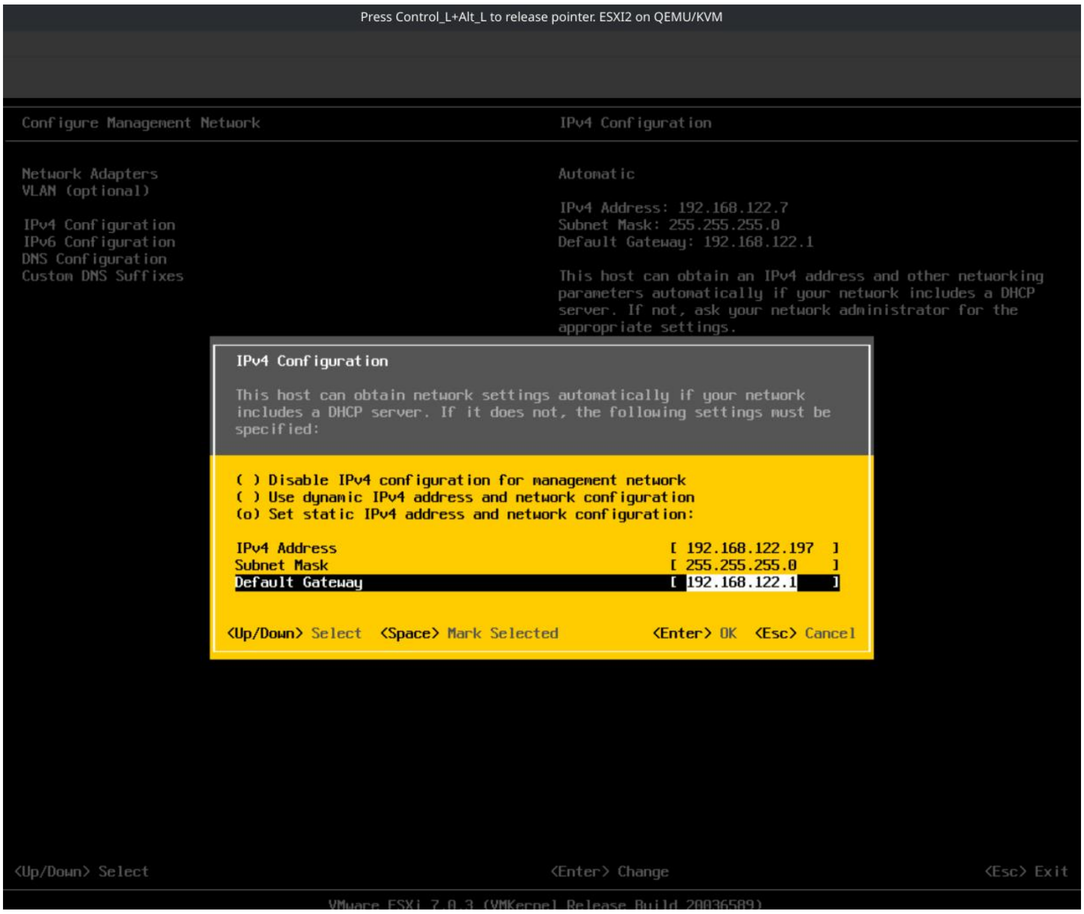
Configuring static IPs for all servers:

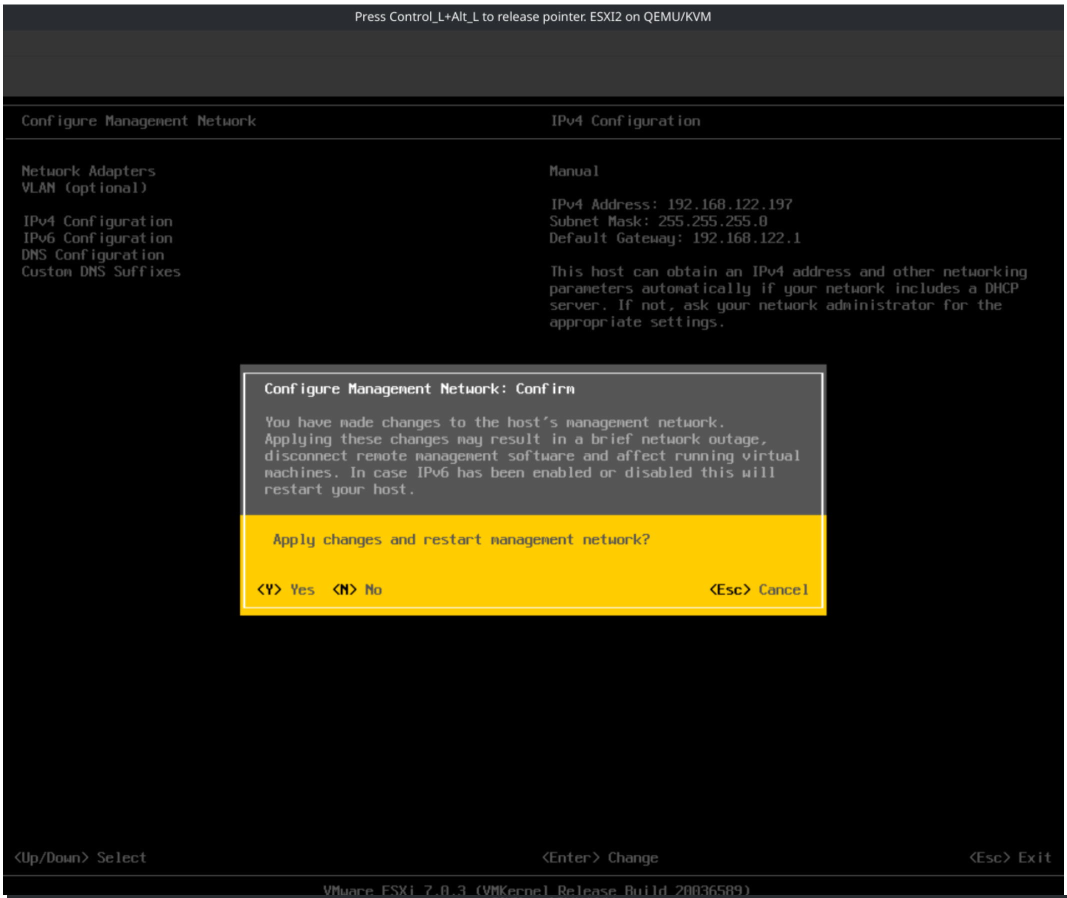
On ESXI servers:

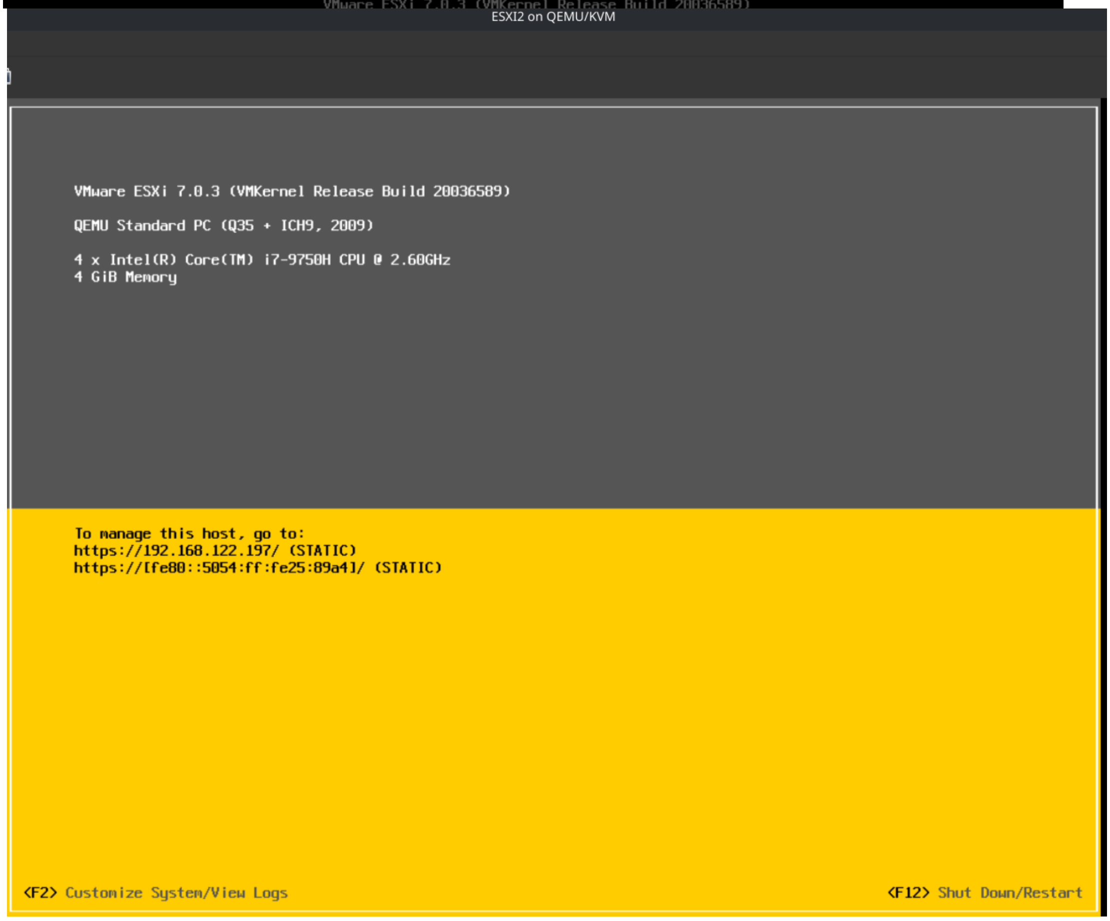




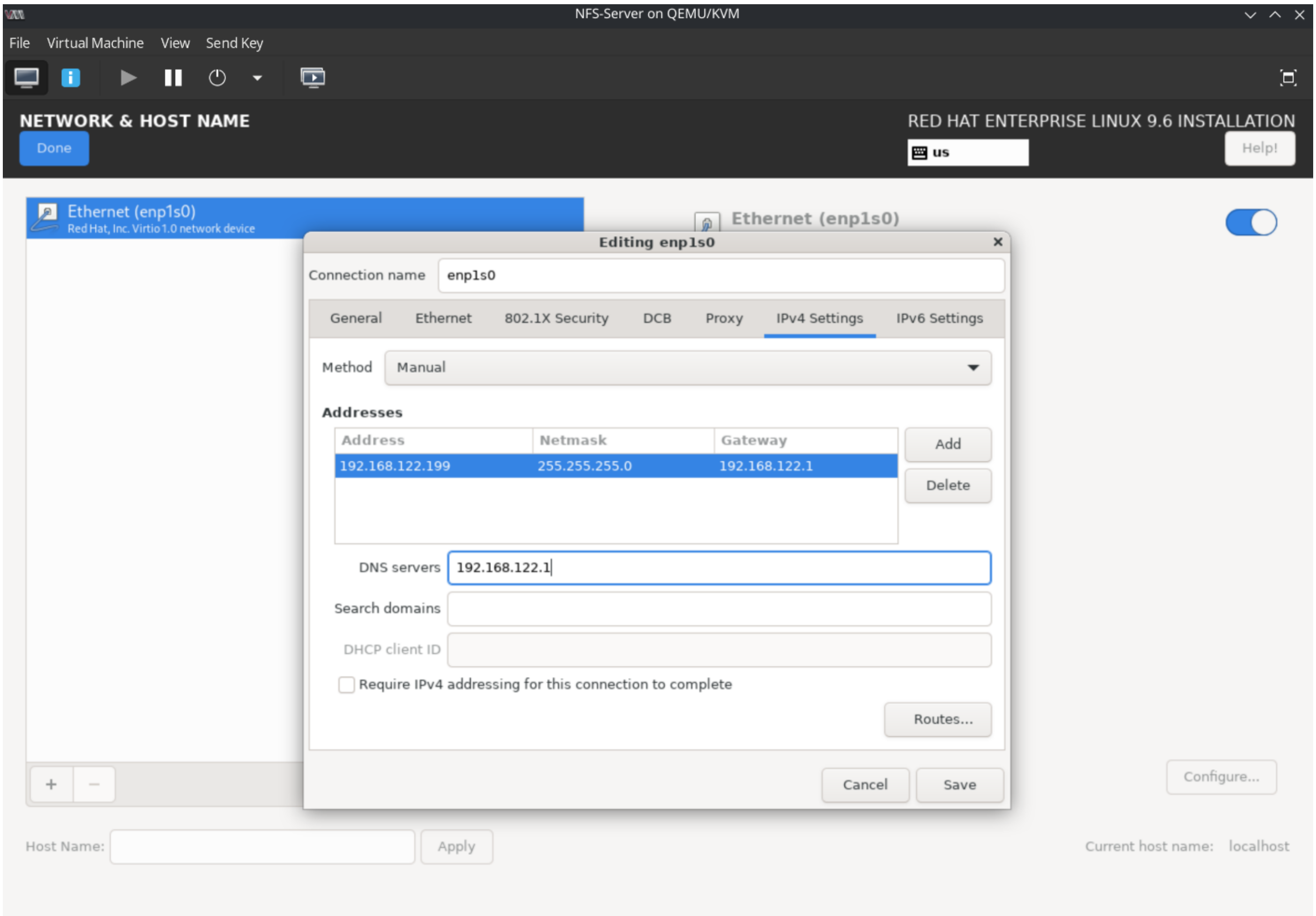

ESXI2 on QEMU/KVM
Configure Management Network
Network Adapters
VLAN (optional)
IPv4 Configuration
IPv6 Configuration
DNS Configuration
Custom DNS Suffixes
IPv4 Configuration
Automatic
IPv4 Address: 192.168.122.7
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.122.1
This host can obtain an IPv4 address and other networking parameters automatically if your network includes a DHCP server. If not, ask your network administrator for the appropriate settings.
<Up/Down> Select
<Enter> Change
<Esc> Exit


Press Control_L+Alt_L to release pointer. ESXI2 on QEMU/KVM
Configure Management Network
Network Adapters
VLAN (optional)
IPv4 Configuration
IPv6 Configuration
DNS Configuration
Custom DNS Suffixes
IPv4 Configuration
Automatic
IPv4 Address: 192.168.122.7
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.122.1
This host can obtain an IPv4 address and other networking parameters automatically if your network includes a DHCP server. If not, ask your network administrator for the appropriate settings.
IPv4 Configuration
This host can obtain network settings automatically if your network includes a DHCP server. If it does not, the following settings must be specified:
( ) Disable IPv4 configuration for management network
( ) Use dynamic IPv4 address and network configuration
(o) Set static IPv4 address and network configuration:
IPv4 Address [192.168.122.197]
Subnet Mask [255.255.255.0]
Default Gateway [192.168.122.1]
<Up/Down> Select <Space> Mark Selected
<Enter> OK <Esc> Cancel
<Up/Down> Select
<Enter> Change
<Esc> Exit



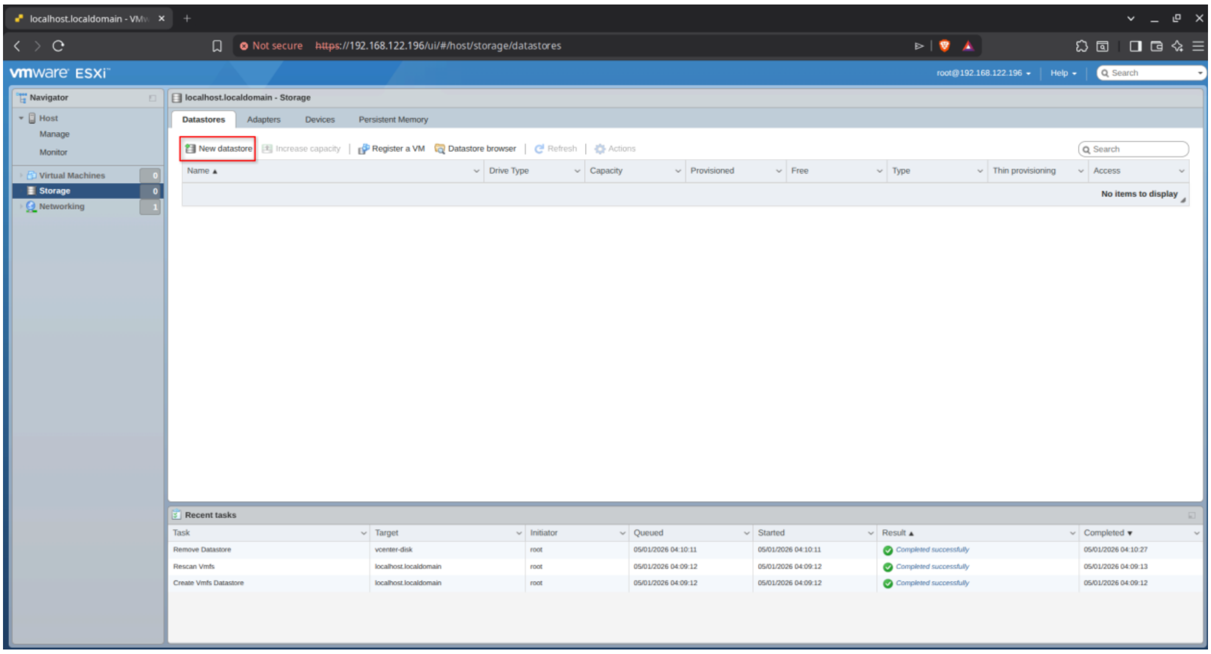


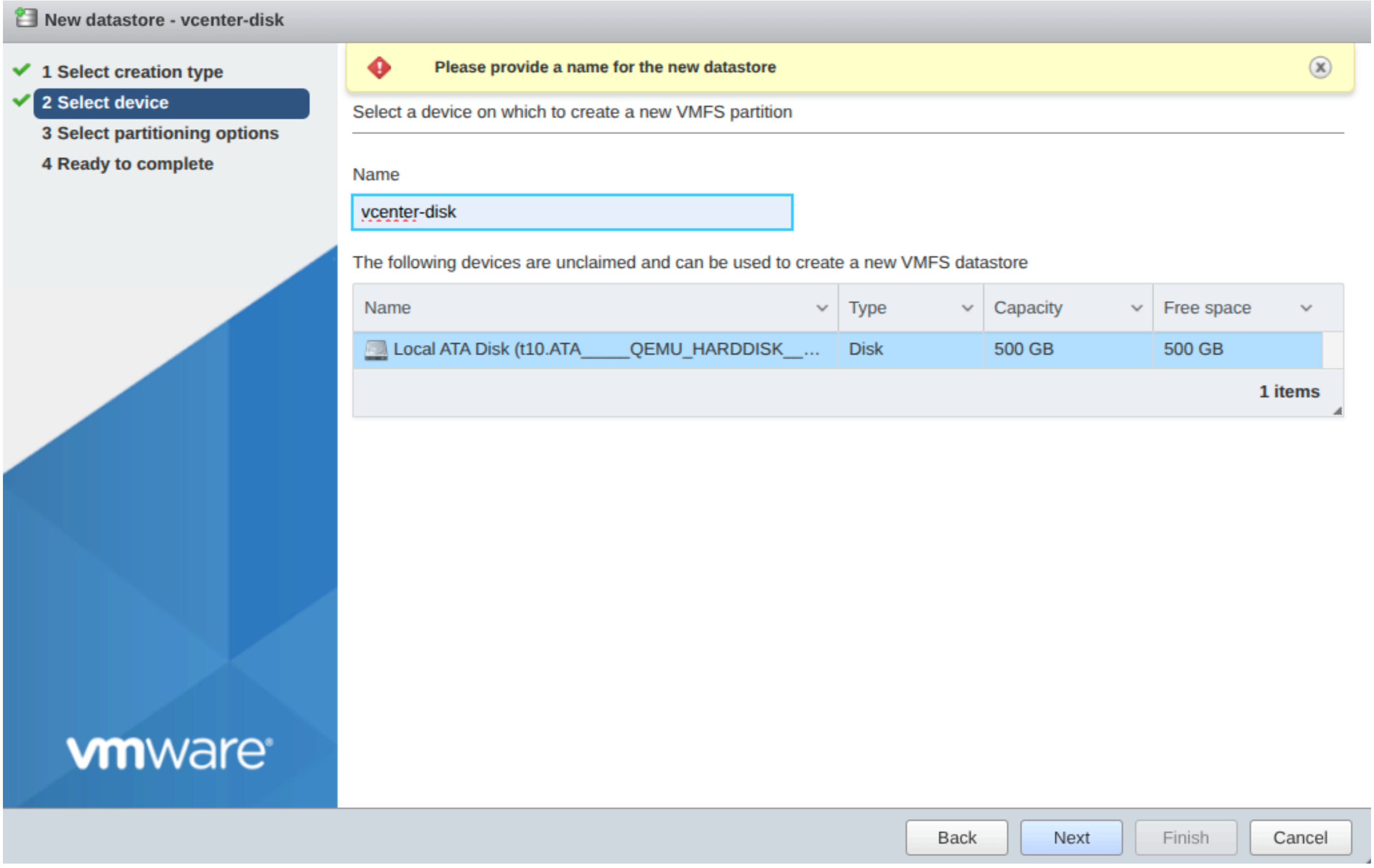
On NFS server:

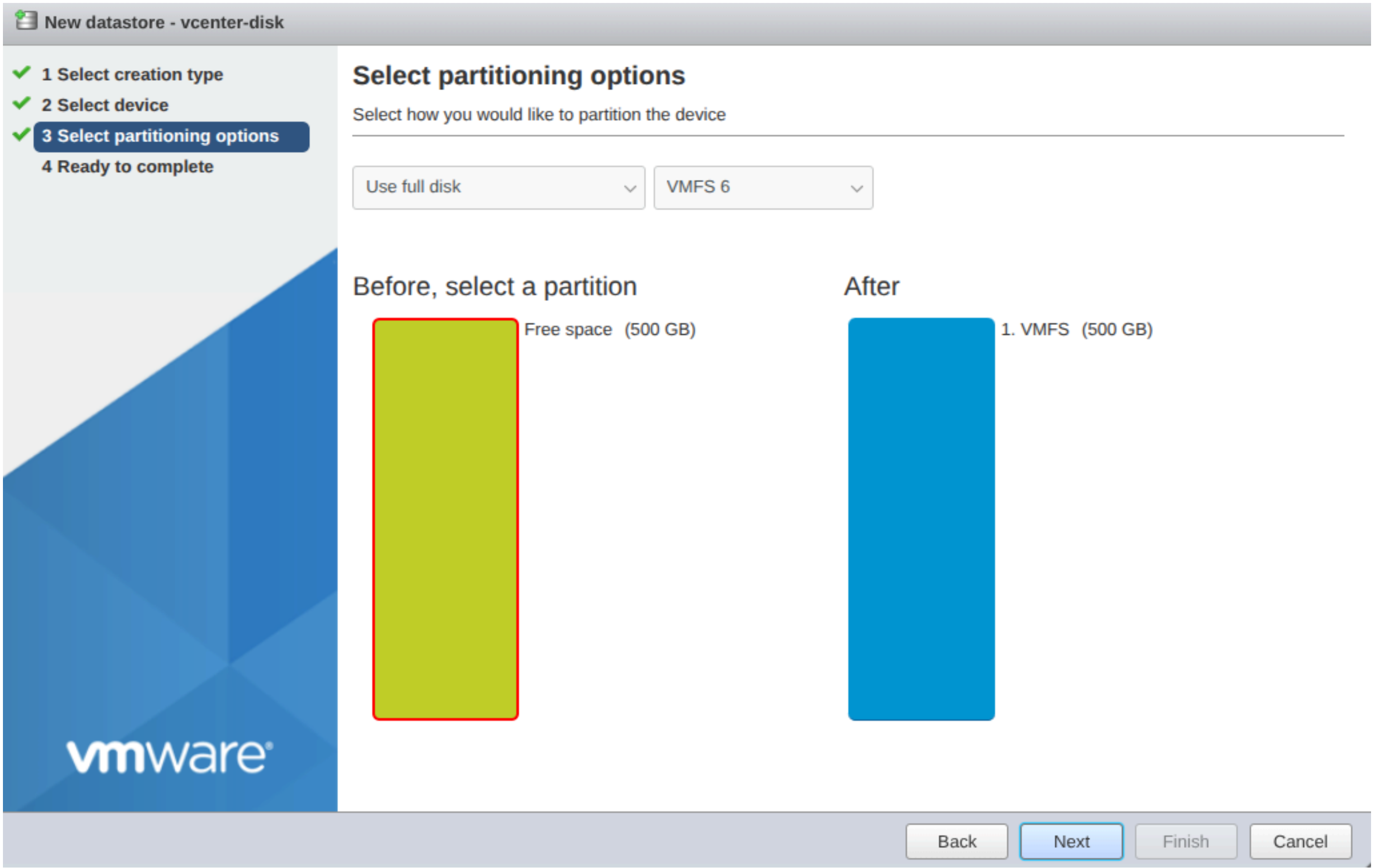


Deploying vCenter appliance:

We first create a data store from the 500GB disk

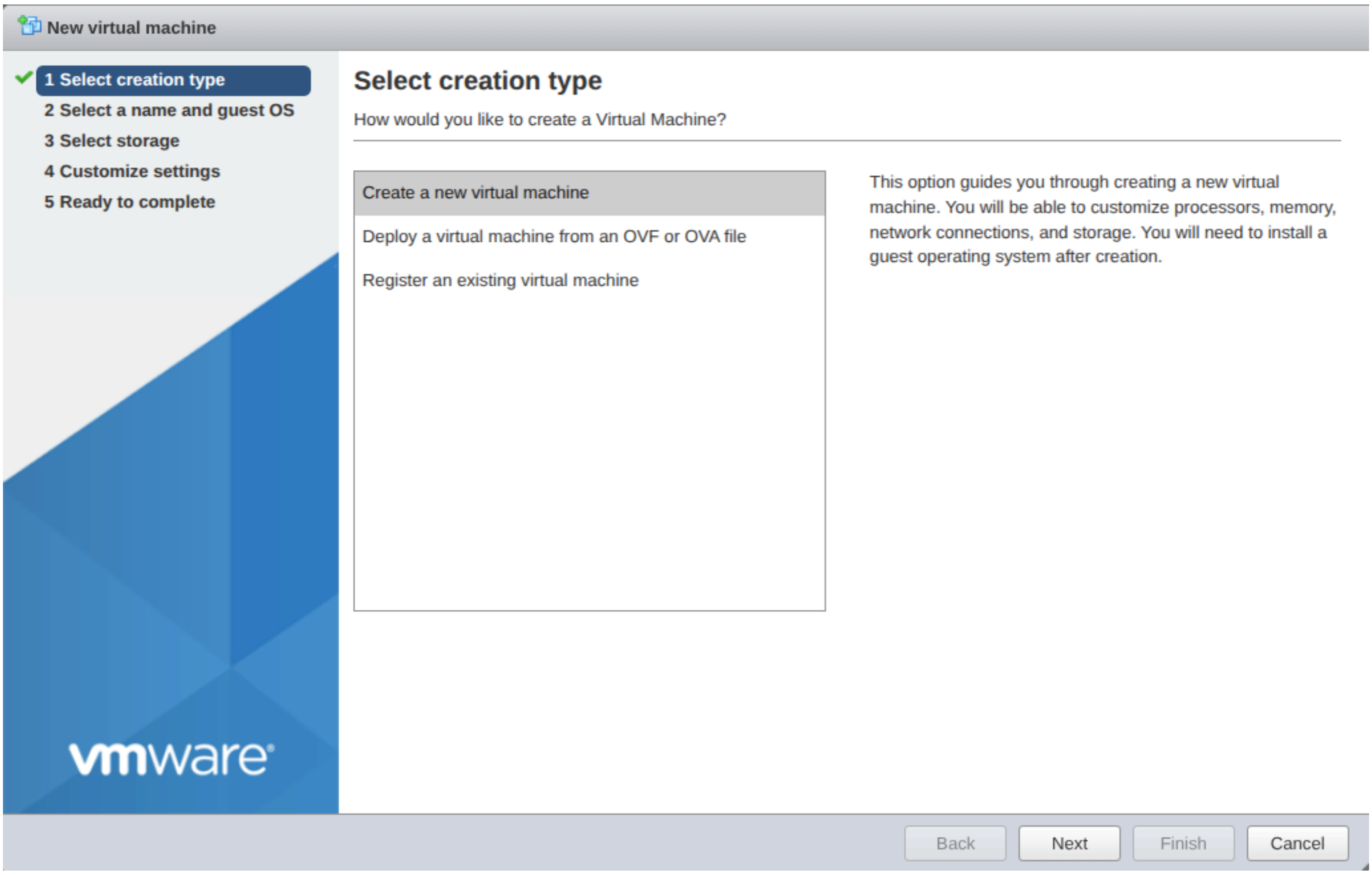


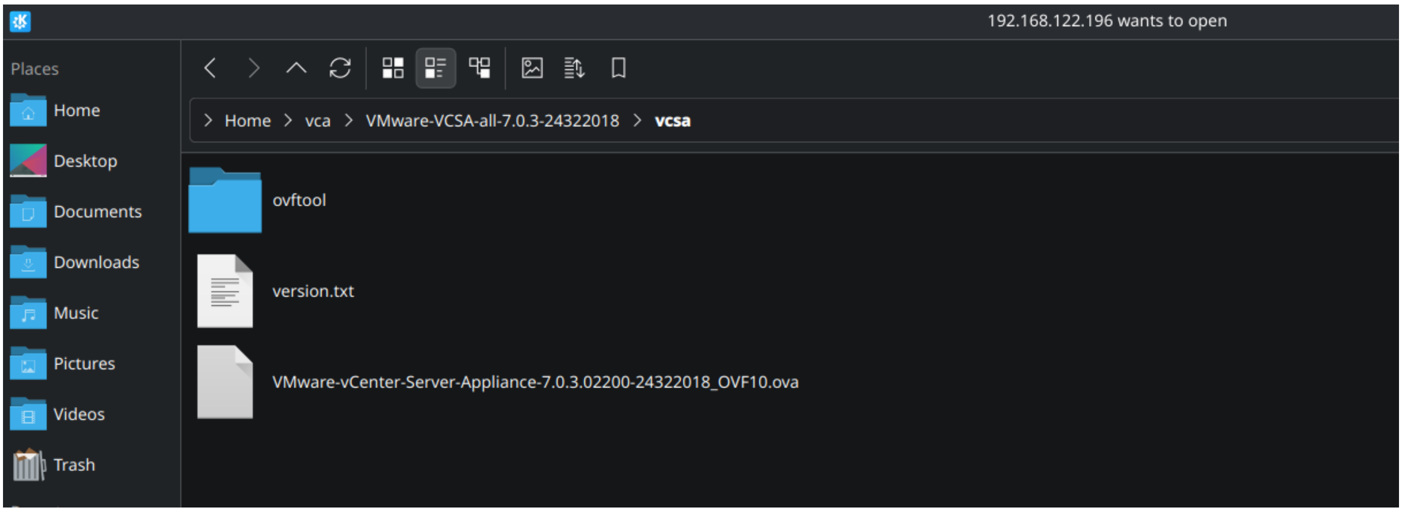


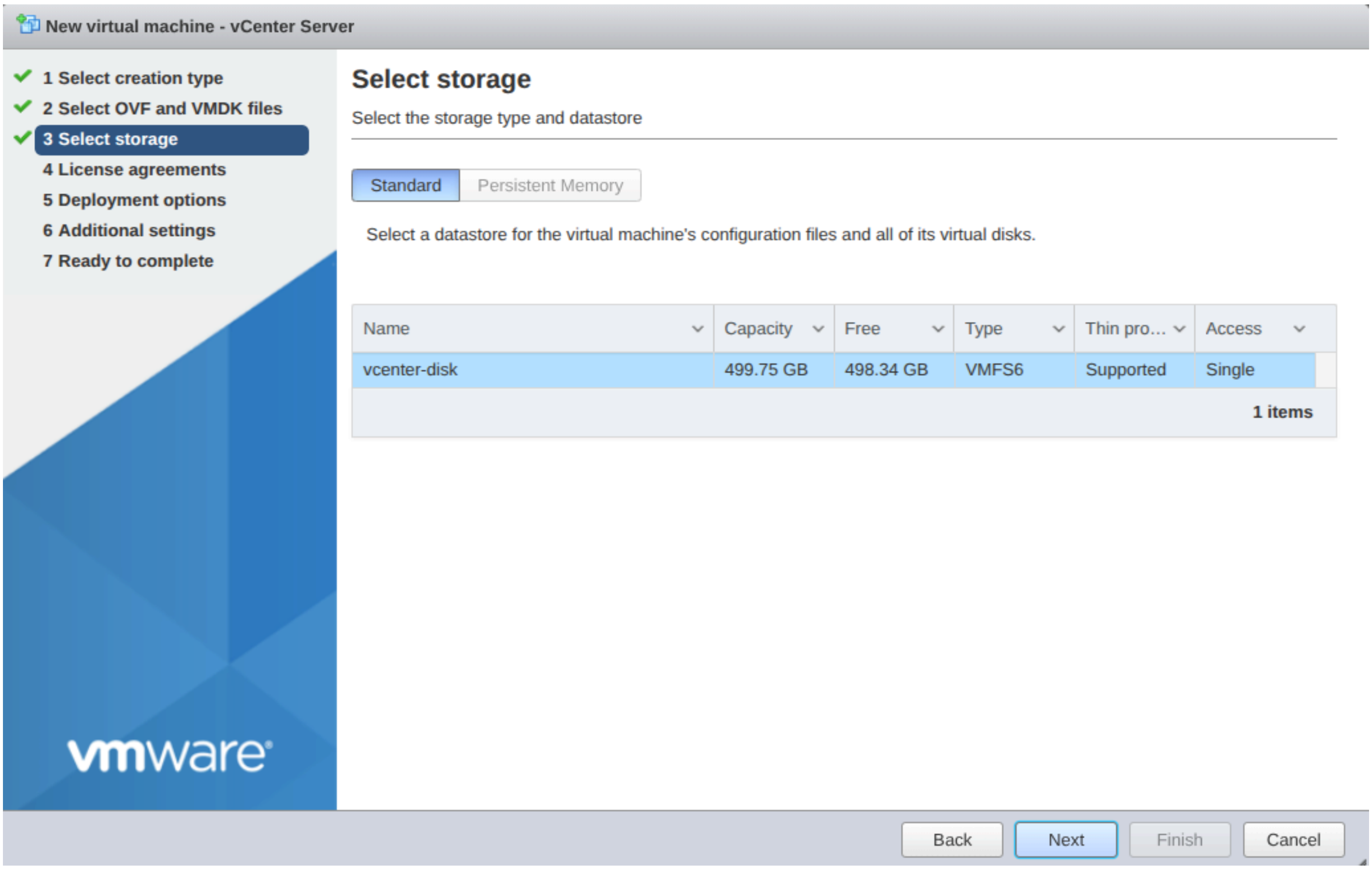


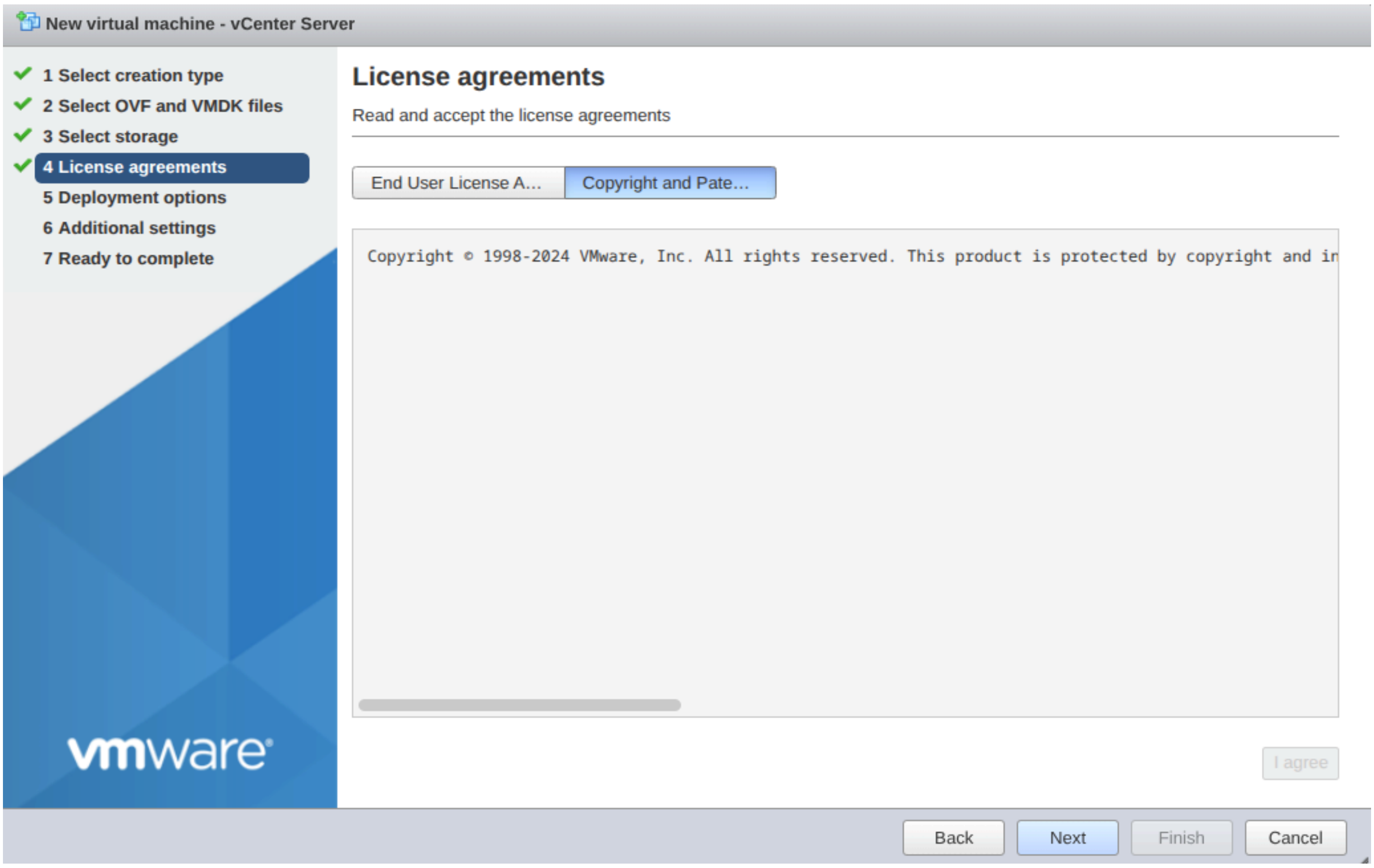
And then we create the VM that will host vServer appliance using the OVA file.

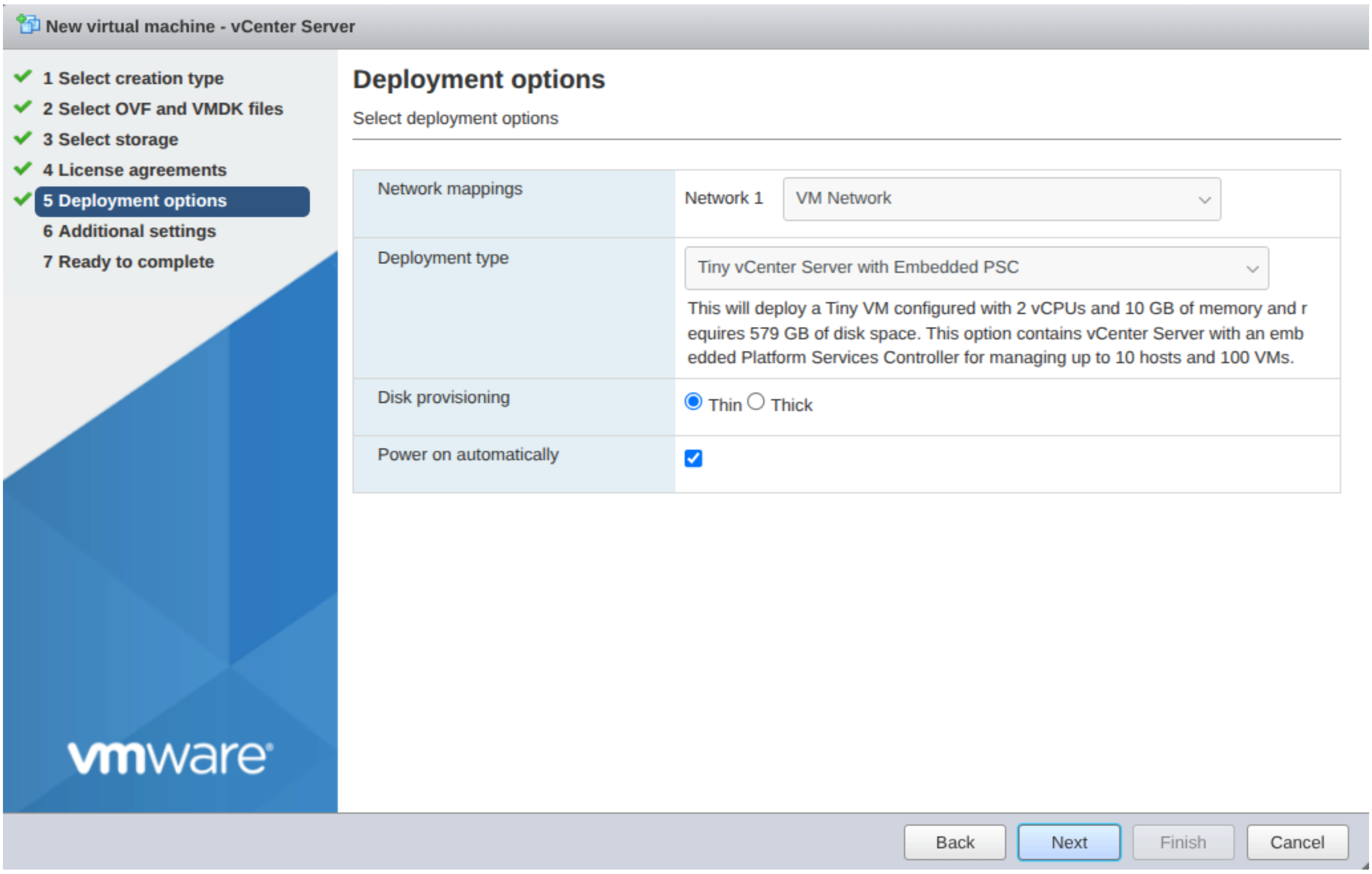
Note that the UI installer for VCA does not work well on modern linux distros.

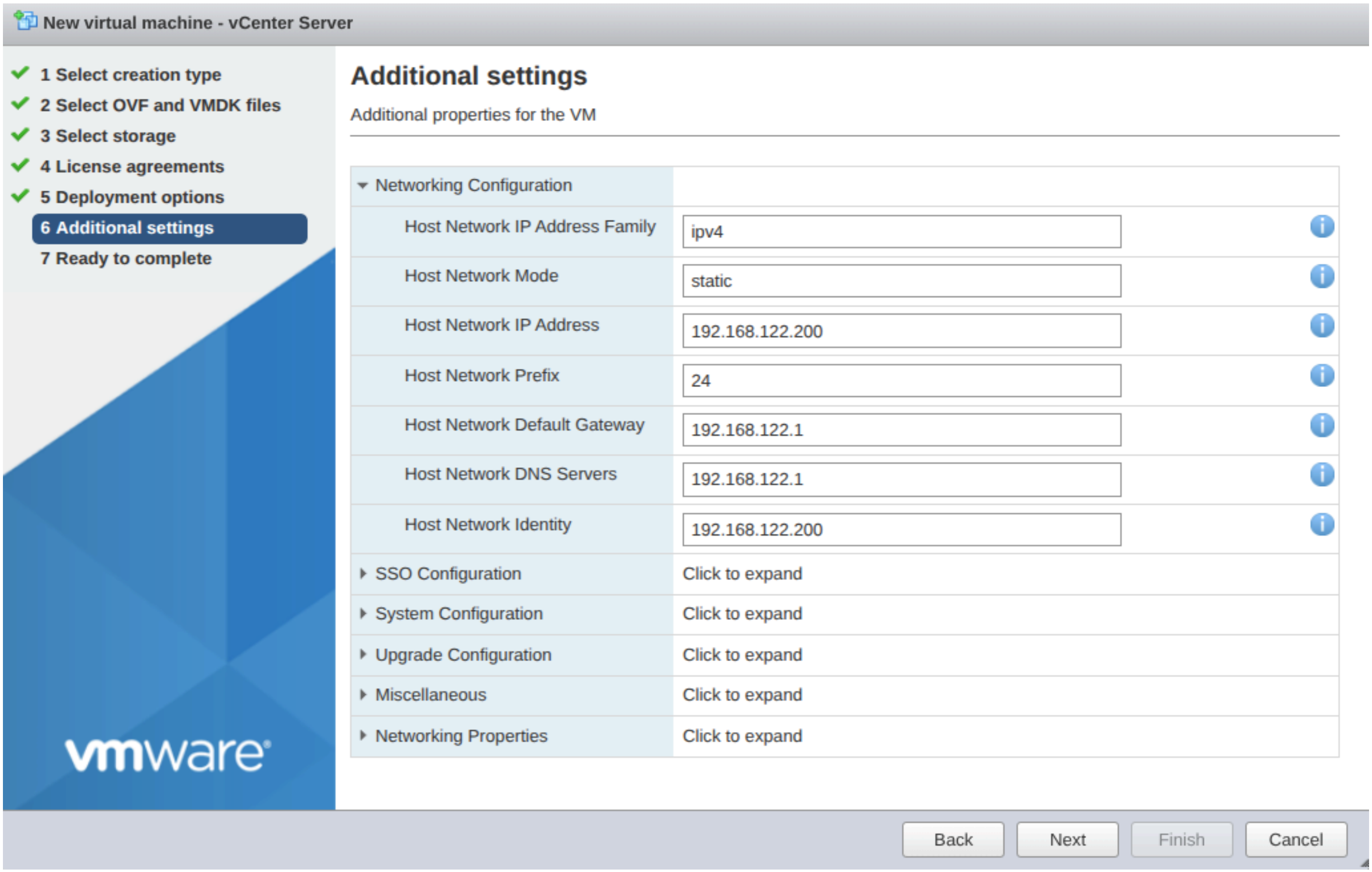


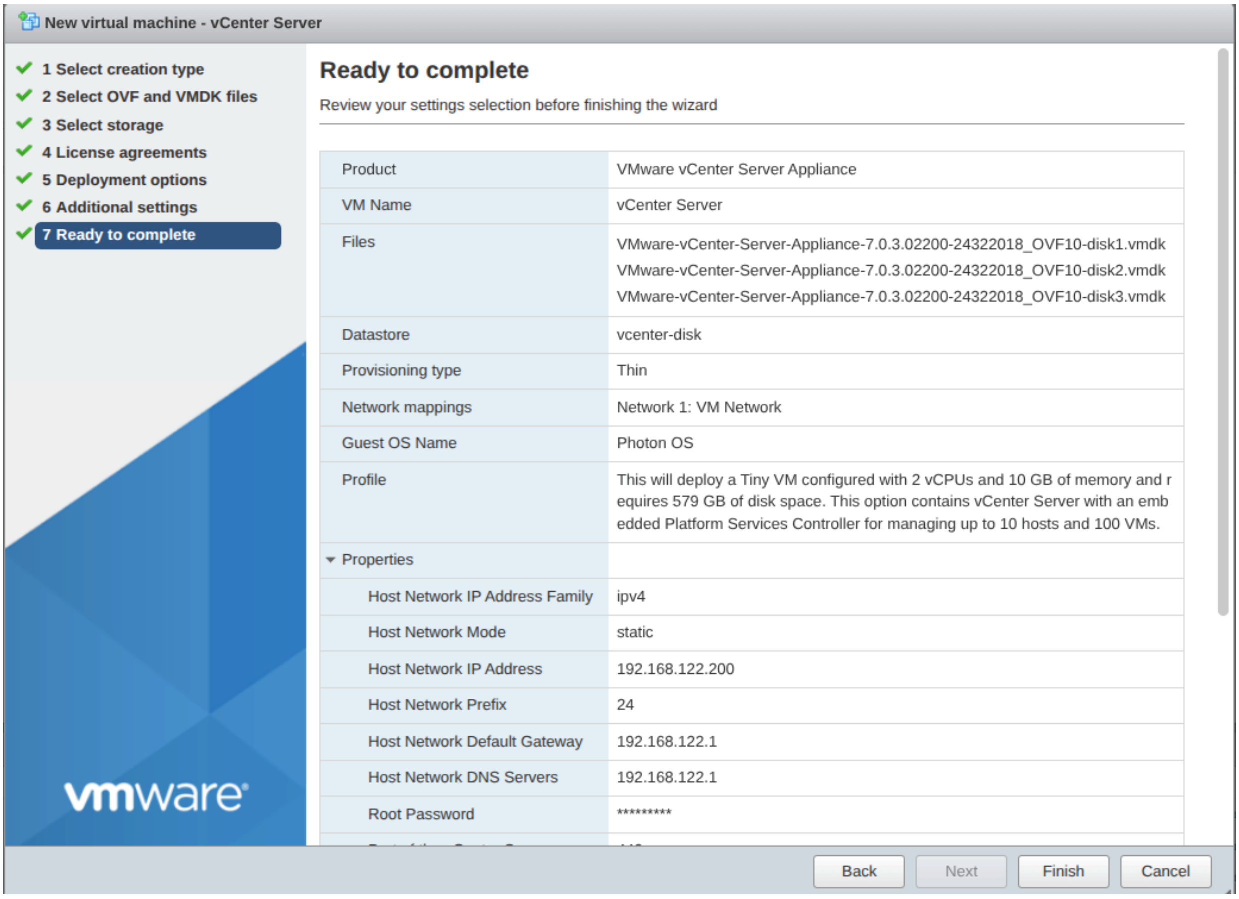


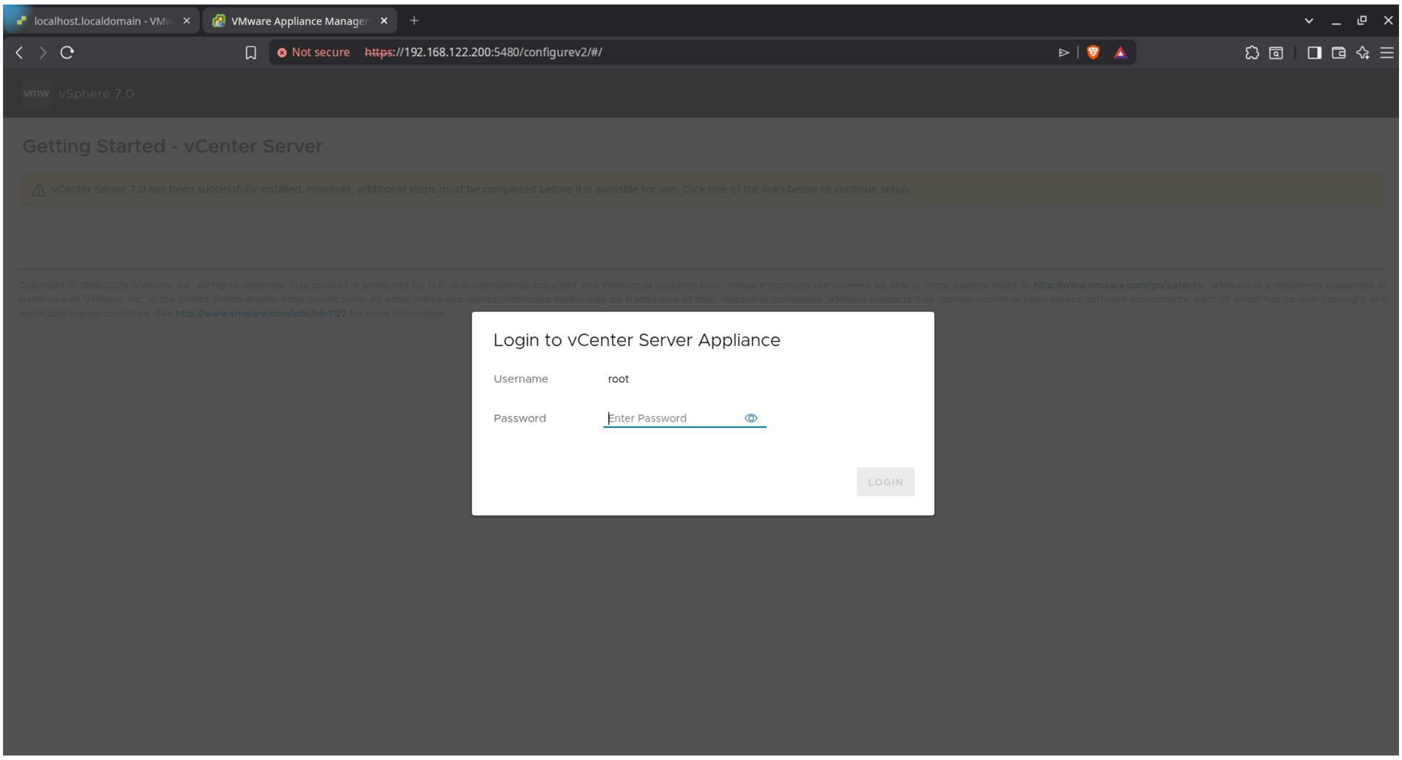

New virtual machine - vCenter Server
1 Select creation type
2 Select OVF and VMDK files
3 Select storage
4 License agreements
5 Deployment options
6 Additional settings
7 Ready to complete
Select storage
Select the storage type and datastore
Standard
Persistent Memory
Select a datastore for the virtual machine's configuration files and all of its virtual disks.
Name
Capacity
Free
Type
Thin pro...
Access
vcenter-disk
499.75 GB
498.34 GB
VMFS6
Supported
Single
1 items
vmware
Back
Next
Finish
Cancel

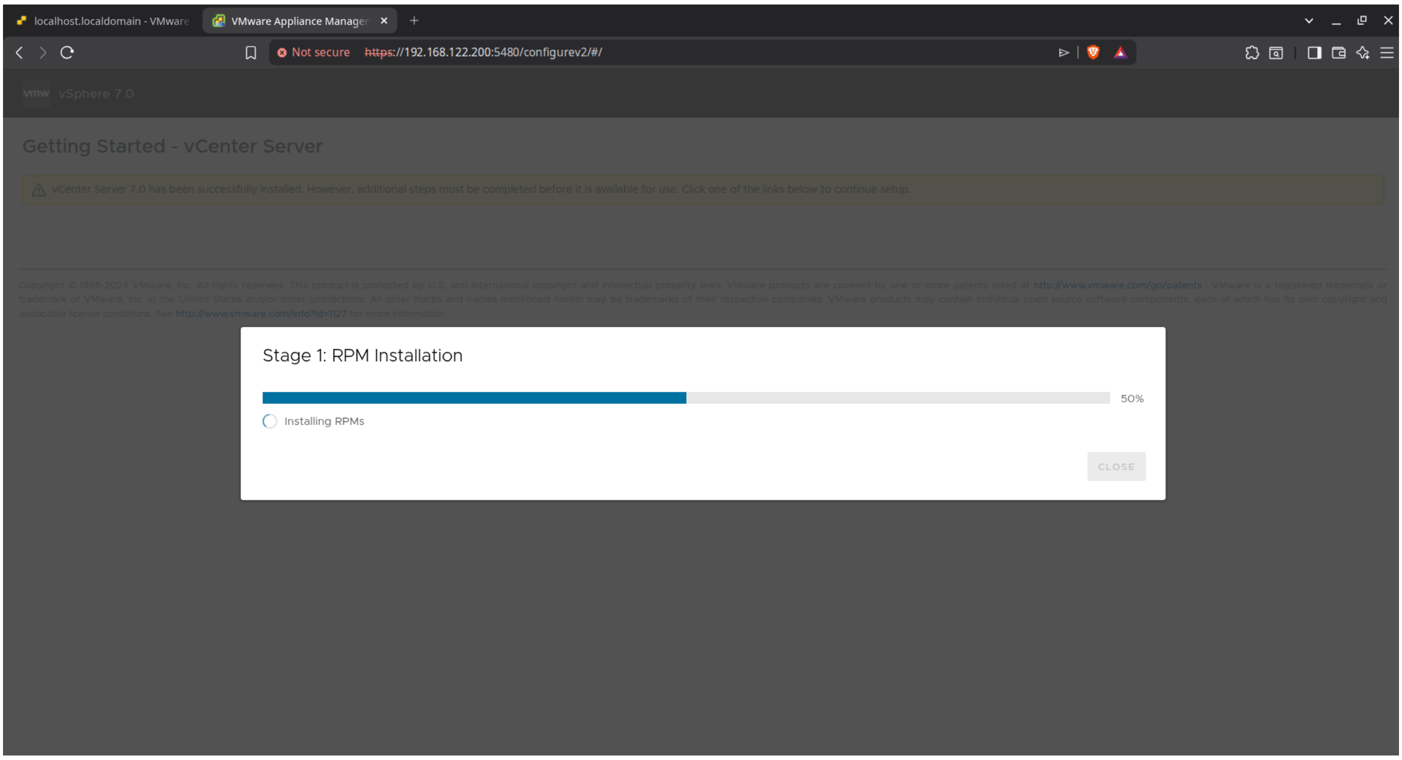

New virtual machine - vCenter Server
1 Select creation type
2 Select OVF and VMDK files
3 Select storage
4 License agreements
5 Deployment options
6 Additional settings
7 Ready to complete
License agreements
Read and accept the license agreements
End User License A...
Copyright and Pate...
Copyright © 1998-2024 VMware, Inc. All rights reserved. This product is protected by copyright and in
I agree
vmware
Back
Next
Finish
Cancel

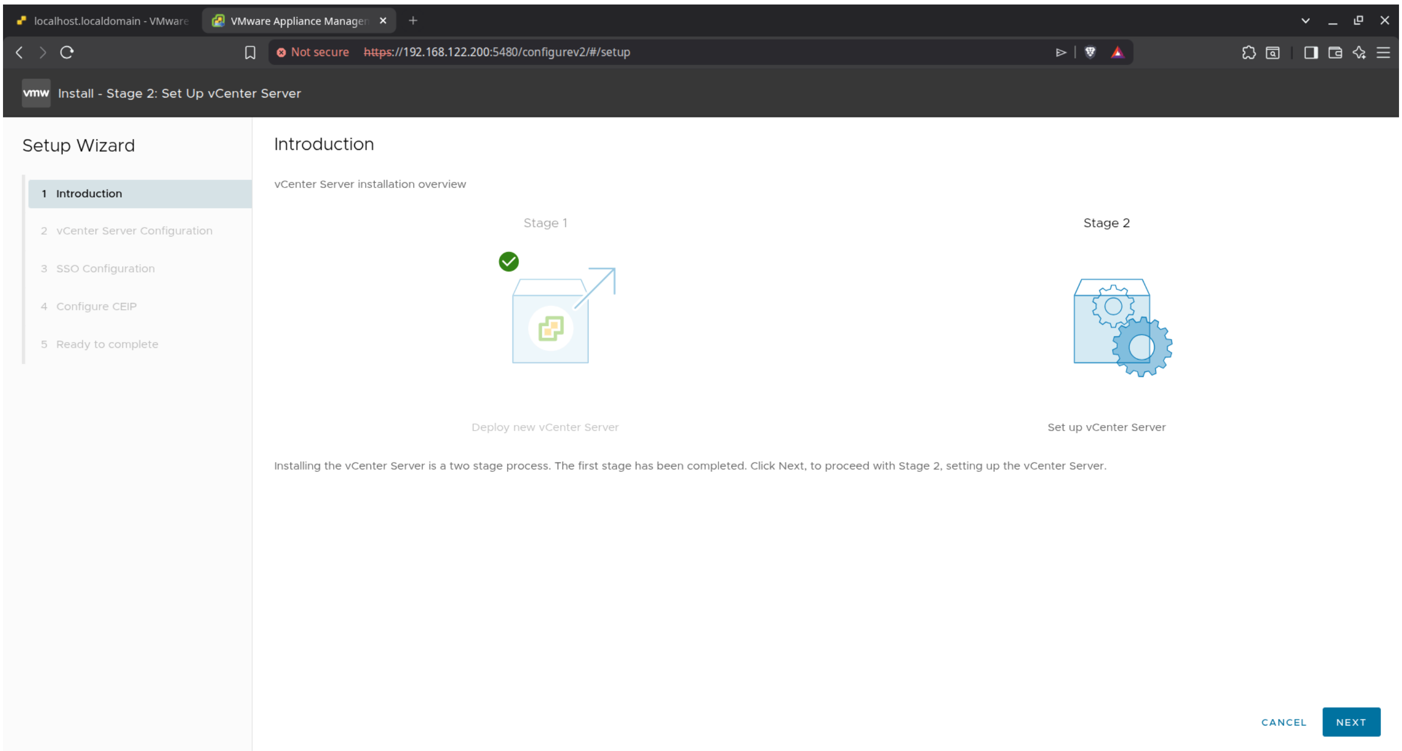

New virtual machine - vCenter Server
1 Select creation type
2 Select OVF and VMDK files
3 Select storage
4 License agreements
5 Deployment options
6 Additional settings
7 Ready to complete
Deployment options
Select deployment options
Network mappings
Network 1
VM Network
Deployment type
Tiny vCenter Server with Embedded PSC
This will deploy a Tiny VM configured with 2 vCPUs and 10 GB of memory and r equires 579 GB of disk space. This option contains vCenter Server with an emb edded Platform Services Controller for managing up to 10 hosts and 100 VMs.
Disk provisioning
Thin
Thick
Power on automatically
Back
Next
Finish
Cancel

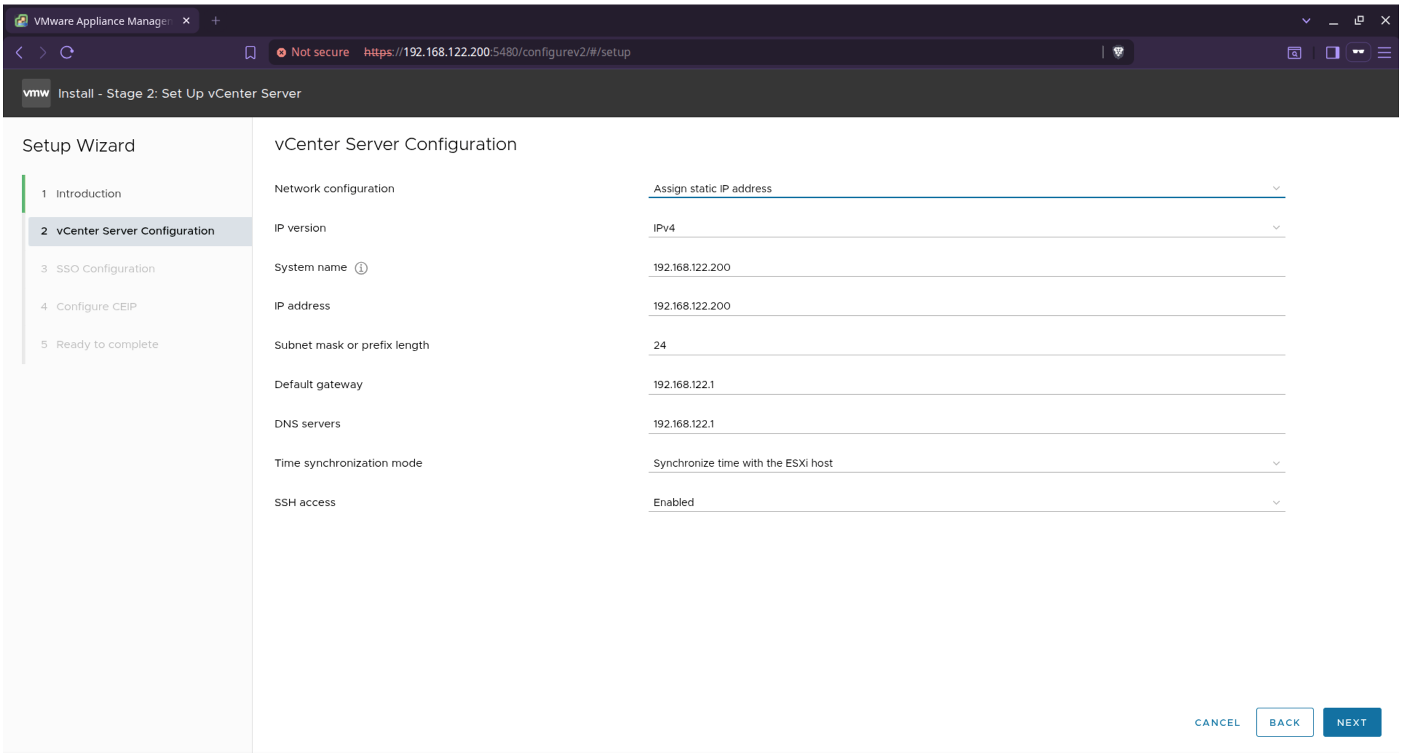

New virtual machine - vCenter Server
1 Select creation type
2 Select OVF and VMDK files
3 Select storage
4 License agreements
5 Deployment options
6 Additional settings
7 Ready to complete
Additional settings
Additional properties for the VM
Networking Configuration
Host Network IP Address Family
ipv4
Host Network Mode
static
Host Network IP Address
192.168.122.200
Host Network Prefix
24
Host Network Default Gateway
192.168.122.1
Host Network DNS Servers
192.168.122.1
Host Network Identity
192.168.122.200
SSO Configuration
Click to expand
System Configuration
Click to expand
Upgrade Configuration
Click to expand
Miscellaneous
Click to expand
Networking Properties
Click to expand
Back
Next
Finish
Cancel

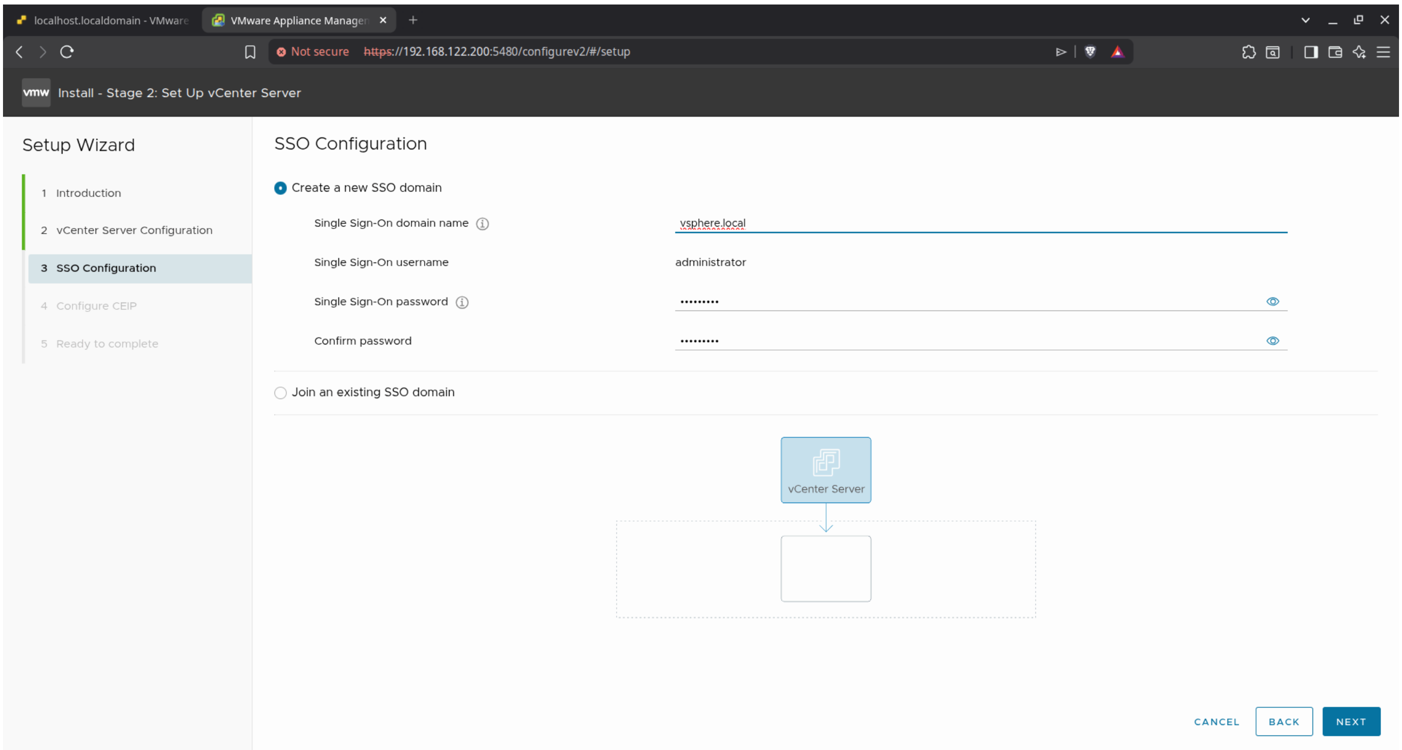

New virtual machine - vCenter Server
1 Select creation type
2 Select OVF and VMDK files
3 Select storage
4 License agreements
5 Deployment options
6 Additional settings
7 Ready to complete
Ready to complete
Review your settings selection before finishing the wizard
Product
VMware vCenter Server Appliance
VM Name
vCenter Server
Files
VMware-vCenter-Server-Appliance-7.0.3.02200-24322018_OVF10-disk1.vmdk
VMware-vCenter-Server-Appliance-7.0.3.02200-24322018_OVF10-disk2.vmdk
VMware-vCenter-Server-Appliance-7.0.3.02200-24322018_OVF10-disk3.vmdk
Datastore
vcenter-disk
Provisioning type
Thin
Network mappings
Network 1: VM Network
Guest OS Name
Photon OS
Profile
This will deploy a Tiny VM configured with 2 vCPUs and 10 GB of memory and r equires 579 GB of disk space. This option contains vCenter Server with an emb edded Platform Services Controller for managing up to 10 hosts and 100 VMs.
Properties
Host Network IP Address Family
ipv4
Host Network Mode
static
Host Network IP Address
192.168.122.200
Host Network Prefix
24
Host Network Default Gateway
192.168.122.1
Host Network DNS Servers
192.168.122.1
Root Password
****
vmware
Back
Next
Finish
Cancel

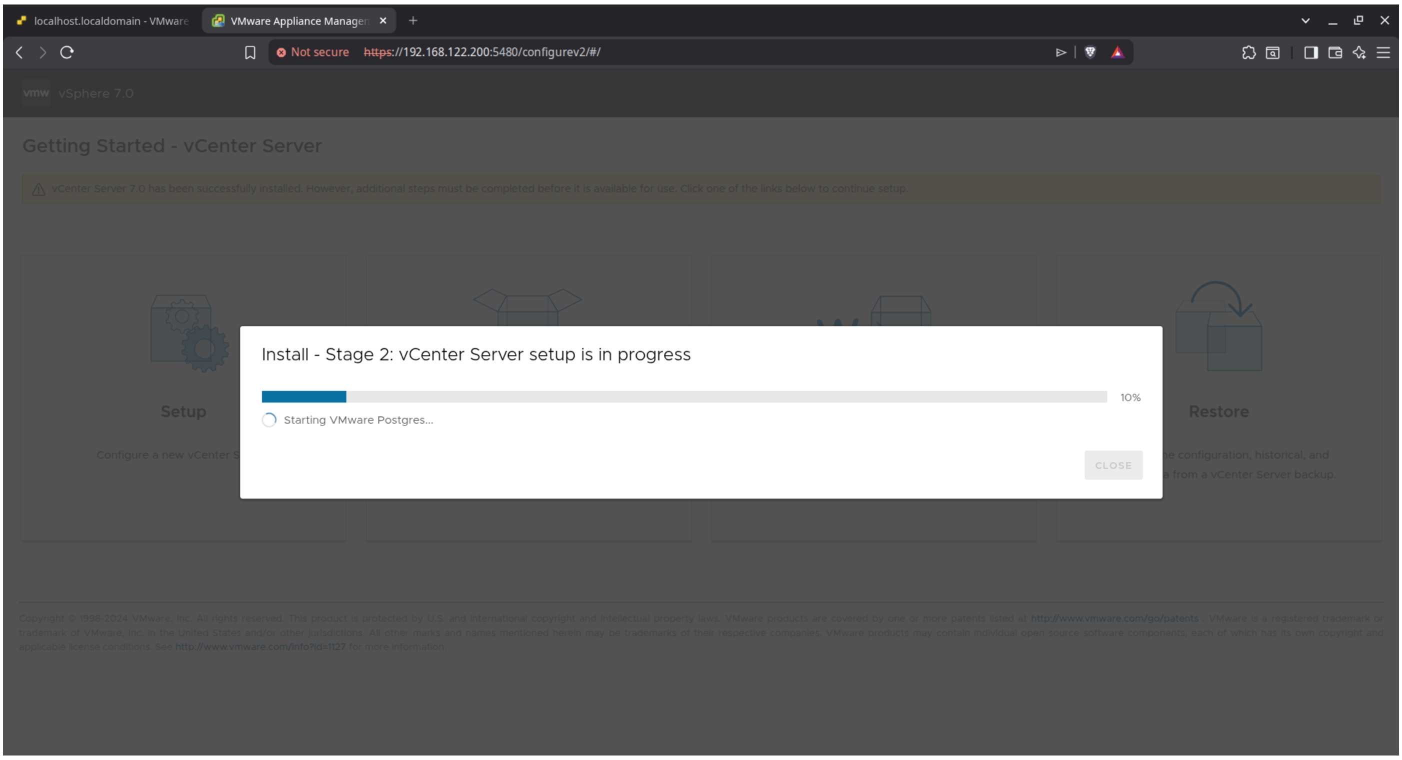

localhost.localdomain - VMw
VMware Appliance Manage
Not secure
https://192.168.122.200:5480/configurev2/#/
vSphere 7.0
Getting Started - vCenter Server
Login to vCenter Server Appliance
Username
root
Password
Enter Password
LOGIN

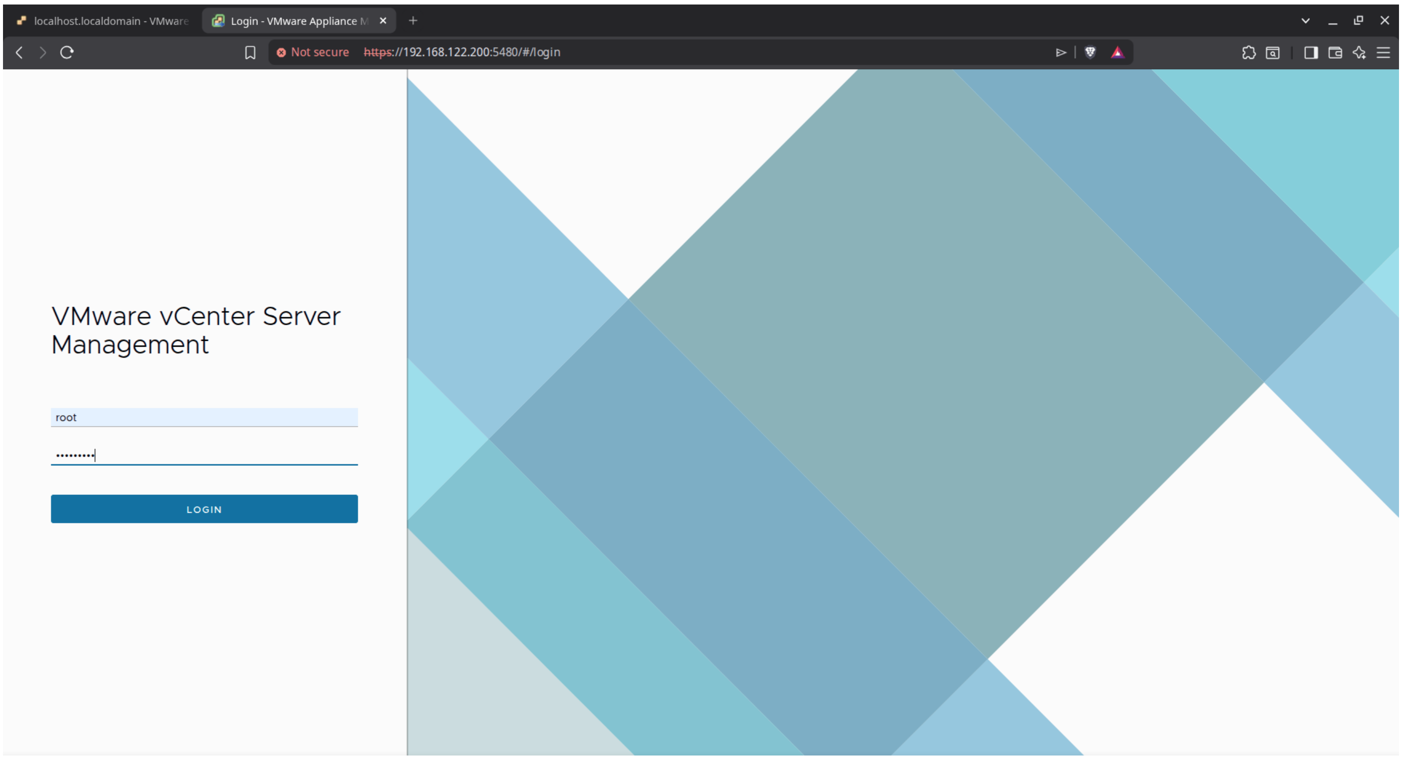

localhost.localdomain - VMware
VMware Appliance Manager
Not secure
https://192.168.122.200:5480/configurev2/#/
vSphere 7.0
Getting Started - vCenter Server
Stage 1: RPM Installation
50%
Installing RPMs
CLOSE

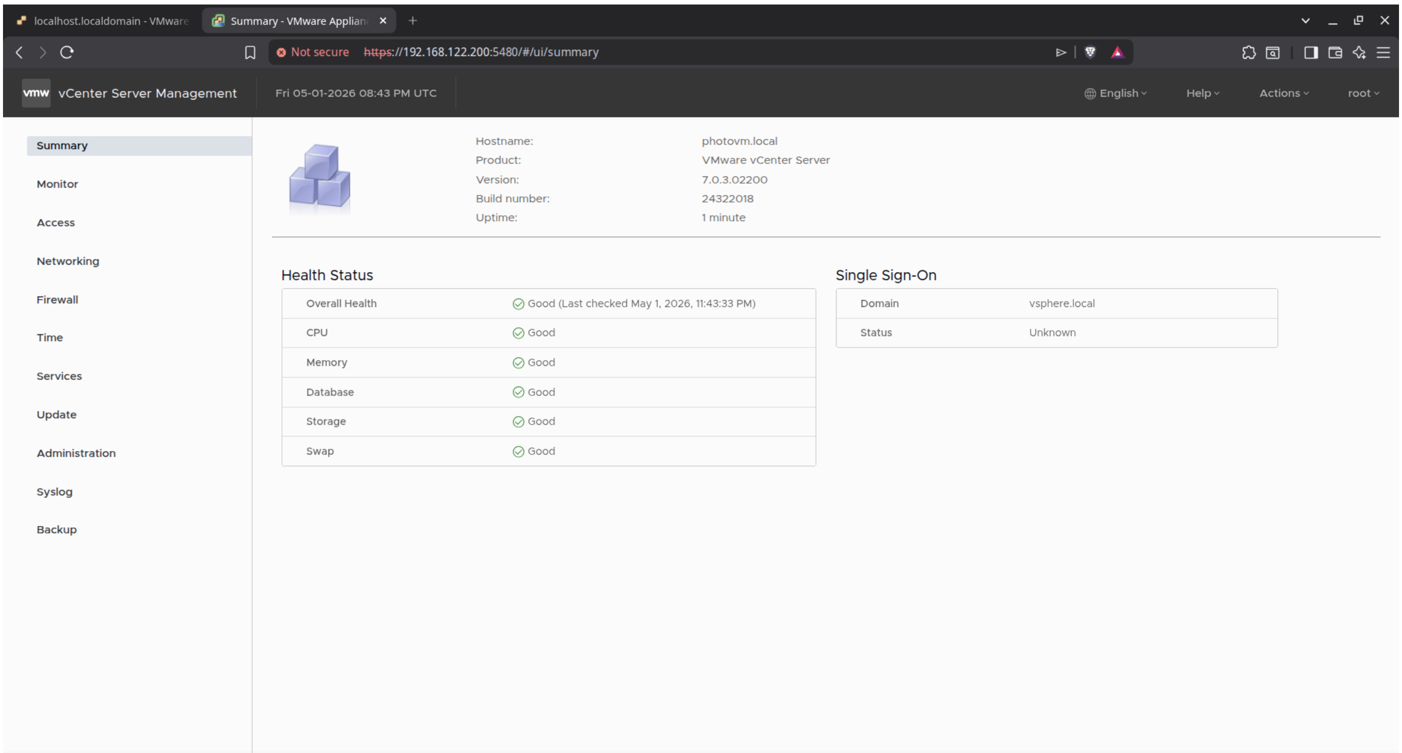

localhost.localdomain - VMware
VMware Appliance Manager
Not secure
https://192.168.122.200:5480/configurev2/#/setup
Install - Stage 2: Set Up vCenter Server
Setup Wizard
1 Introduction
2 vCenter Server Configuration
3 SSO Configuration
4 Configure CEIP
5 Ready to complete
Introduction
vCenter Server installation overview
Stage 1
Stage 2
Deploy new vCenter Server
Set up vCenter Server
Installing the vCenter Server is a two stage process. The first stage has been completed. Click Next, to proceed with Stage 2, setting up the vCenter Server.
CANCEL
NEXT


VMware Appliance Management
Not secure https://192.168.122.200:5480/configurev2/#/setup
Install - Stage 2: Set Up vCenter Server
Setup Wizard
1 Introduction
2 vCenter Server Configuration
3 SSO Configuration
4 Configure CEIP
5 Ready to complete
vCenter Server Configuration
Network configuration
Assign static IP address
IP version
IPv4
System name
192.168.122.200
IP address
192.168.122.200
Subnet mask or prefix length
24
Default gateway
192.168.122.1
DNS servers
192.168.122.1
Time synchronization mode
Synchronize time with the ESXi host
SSH access
Enabled
CANCEL
BACK
NEXT

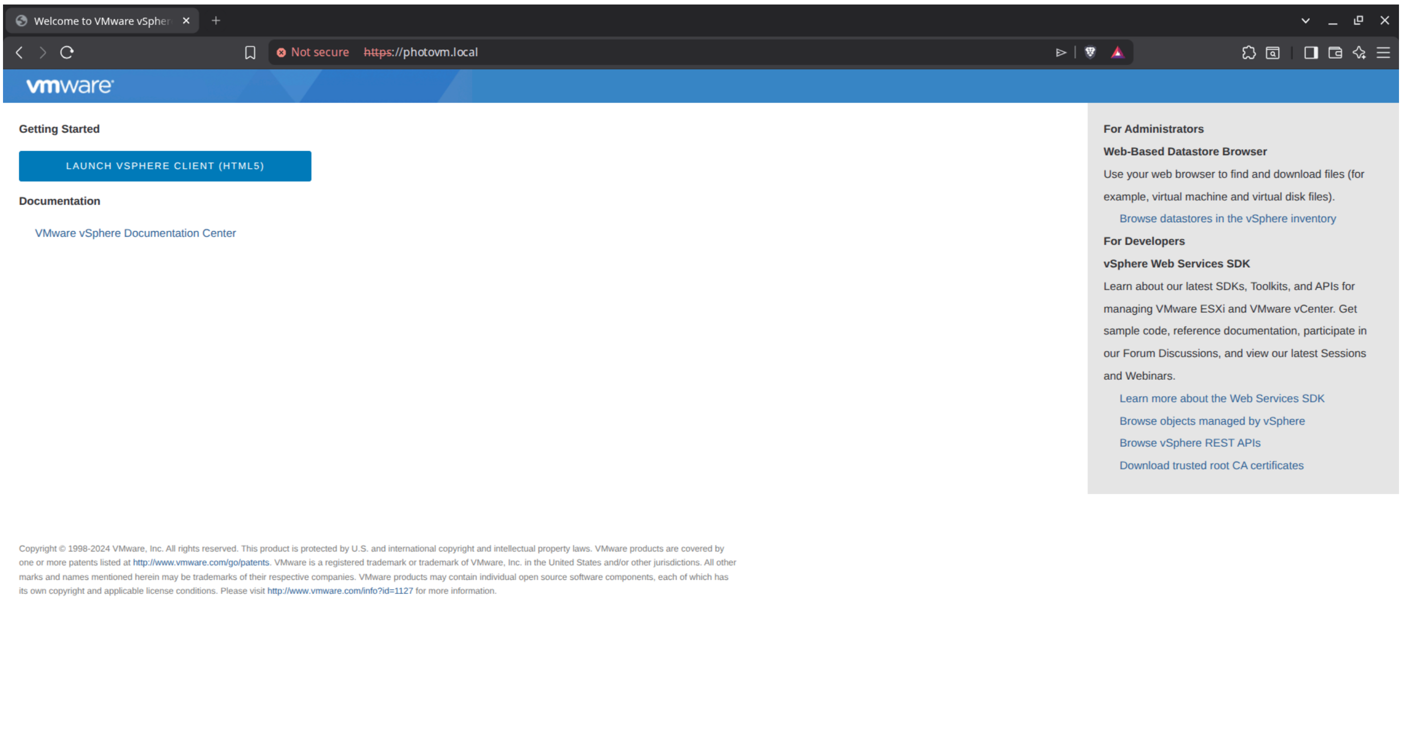

localhost.localdomain - VMware
VMware Appliance Management
Not secure https://192.168.122.200:5480/configurev2/#/setup
Install - Stage 2: Set Up vCenter Server
Setup Wizard
1 Introduction
2 vCenter Server Configuration
3 SSO Configuration
4 Configure CEIP
5 Ready to complete
SSO Configuration
Create a new SSO domain
Single Sign-On domain name
vsphere.local
Single Sign-On username
administrator
Single Sign-On password
Confirm password
Join an existing SSO domain
vCenter Server
CANCEL
BACK
NEXT

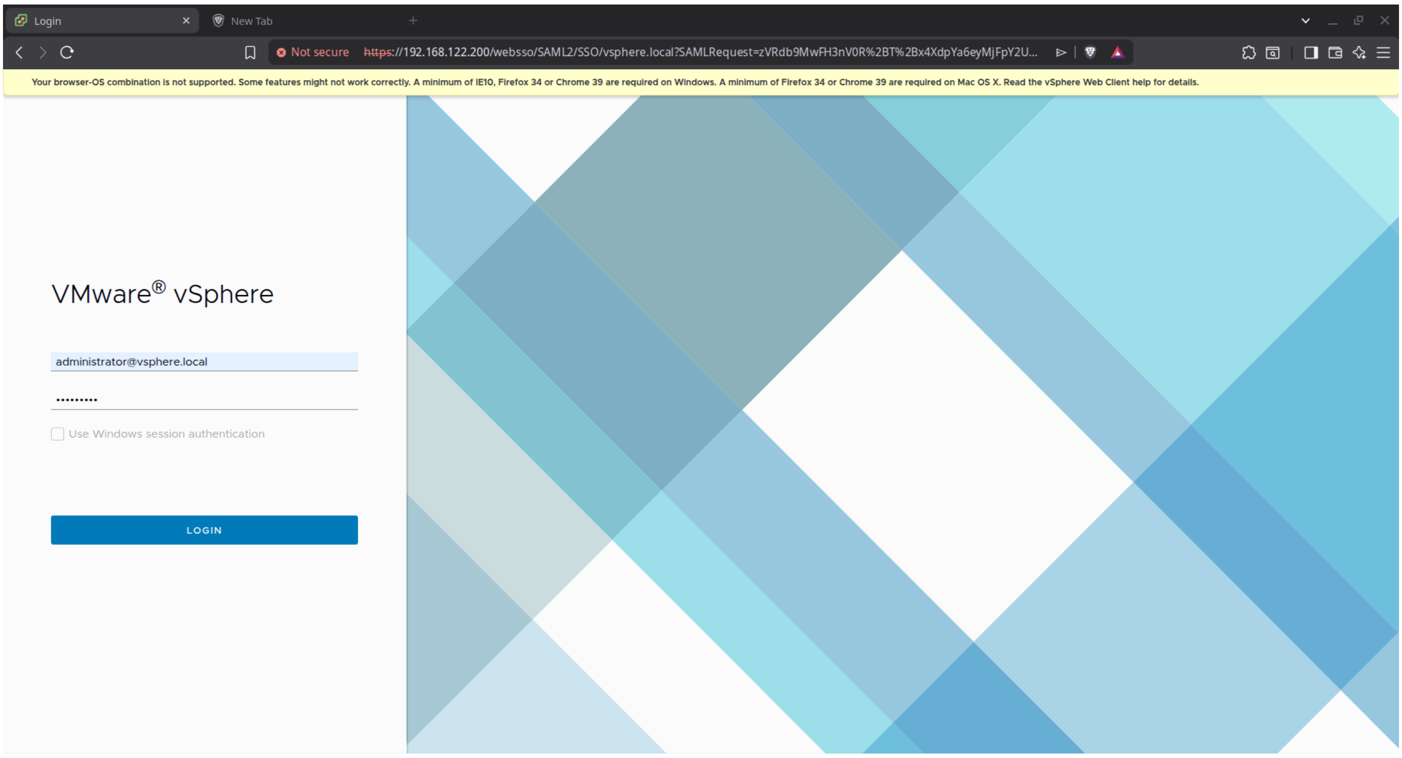

localhost.localdomain - VMware
VMware Appliance Manager
Not secure
https://192.168.122.200:5480/configurev2/#/
Install - Stage 2: vCenter Server setup is in progress
10%
Starting VMware Postgres...
CLOSE

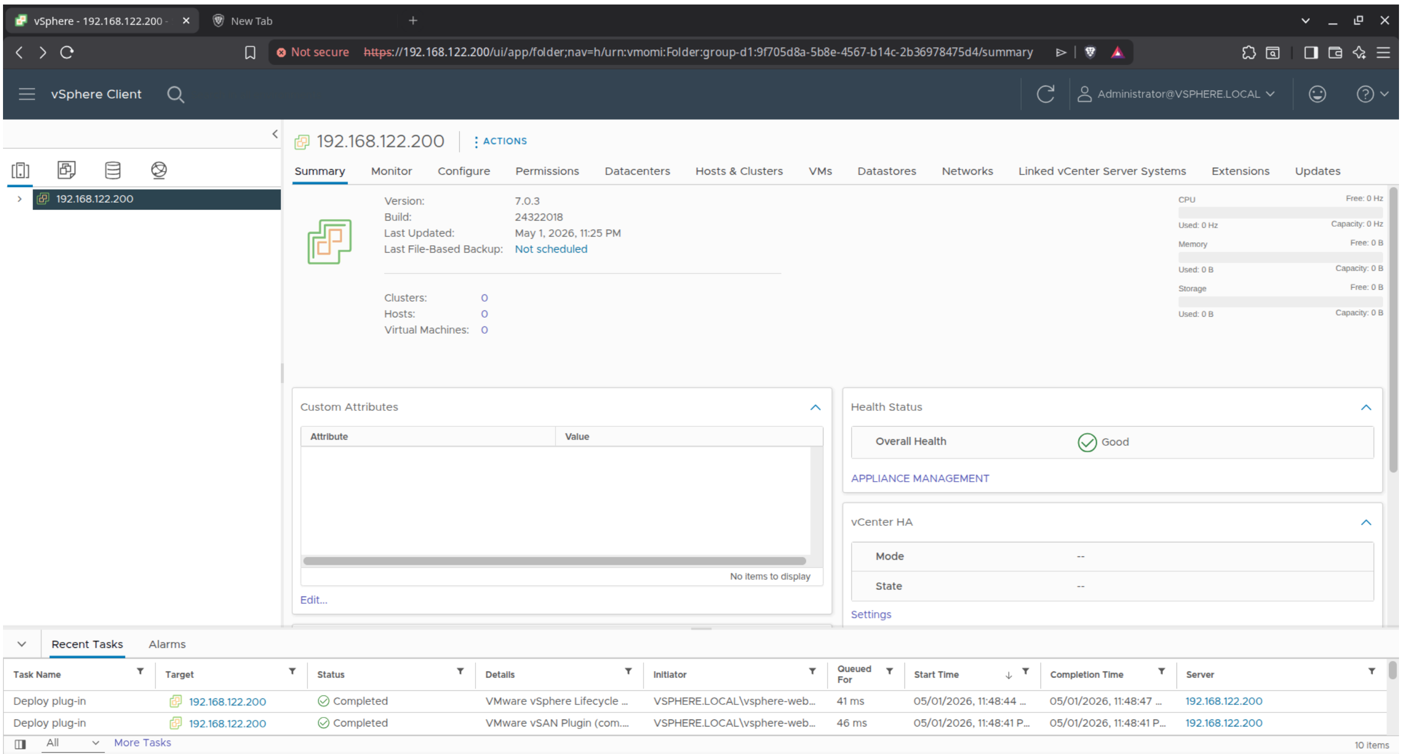

localhost.localdomain - VMware
Login - VMware Appliance M
Not secure
https://192.168.122.200:5480/#/login
VMware vCenter Server Management
root
LOGIN

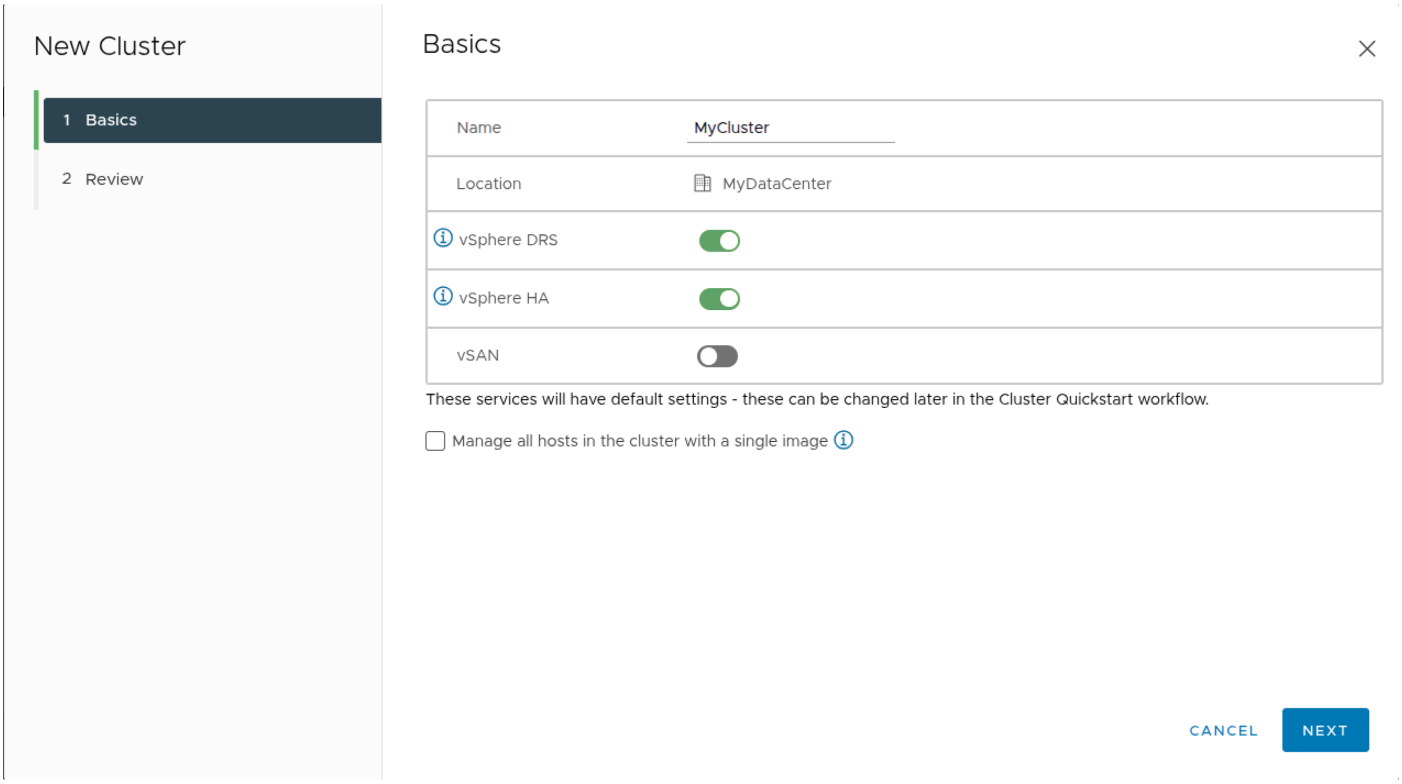


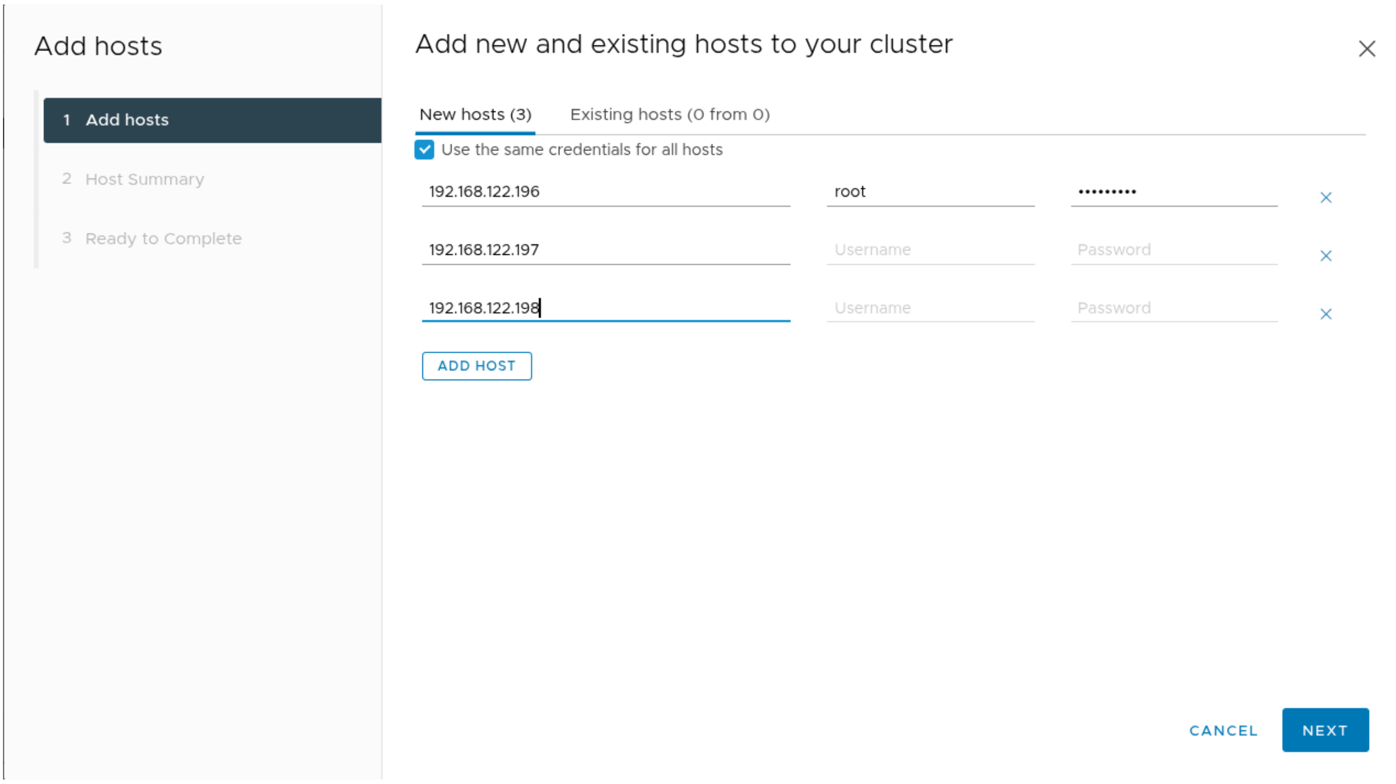
Adding ESXI hosts to Cluster:

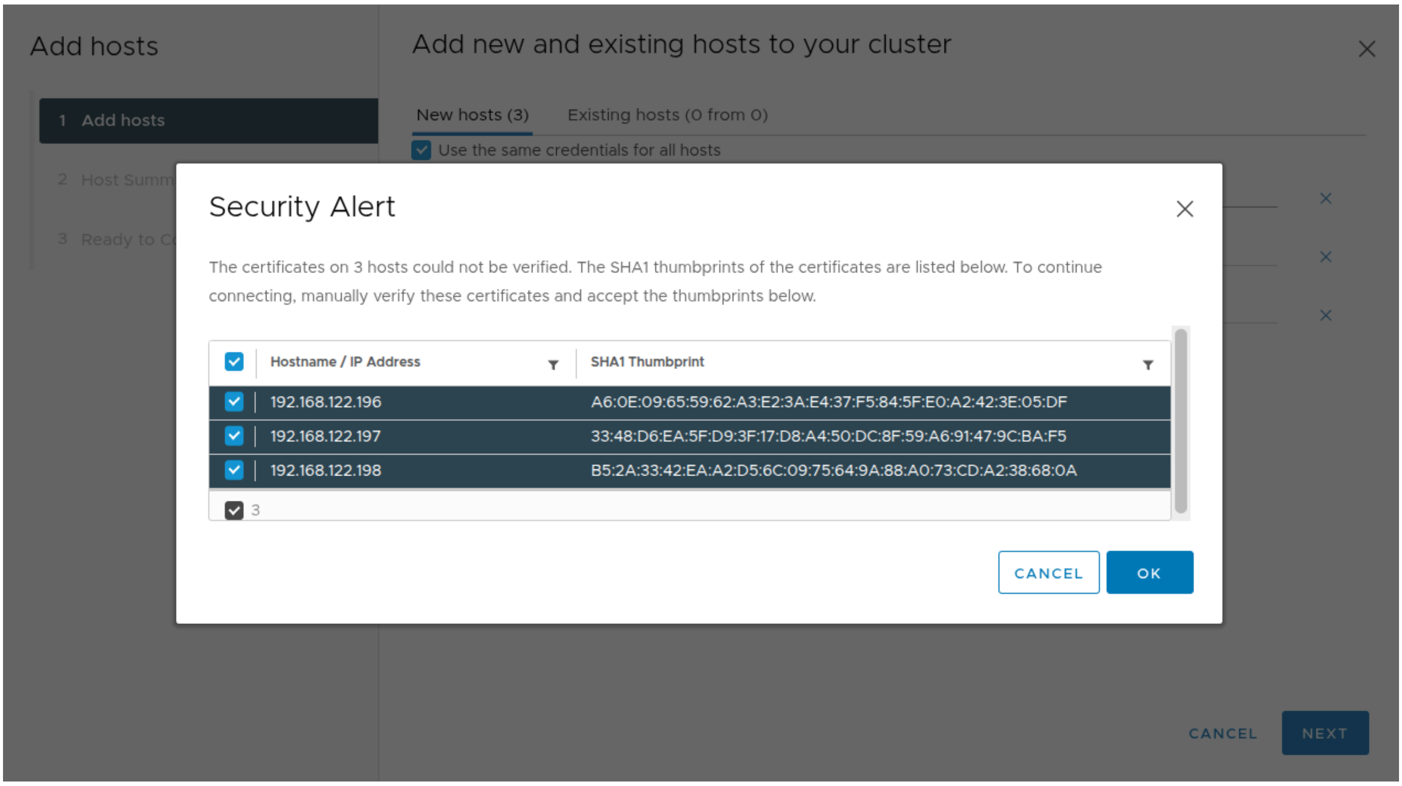


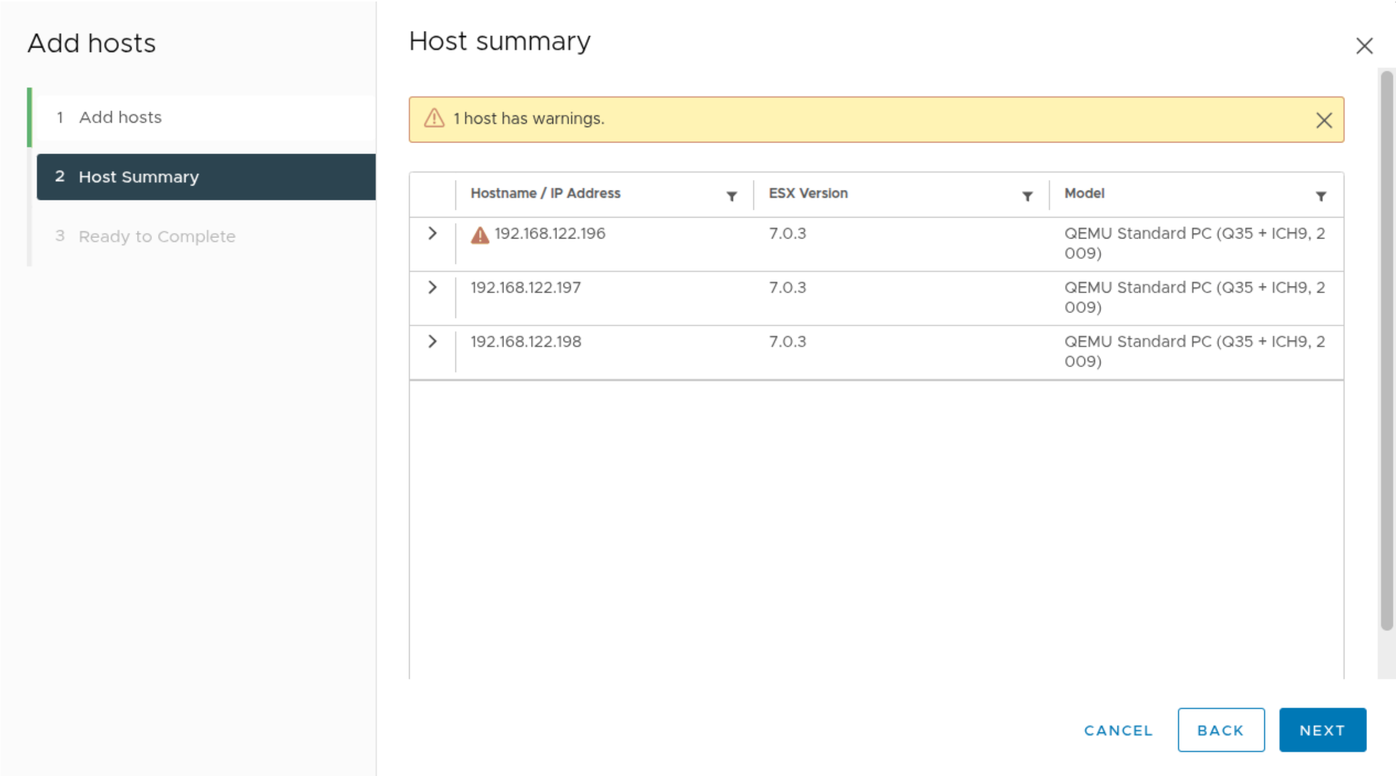

Login
New Tab
Not secure https://192.168.122.200/websso/SAML2/SSO/vsphere.local?SAMLRequest=zVRdb9MwFH3nV0R%2BT%2Bx4XdpYa6eyMjFpY2U...
Your browser-OS combination is not supported. Some features might not work correctly. A minimum of IE10, Firefox 34 or Chrome 39 are required on Windows. A minimum of Firefox 34 or Chrome 39 are required on Mac OS X. Read the vSphere Web Client help for details.
VMware® vSphere
administrator@vsphere.local
Use Windows session authentication
LOGIN

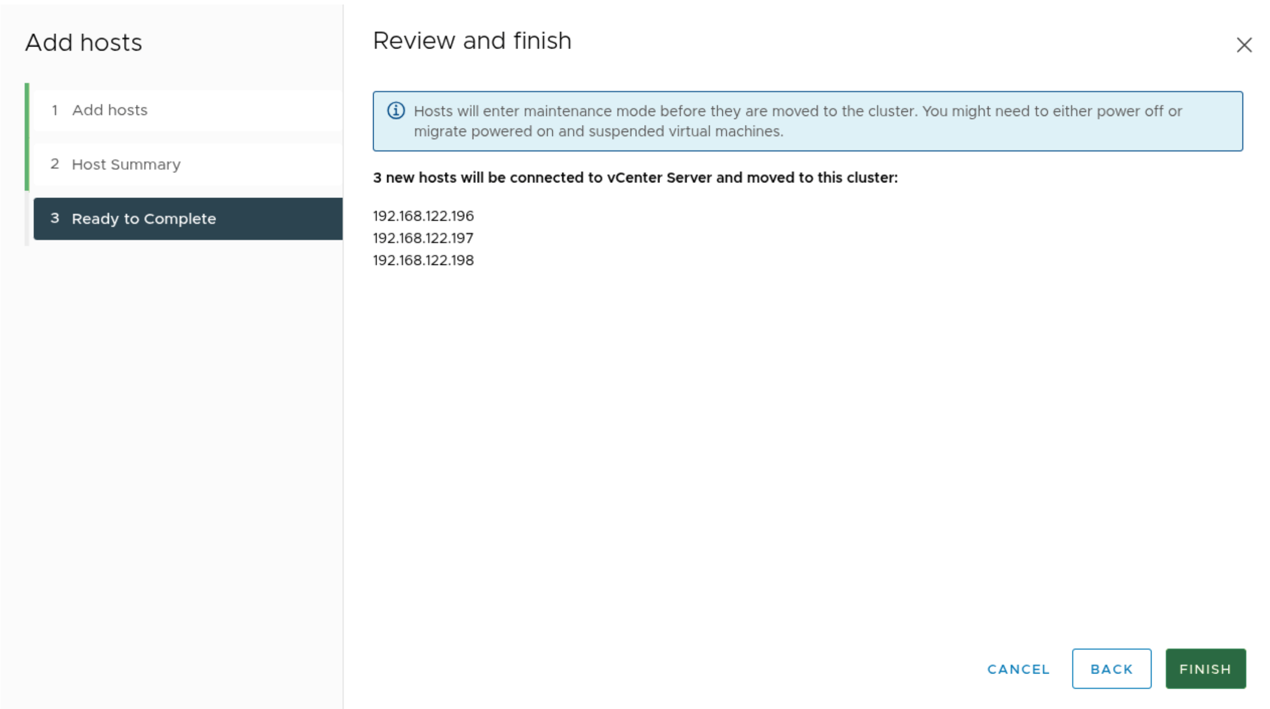

vSphere - 192.168.122.200
New Tab
Not secure https://192.168.122.200/ui/app/folder;nav=h/urn:vmomi:Folder:group-d1:9f705d8a-5b8e-4567-b14c-2b36978475d4/summary
vSphere Client
Administrator@VSPHERE.LOCAL
192.168.122.200
ACTIONS
Summary Monitor Configure Permissions Datacenters Hosts & Clusters VMs Datastores Networks Linked vCenter Server Systems Extensions Updates
Version: 7.0.3
Build: 24322018
Last Updated: May 1, 2026, 11:25 PM
Last File-Based Backup: Not scheduled
Clusters: 0
Hosts: 0
Virtual Machines: 0
Custom Attributes
Attribute
Value
No items to display
Edit...
Health Status
Overall Health
Good
APPLIANCE MANAGEMENT
vCenter HA
Mode --
State --
Settings
Recent Tasks
Alarms
Task Name
Target
Status
Details
Initiator
Queued For
Start Time
Completion Time
Server
Deploy plug-in 192.168.122.200 Completed VMware vSphere Lifecycle ... VSPHERE.LOCAL\vsphere-web... 41 ms 05/01/2026, 11:48:44 ... 05/01/2026, 11:48:47 ... 192.168.122.200
Deploy plug-in 192.168.122.200 Completed VMware vSAN Plugin (com.... VSPHERE.LOCAL\vsphere-web... 46 ms 05/01/2026, 11:48:41 P... 05/01/2026, 11:48:41 P... 192.168.122.200
All
More Tasks
10 items


New Cluster
1 Basics
2 Review
Basics
Name
MyCluster
Location
MyDataCenter
vSphere DRS
vSphere HA
vSAN
These services will have default settings - these can be changed later in the Cluster Quickstart workflow.
Manage all hosts in the cluster with a single image
CANCEL
NEXT


Add hosts
1 Add hosts
2 Host Summary
3 Ready to Complete
Add new and existing hosts to your cluster
New hosts (3)
Existing hosts (0 from 0)
Use the same credentials for all hosts
192.168.122.196
root
192.168.122.197
Username
Password
192.168.122.198
Username
Password
ADD HOST
CANCEL
NEXT

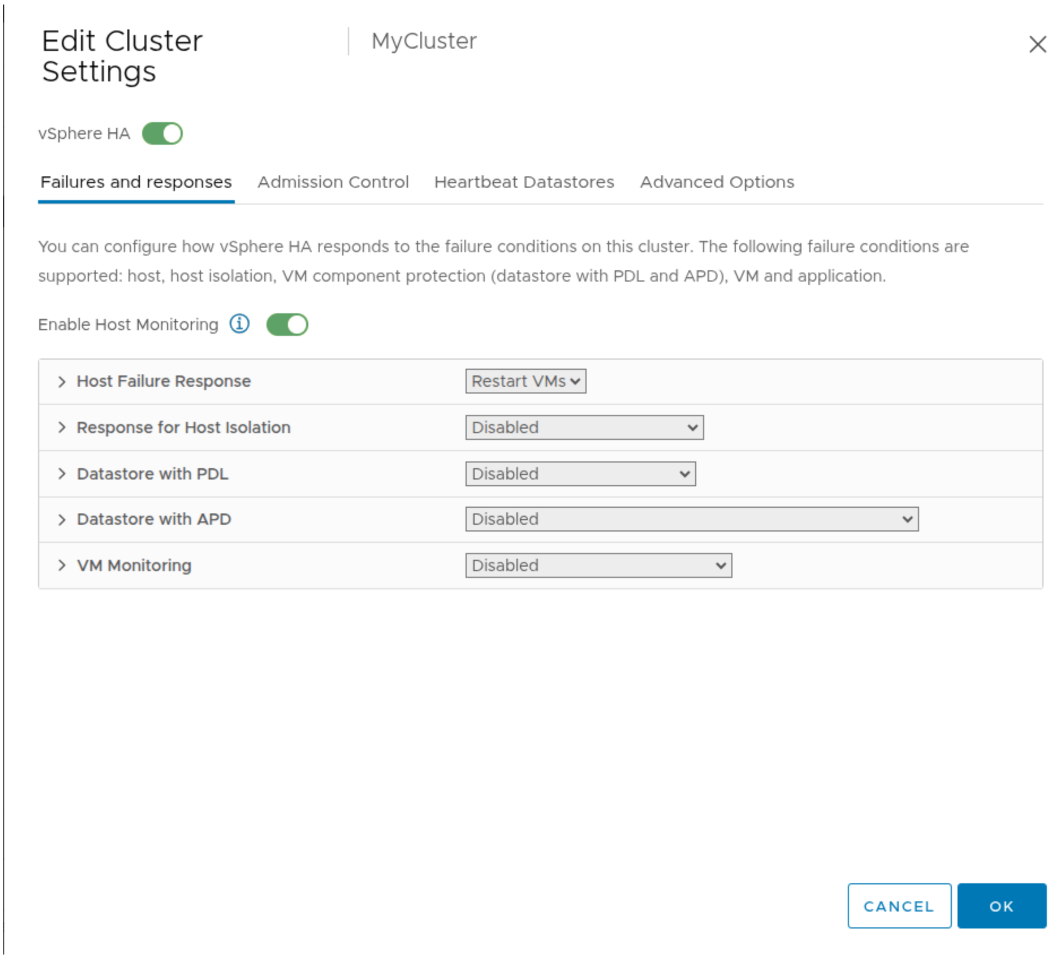

Add hosts
1 Add hosts
Add new and existing hosts to your cluster
New hosts (3)
Existing hosts (0 from 0)
Use the same credentials for all hosts
Security Alert
The certificates on 3 hosts could not be verified. The SHA1 thumbprints of the certificates are listed below. To continue connecting, manually verify these certificates and accept the thumbprints below.
Hostname / IP Address
SHA1 Thumbprint
192.168.122.196
A6:0E:09:65:59:62:A3:E2:3A:E4:37:F5:84:5F:E0:A2:42:3E:05:DF
192.168.122.197
33:48:D6:EA:5F:D9:3F:17:D8:A4:50:DC:8F:59:A6:91:47:9C:BA:F5
192.168.122.198
B5:2A:33:42:EA:A2:D5:6C:09:75:64:9A:88:A0:73:CD:A2:38:68:0A
3
CANCEL
OK
CANCEL
NEXT


Add hosts
1 Add hosts
2 Host Summary
3 Ready to Complete
Host summary
1 host has warnings.
Hostname / IP Address
ESX Version
Model
192.168.122.196
7.0.3
QEMU Standard PC (Q35 + ICH9, 2009)
192.168.122.197
7.0.3
QEMU Standard PC (Q35 + ICH9, 2009)
192.168.122.198
7.0.3
QEMU Standard PC (Q35 + ICH9, 2009)
CANCEL
BACK
NEXT


Add hosts
1 Add hosts
2 Host Summary
3 Ready to Complete
Review and finish
Hosts will enter maintenance mode before they are moved to the cluster. You might need to either power off or migrate powered on and suspended virtual machines.
3 new hosts will be connected to vCenter Server and moved to this cluster:
192.168.122.196
192.168.122.197
192.168.122.198
CANCEL
BACK
FINISH

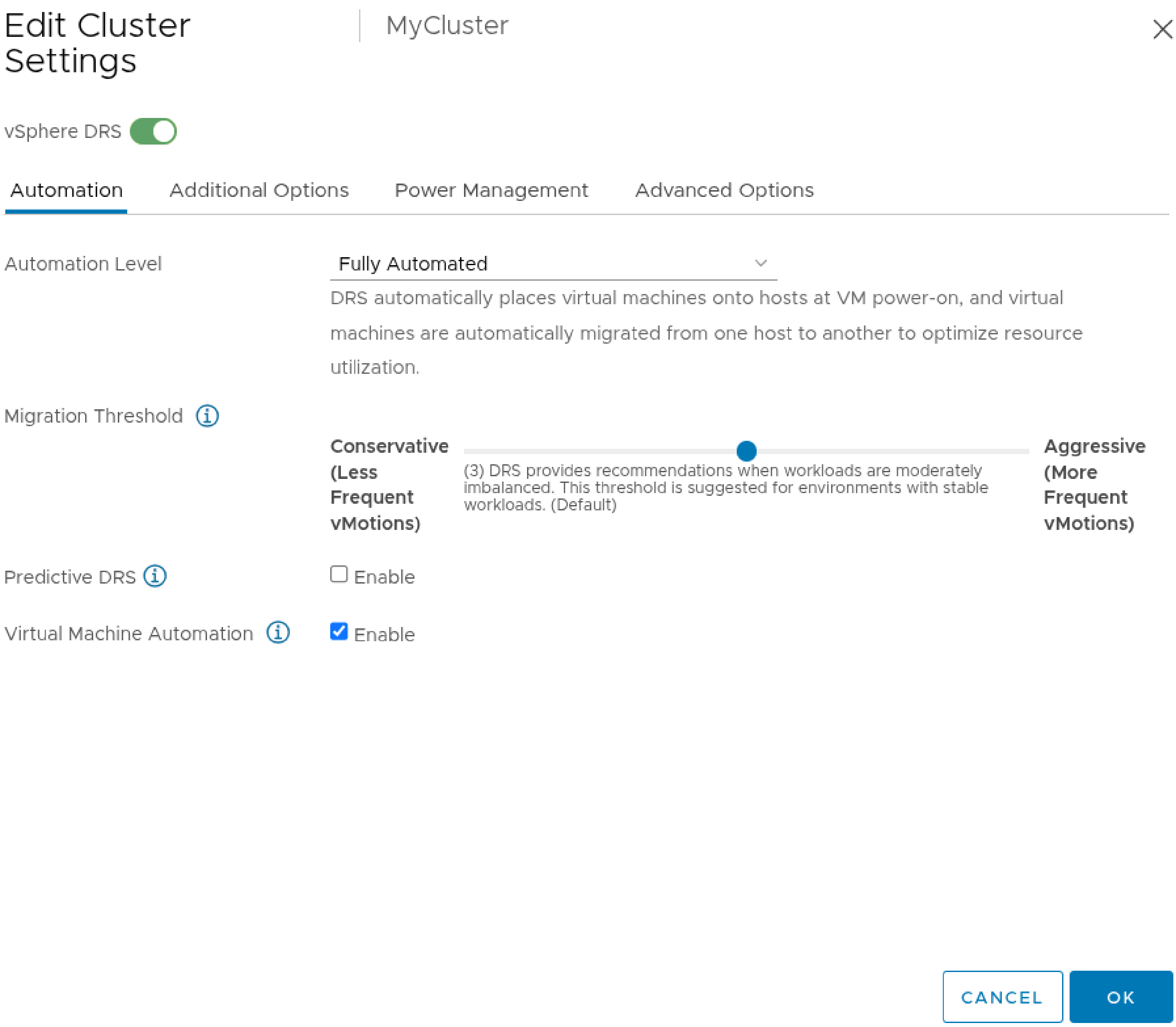
Configuring HA for the vSphere cluster:

Configuring the High Availability (HA) settings for the cluster. We enable "Host Monitoring" and set the "Host Failure Response" to Restart VMs, ensuring that if a physical server fails, its virtual machines will automatically reboot on another healthy host.



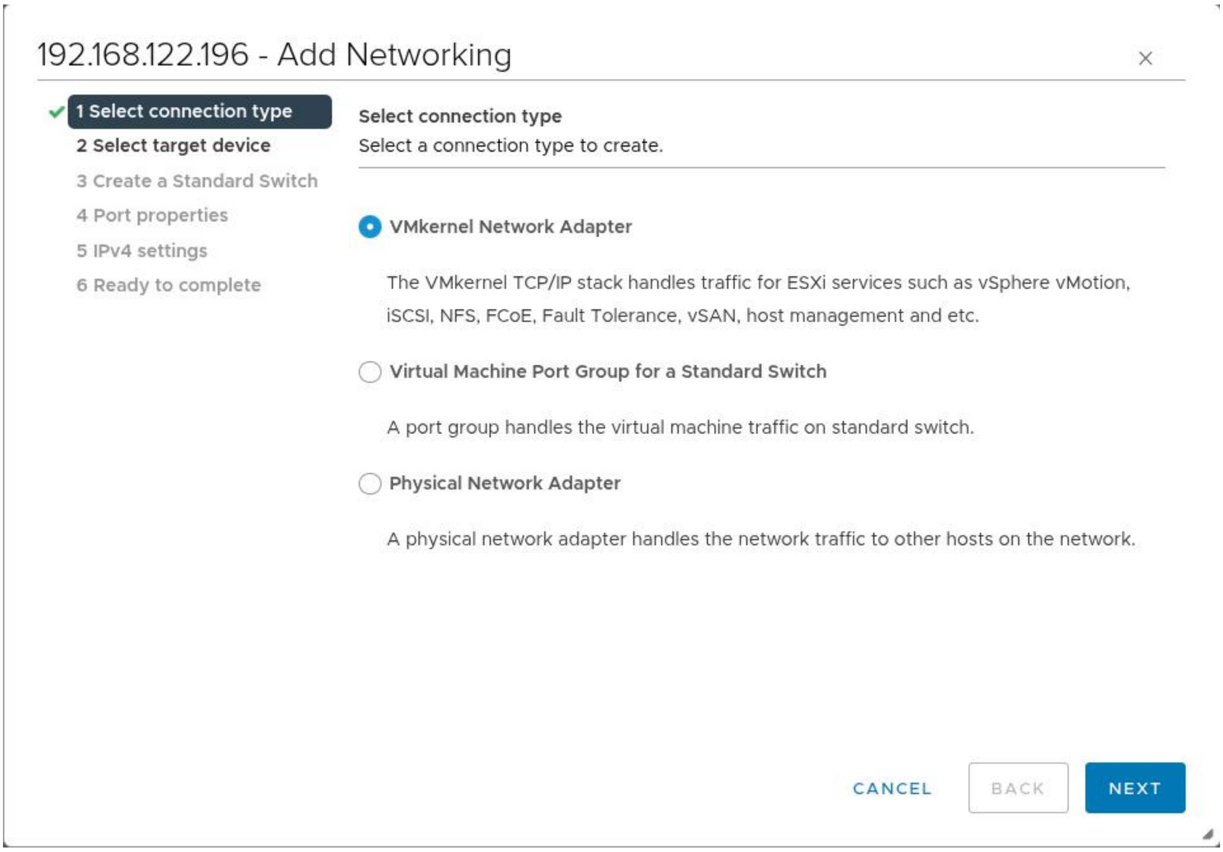
Configuring vSphere DRS:

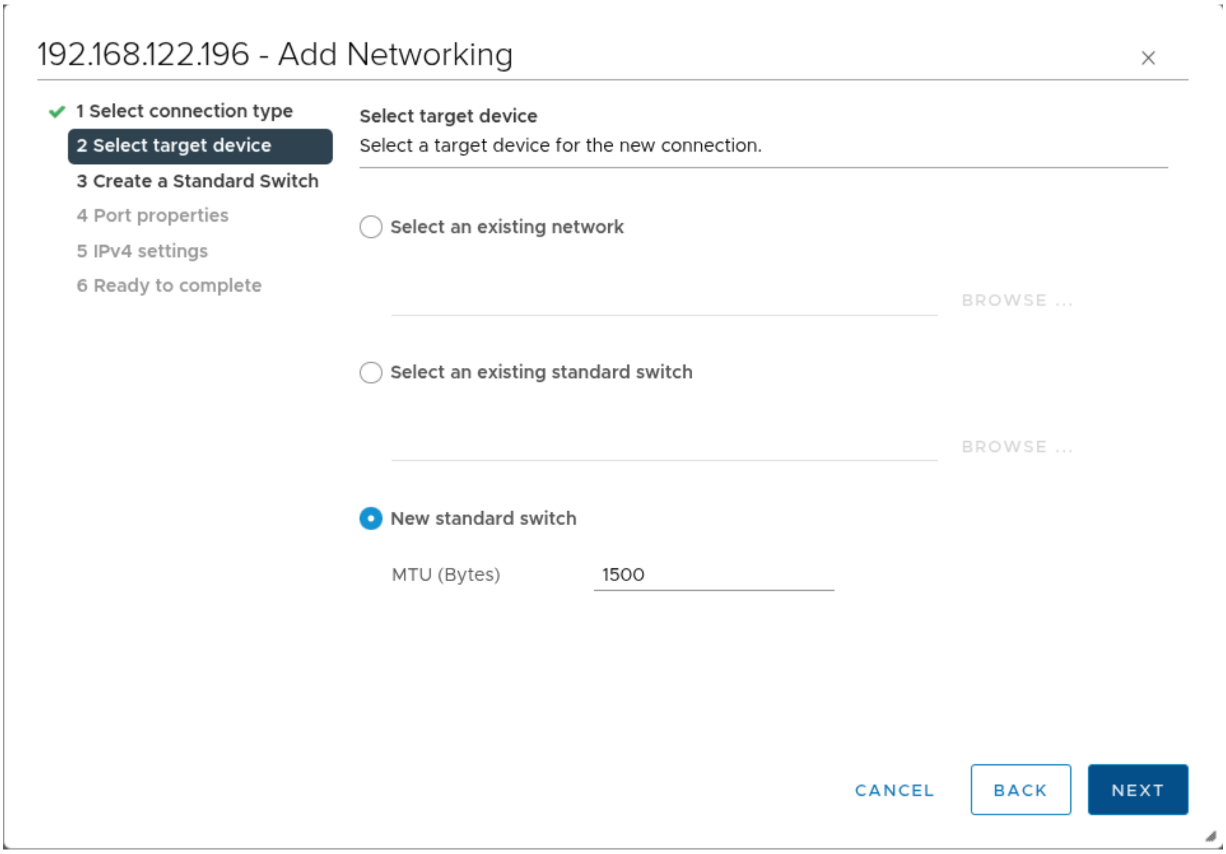
Activating the Distributed Resource Scheduler (DRS) to automate workload balancing. The automation level is set to Fully Automated, allowing vSphere to live-migrate VMs (via vMotion) between hosts to optimize CPU and Memory utilization without manual intervention

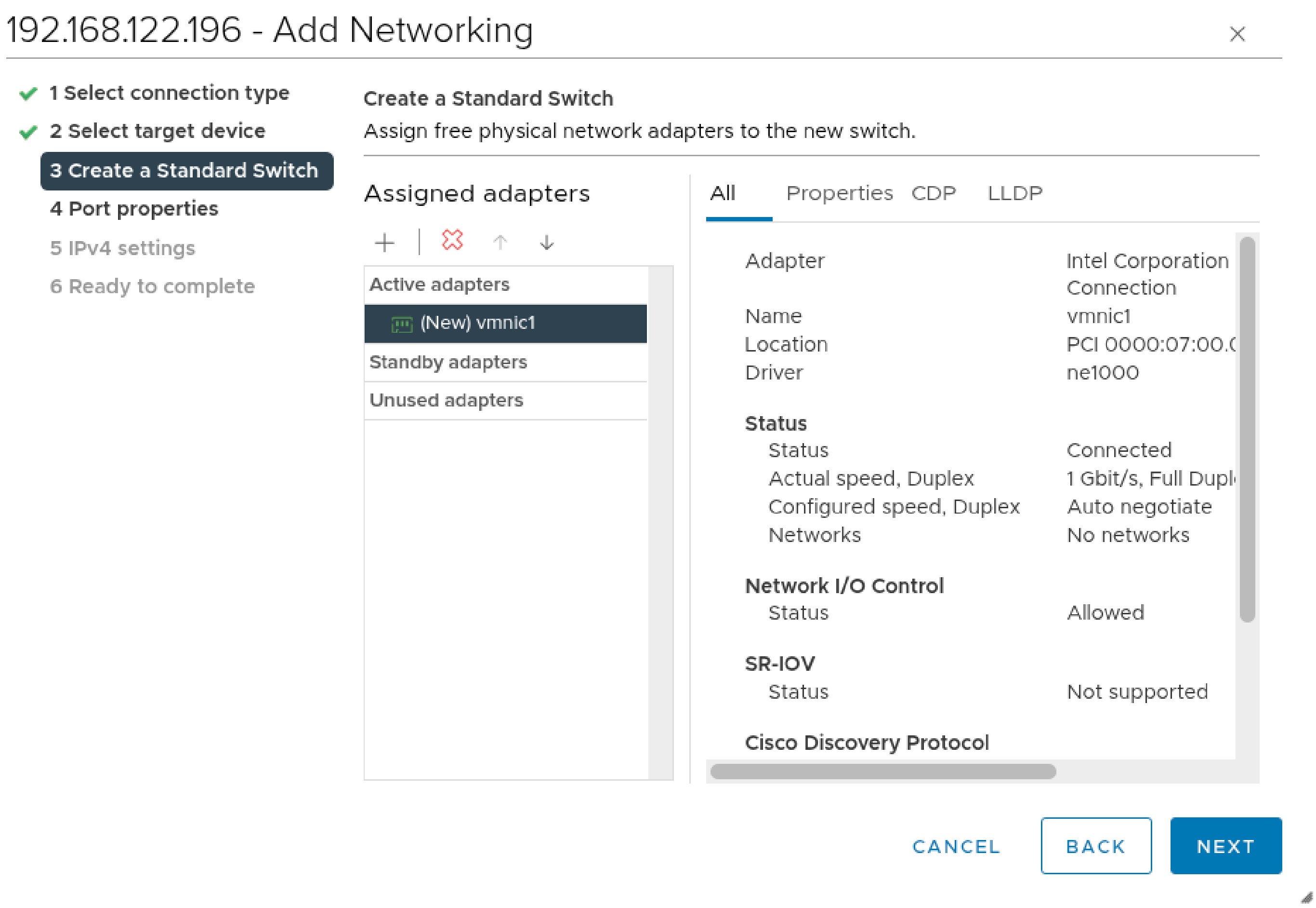


Network Configuration:

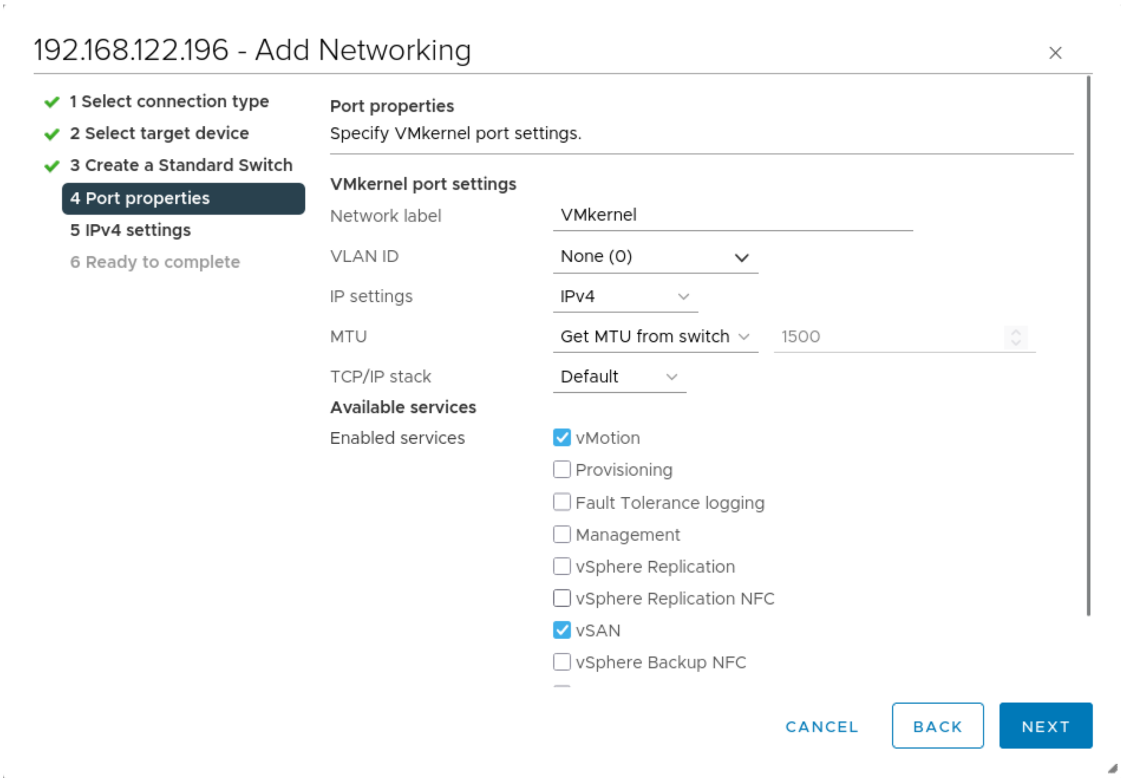
Creating a new Standard Switch (vSwitch1) on host 192.168.122.196 (VCA host) and assigning a free physical adapter (vmnic1) to handle specialized traffic.

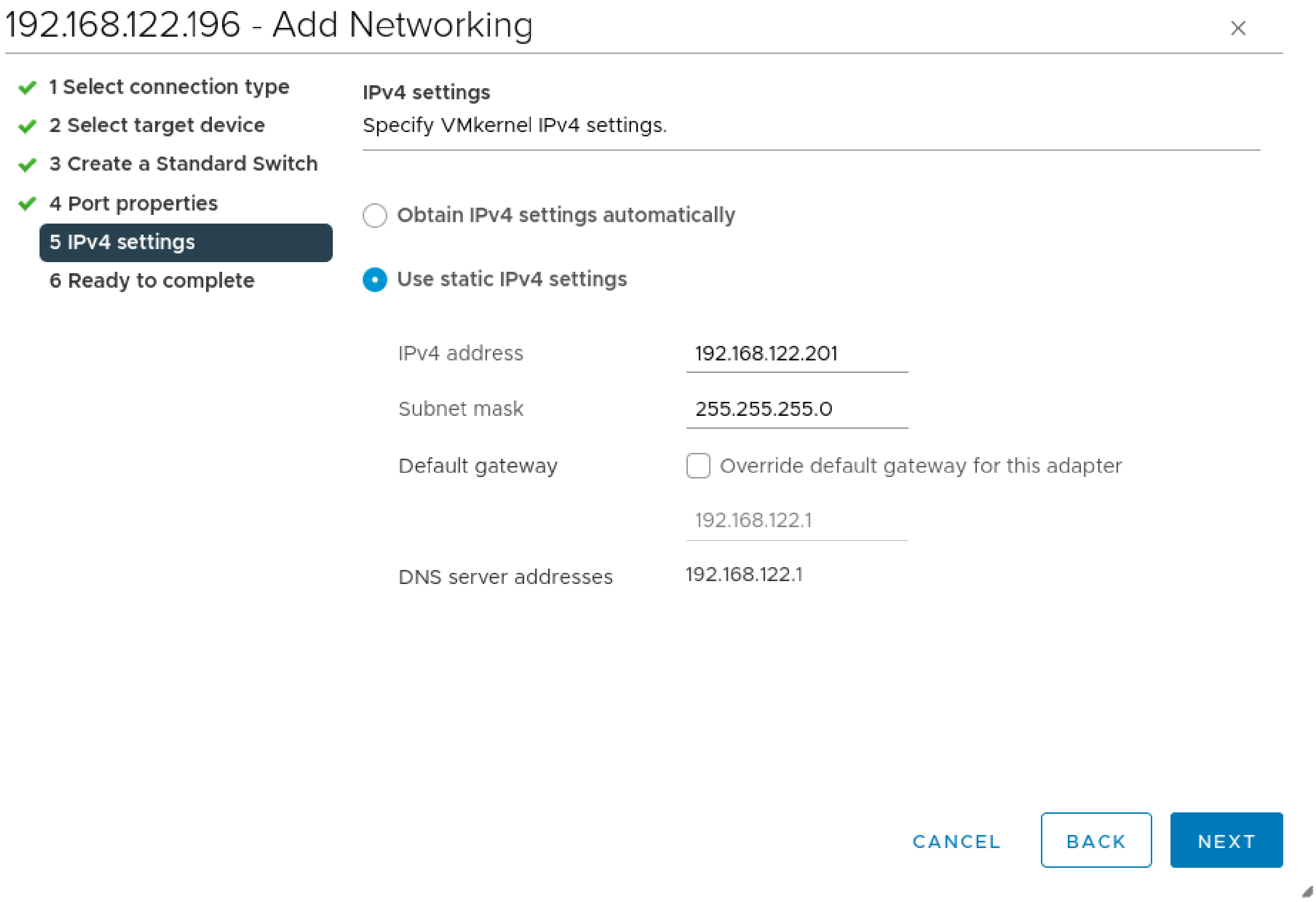

192.168.122.196 - Add Networking
1 Select connection type
2 Select target device
3 Create a Standard Switch
4 Port properties
5 IPv4 settings
6 Ready to complete
Select connection type
Select a connection type to create.
VMkernel Network Adapter
The VMkernel TCP/IP stack handles traffic for ESXi services such as vSphere vMotion, iSCSI, NFS, FCoE, Fault Tolerance, vSAN, host management and etc.
Virtual Machine Port Group for a Standard Switch
A port group handles the virtual machine traffic on standard switch.
Physical Network Adapter
A physical network adapter handles the network traffic to other hosts on the network.
CANCEL
BACK
NEXT


192.168.122.196 - Add Networking
1 Select connection type
2 Select target device
3 Create a Standard Switch
4 Port properties
5 IPv4 settings
6 Ready to complete
Select target device
Select a target device for the new connection.
Select an existing network
BROWSE ...
Select an existing standard switch
BROWSE ...
New standard switch
MTU (Bytes)
1500
CANCEL
BACK
NEXT

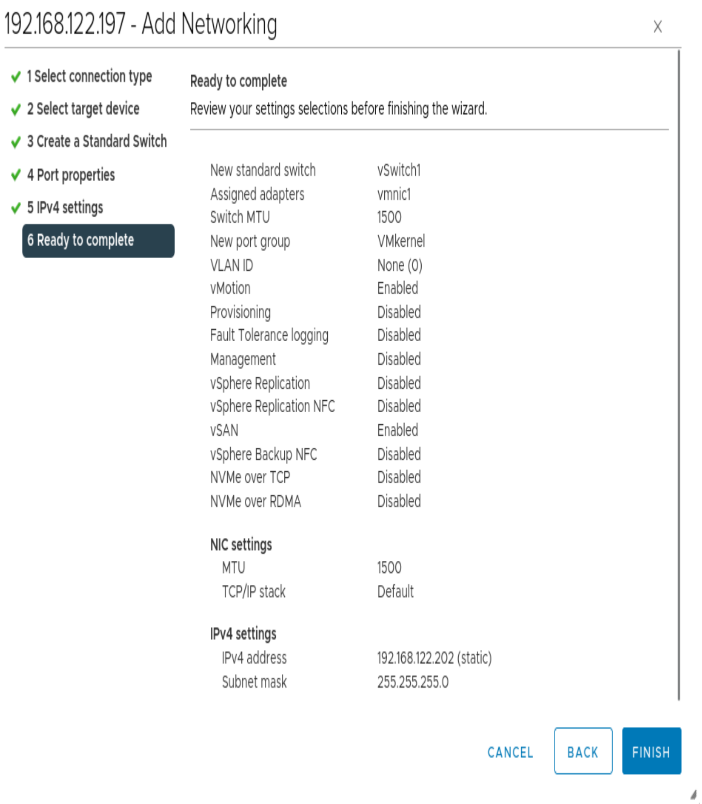


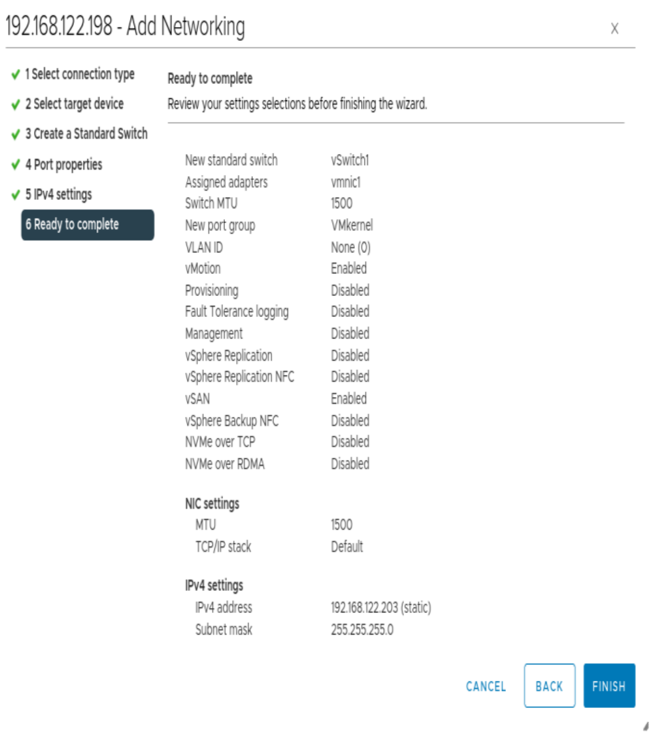
by Configuring the VMkernel port properties. We enable specific services like vMotion (for live migration) and vSAN to sure the host can communicate effectively within the cluster.





We will do a similar configuration to the remaining 2 ESXi hosts





NFS Shared Storage Setup:

This phase involves preparing a Linux-based (RHEL) server to act as the NFS server. This server will host the files for the vSphere datastore.

```
[adham@NFS-Server ~]$ sudo dnf install nfs-utils
Updating Subscription Management repositories.
Red Hat Enterprise Linux 9 for x86_64 - BaseOS (RPMs)                                        918 kB/s | 111 MB     02:03
Red Hat Enterprise Linux 9 for x86_64 - AppStream (RPMs)                                     1.4 MB/s |  88 MB     01:01
Last metadata expiration check: 0:00:13 ago on Sat 02 May 2026 01:32:32 AM EEST.
Dependencies resolved.
==========================================================================================================================
 Package                      Architecture       Version                    Repository                              Size
==========================================================================================================================
Installing:
 nfs-utils                    x86_64             1:2.5.4-38.el9_7.3         rhel-9-for-x86_64-baseos-rpms          469 k
Upgrading:
 selinux-policy               noarch             38.1.65-1.el9_7.1          rhel-9-for-x86_64-baseos-rpms           43 k
 selinux-policy-targeted      noarch             38.1.65-1.el9_7.1          rhel-9-for-x86_64-baseos-rpms          6.9 M
Installing dependencies:
 gssproxy                     x86_64             0.8.4-7.el9                rhel-9-for-x86_64-baseos-rpms          114 k
 libev                        x86_64             4.33-6.el9                 rhel-9-for-x86_64-baseos-rpms           55 k
 libnfsidmap                  x86_64             1:2.5.4-38.el9_7.3         rhel-9-for-x86_64-baseos-rpms           68 k
 libverto-libev               x86_64             0.3.2-3.el9                rhel-9-for-x86_64-baseos-rpms           15 k
 rpcbind                      x86_64             1.2.6-7.el9                rhel-9-for-x86_64-baseos-rpms           62 k
 sssd-nfs-idmap               x86_64             2.9.6-4.el9                rhel-9-for-x86_64-baseos-rpms           41 k

Transaction Summary
==========================================================================================================================
Install  7 Packages
Upgrade  2 Packages

Total download size: 7.8 M
Is this ok [y/N]:
```

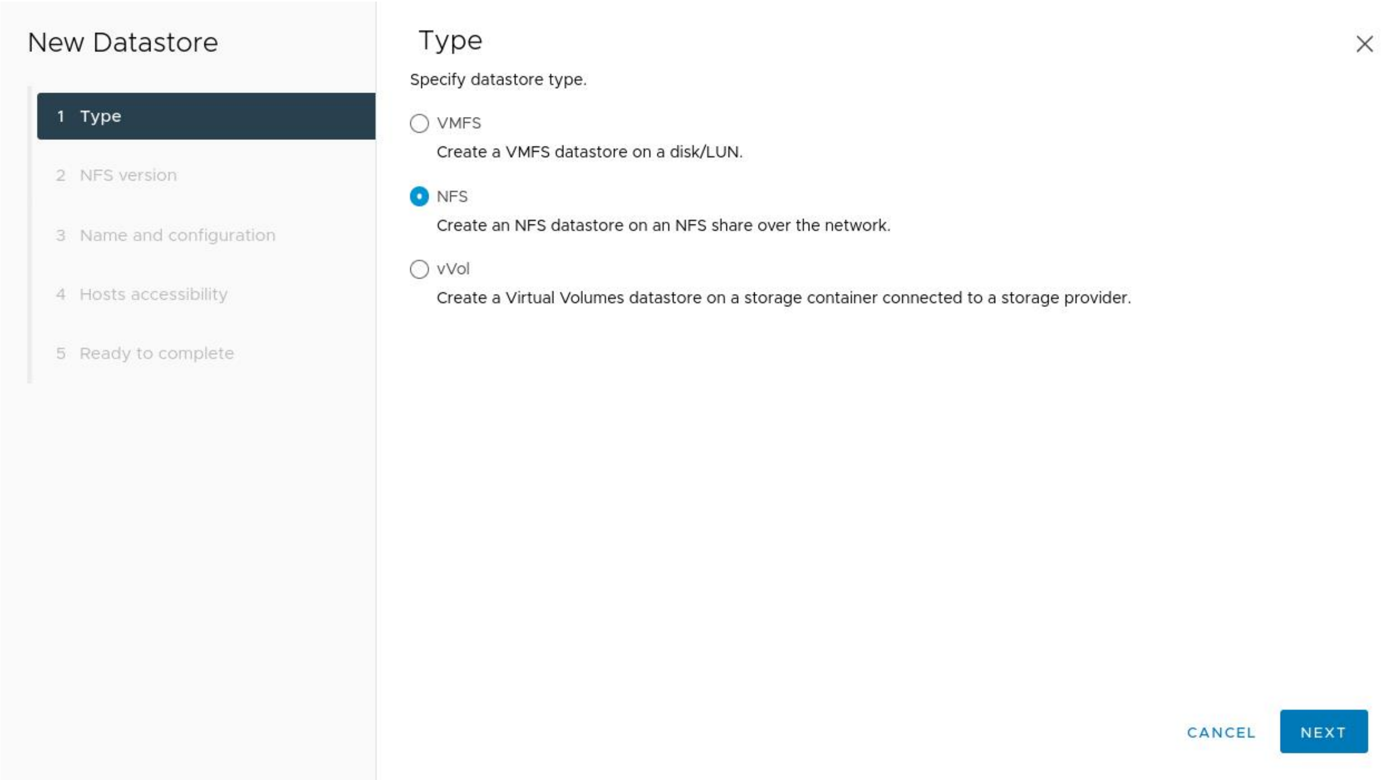
```
[adham@NFS-Server ~]$ sudo su
[sudo] password for adham:
[root@NFS-Server adham]# firewall-cmd --permanent --add-service=nfs --add-service=rpc-bind --add-service=mountd
success
```

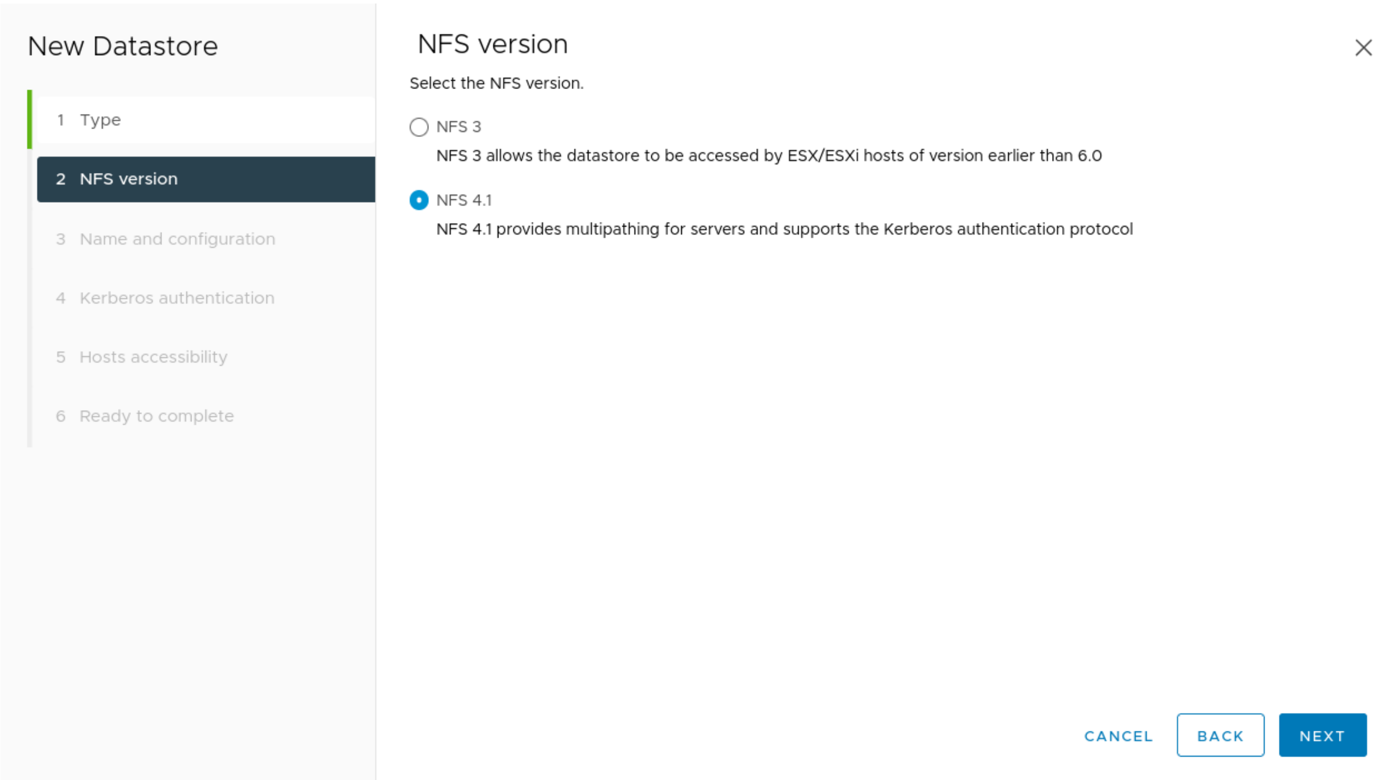
```
[root@NFS-Server adham]# mkdir -p /shared
[root@NFS-Server adham]# cd /shared/
[root@NFS-Server shared]# nano /etc/exports
[root@NFS-Server shared]# systemctl restart nfs
Failed to restart nfs.service: Unit nfs.service not found.
[root@NFS-Server shared]# systemctl restart nfs-server.service
[root@NFS-Server shared]# echo "Hello ITI" >> file1.txt
```

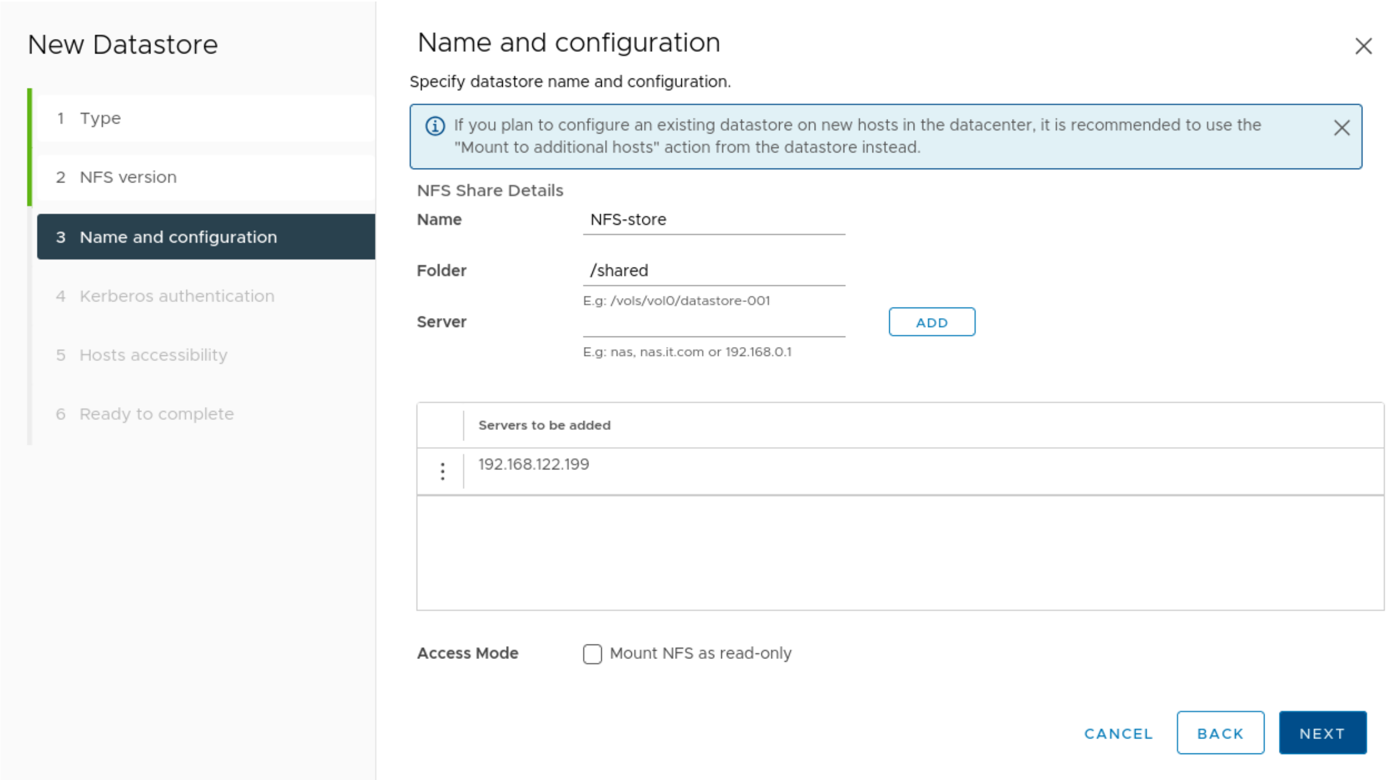
```
  GNU nano 5.6.1                                        /etc/exports
/shared 192.168.122.0/24(rw,sync,no_root_squash)
```

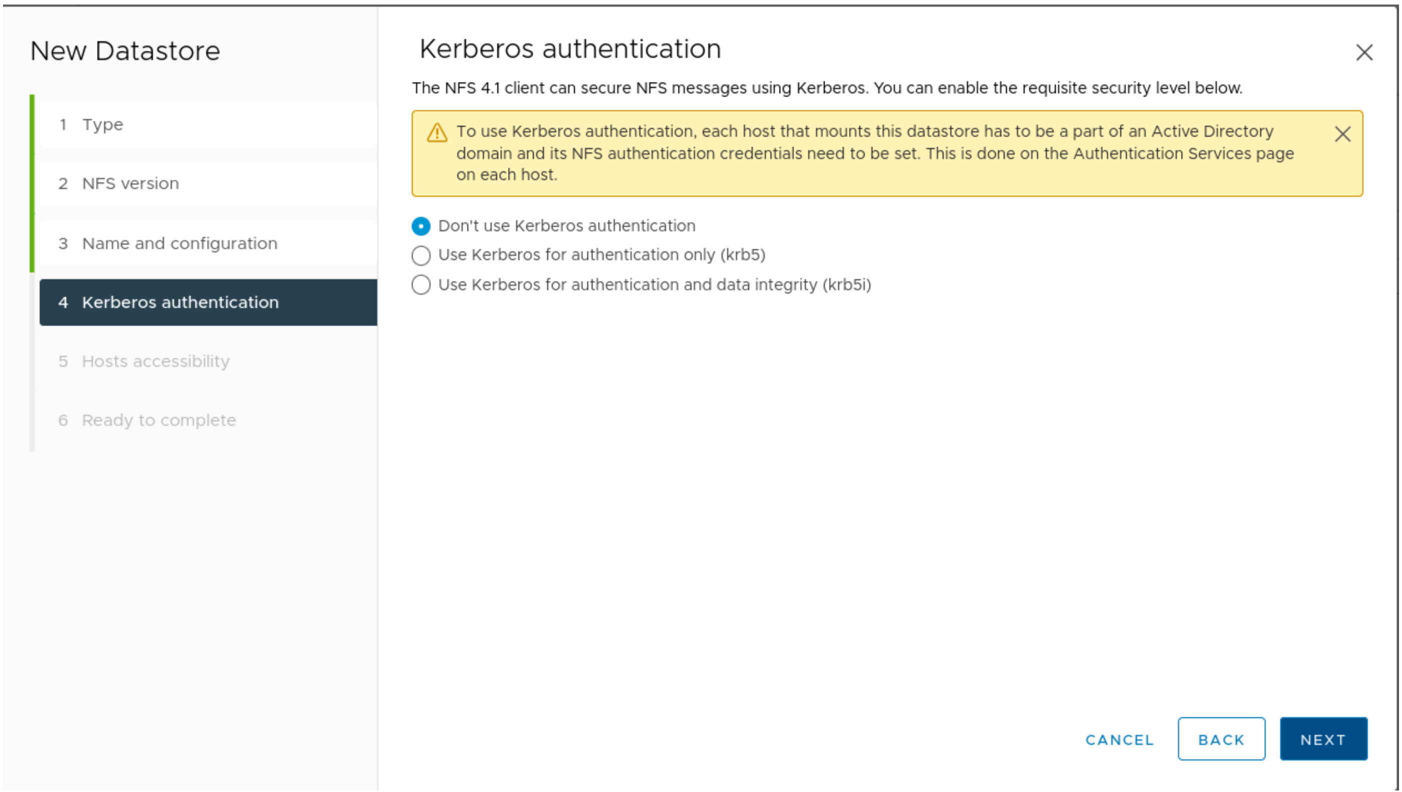
```
[root@NFS-Server shared]# ls -la
total 4
drwxr-xr-x.  2 nobody nobody  23 May  2 01:42 .
dr-xr-xr-x. 19 root   root   249 May  2 01:38 ..
-rw-r--r--.  1 nobody nobody  10 May  2 01:42 file1.txt
[root@NFS-Server shared]# chmod ug+w -R /shared
[root@NFS-Server shared]# ls -la
total 4
drwxrwxr-x.  2 nobody nobody  23 May  2 01:42 .
dr-xr-xr-x. 19 root   root   249 May  2 01:38 ..
-rw-rw-r--.  1 nobody nobody  10 May  2 01:42 file1.txt
```

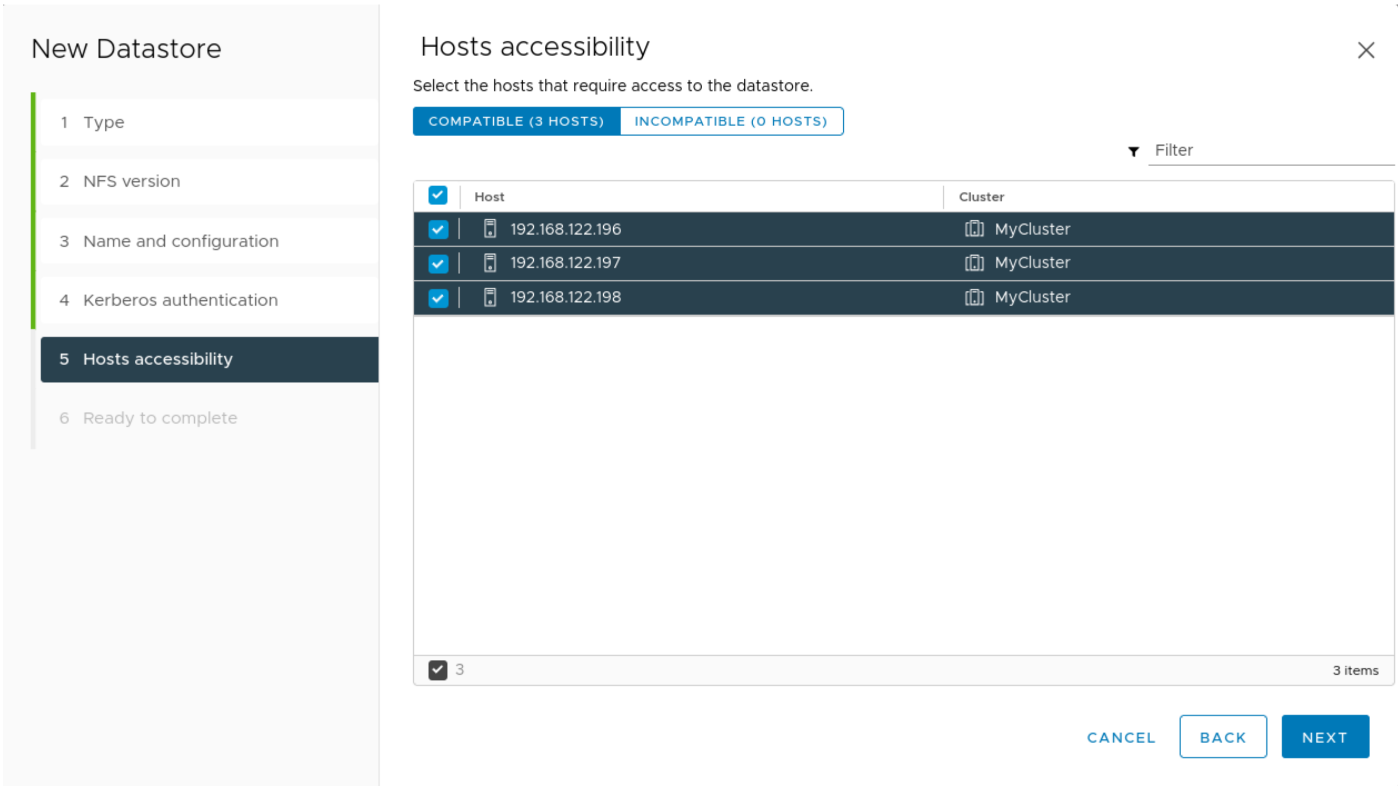
In the vSphere client we right click on the datacenter —> storage —> new datastore.

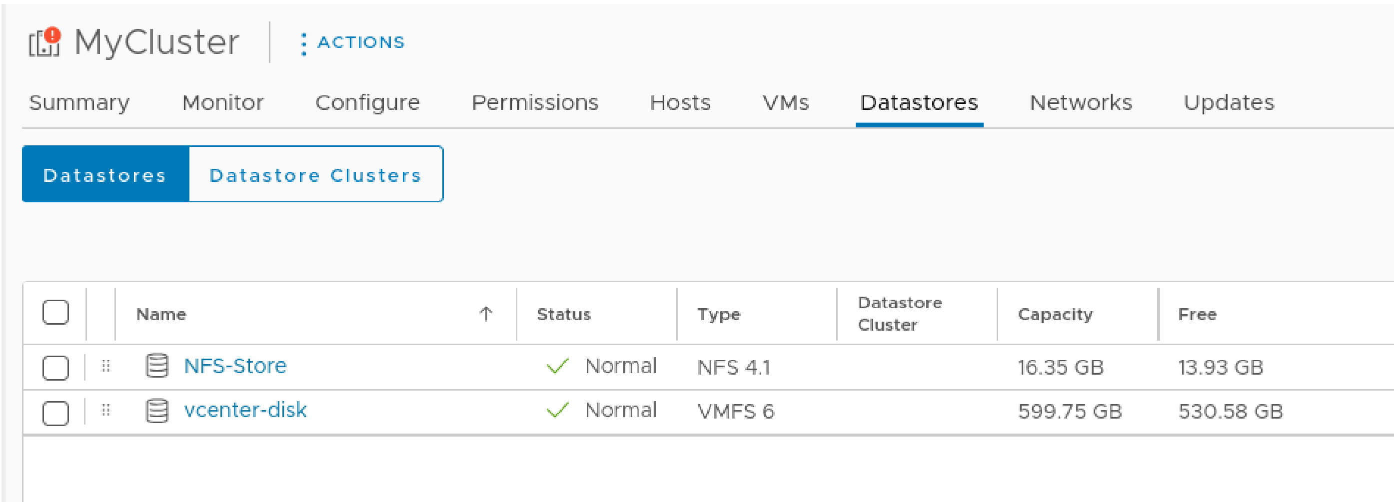

New Datastore
1 Type
2 NFS version
3 Name and configuration
4 Hosts accessibility
5 Ready to complete
Type
Specify datastore type.
VMFS
Create a VMFS datastore on a disk/LUN.
NFS
Create an NFS datastore on an NFS share over the network.
vVol
Create a Virtual Volumes datastore on a storage container connected to a storage provider.
CANCEL
NEXT

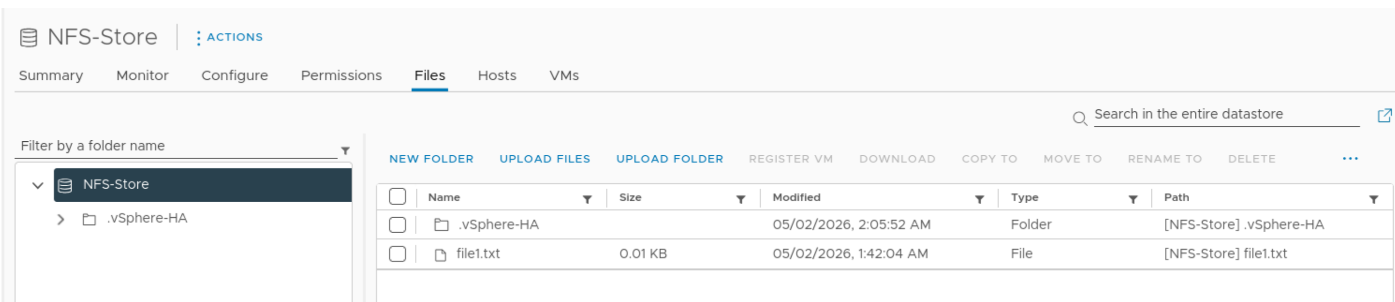

New Datastore
1 Type
2 NFS version
3 Name and configuration
4 Kerberos authentication
5 Hosts accessibility
6 Ready to complete
NFS version
Select the NFS version.
NFS 3
NFS 3 allows the datastore to be accessed by ESX/ESXi hosts of version earlier than 6.0
NFS 4.1
NFS 4.1 provides multipathing for servers and supports the Kerberos authentication protocol
CANCEL
BACK
NEXT


New Datastore
1 Type
2 NFS version
3 Name and configuration
4 Kerberos authentication
5 Hosts accessibility
6 Ready to complete
Name and configuration
Specify datastore name and configuration.
If you plan to configure an existing datastore on new hosts in the datacenter, it is recommended to use the "Mount to additional hosts" action from the datastore instead.
NFS Share Details
Name
NFS-store
Folder
/shared
E.g: /vols/vol0/datastore-001
Server
ADD
E.g: nas, nas.it.com or 192.168.0.1
Servers to be added
192.168.122.199
Access Mode
Mount NFS as read-only
CANCEL
BACK
NEXT


New Datastore
1 Type
2 NFS version
3 Name and configuration
4 Kerberos authentication
5 Hosts accessibility
6 Ready to complete
Kerberos authentication
The NFS 4.1 client can secure NFS messages using Kerberos. You can enable the requisite security level below.
To use Kerberos authentication, each host that mounts this datastore has to be a part of an Active Directory domain and its NFS authentication credentials need to be set. This is done on the Authentication Services page on each host.
Don't use Kerberos authentication
Use Kerberos for authentication only (krb5)
Use Kerberos for authentication and data integrity (krb5i)
CANCEL
BACK
NEXT

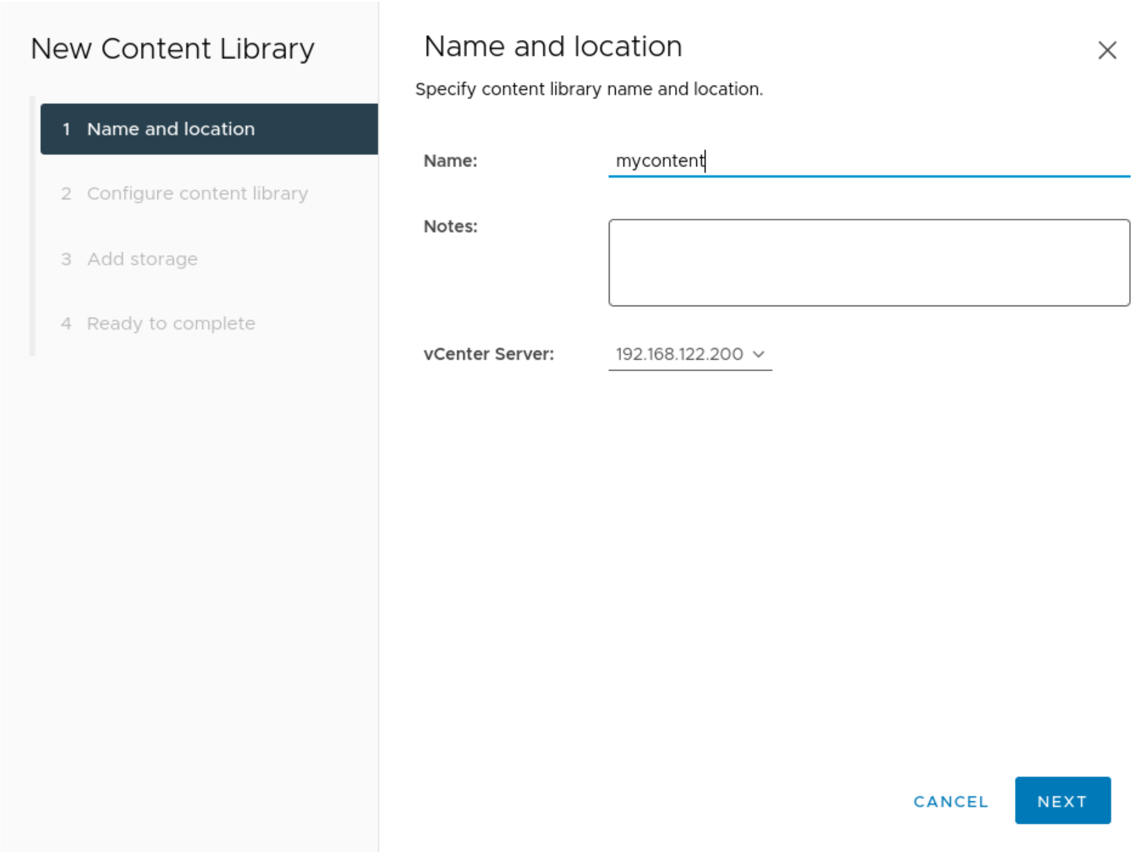

New Datastore
1 Type
2 NFS version
3 Name and configuration
4 Kerberos authentication
5 Hosts accessibility
6 Ready to complete
Hosts accessibility
Select the hosts that require access to the datastore.
COMPATIBLE (3 HOSTS)
INCOMPATIBLE (0 HOSTS)
Filter
Host
Cluster
192.168.122.196
MyCluster
192.168.122.197
MyCluster
192.168.122.198
MyCluster
3
3 items
CANCEL
BACK
NEXT

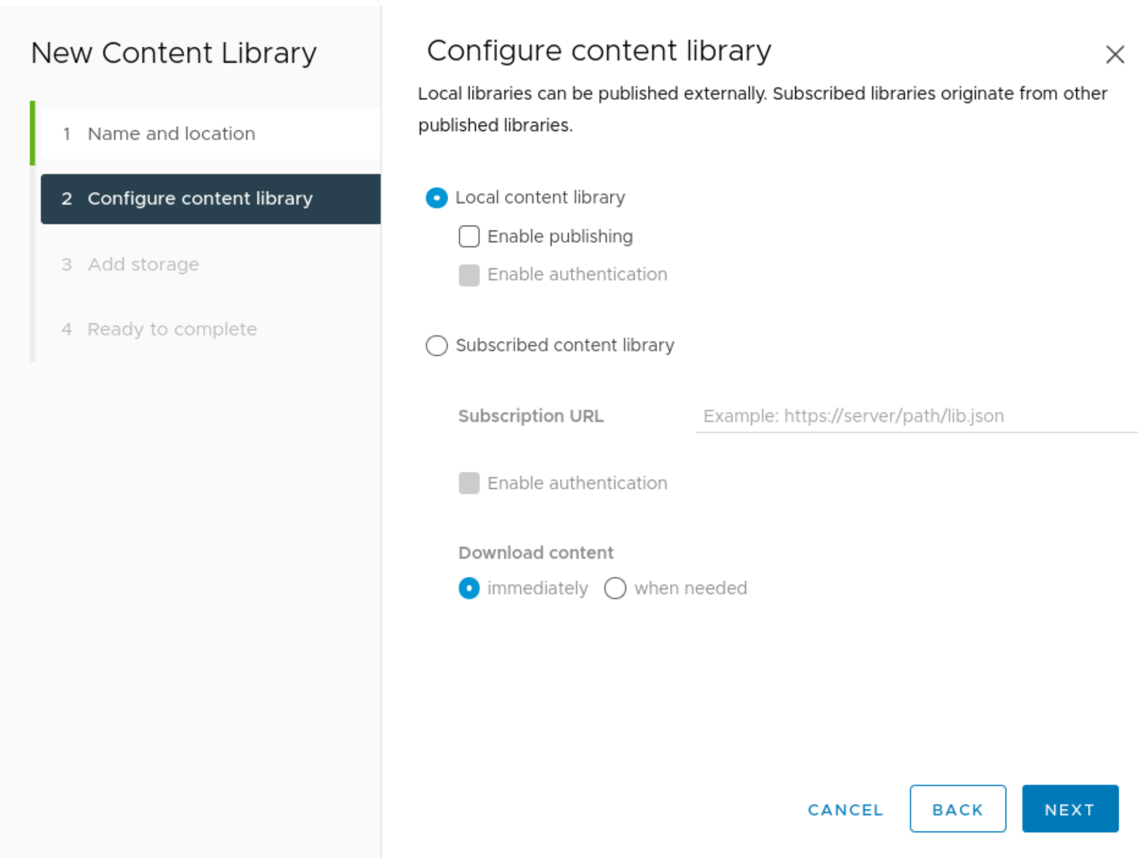

MyCluster
ACTIONS
Summary
Monitor
Configure
Permissions
Hosts
VMs
Datastores
Networks
Updates
Datastores
Datastore Clusters
Name
Status
Type
Datastore Cluster
Capacity
Free
NFS-Store
Normal
NFS 4.1
16.35 GB
13.93 GB
vcenter-disk
Normal
VMFS 6
599.75 GB
530.58 GB

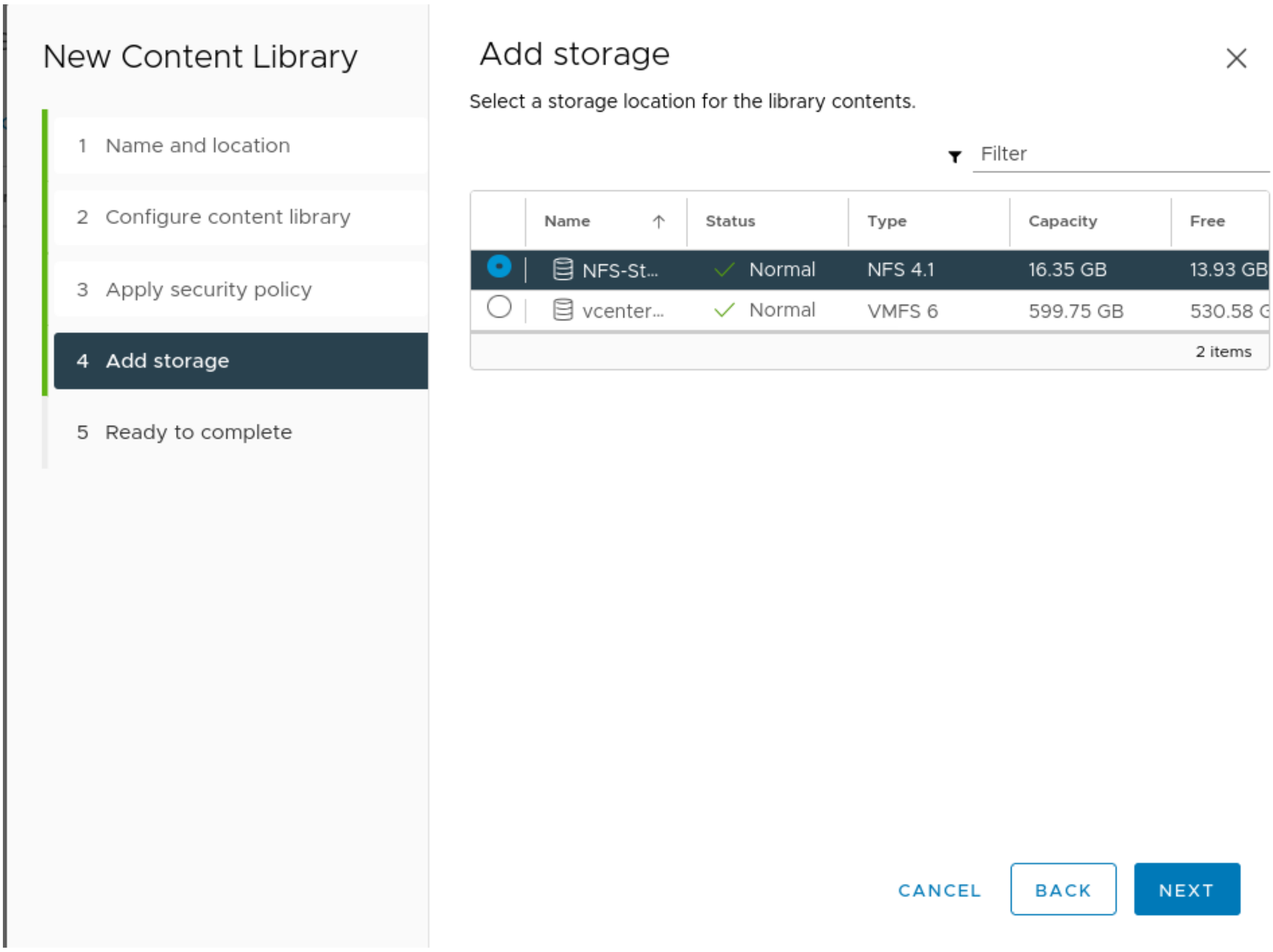

NFS-Store
ACTIONS
Summary
Monitor
Configure
Permissions
Files
Hosts
VMs
Search in the entire datastore
Filter by a folder name
NFS-Store
.vSphere-HA
NEW FOLDER
UPLOAD FILES
UPLOAD FOLDER
REGISTER VM
DOWNLOAD
COPY TO
MOVE TO
RENAME TO
DELETE
Name
Size
Modified
Type
Path
.vSphere-HA
05/02/2026, 2:05:52 AM
Folder
[NFS-Store] .vSphere-HA
file1.txt
0.01 KB
05/02/2026, 1:42:04 AM
File
[NFS-Store] file1.txt

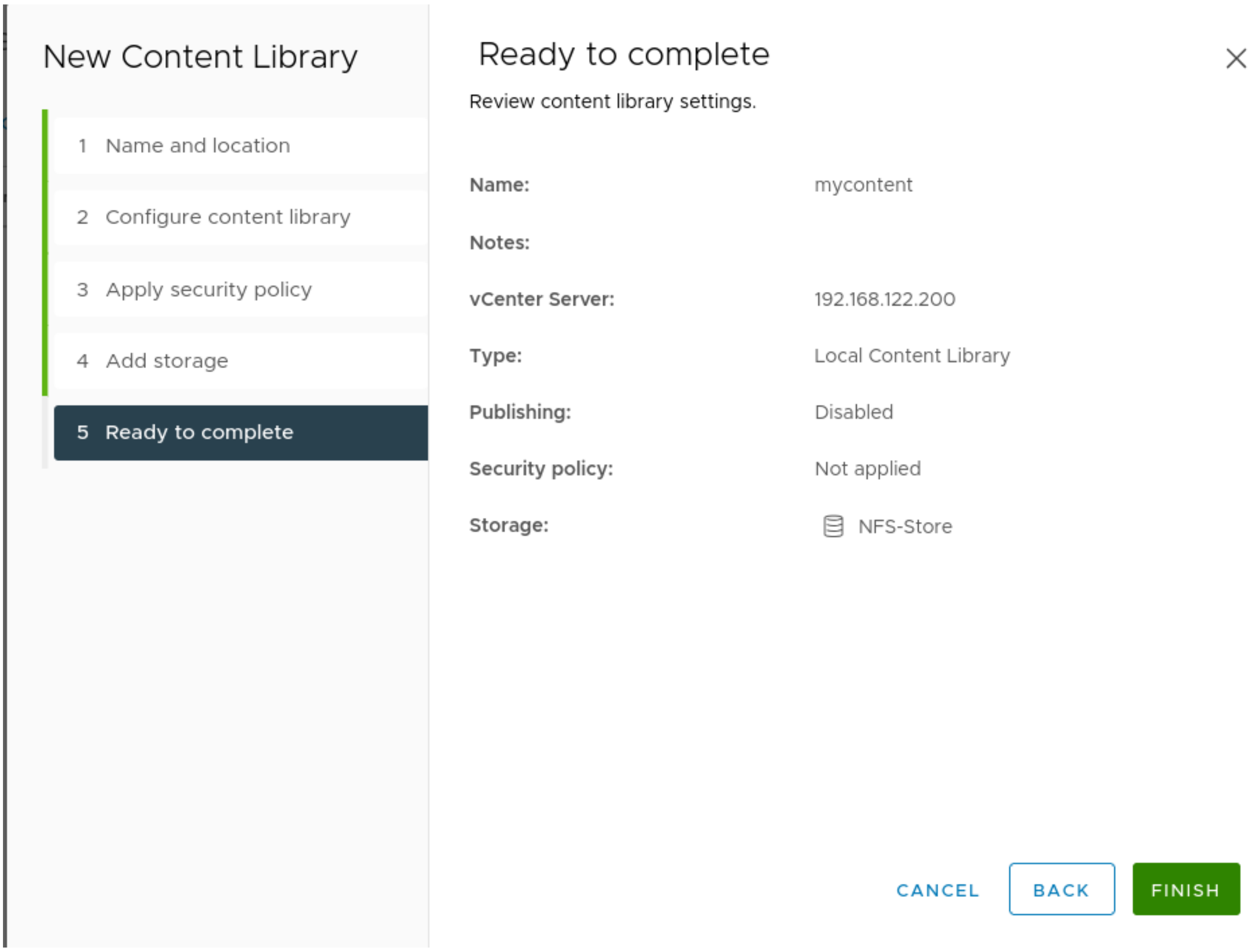
Creating Content Library:

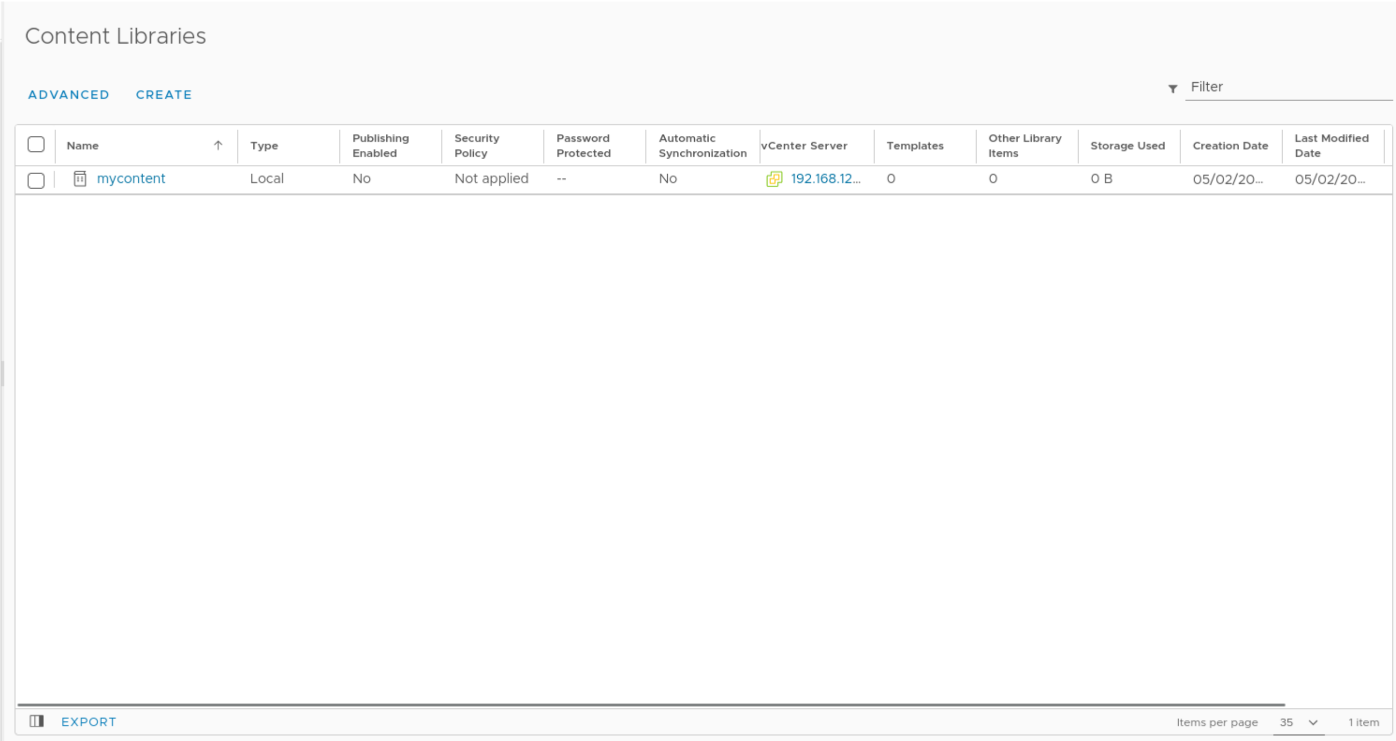
First, you initiate the "New Content Library" wizard from the vSphere Client. This involves naming the library and configuring its type (e.g., a local library).



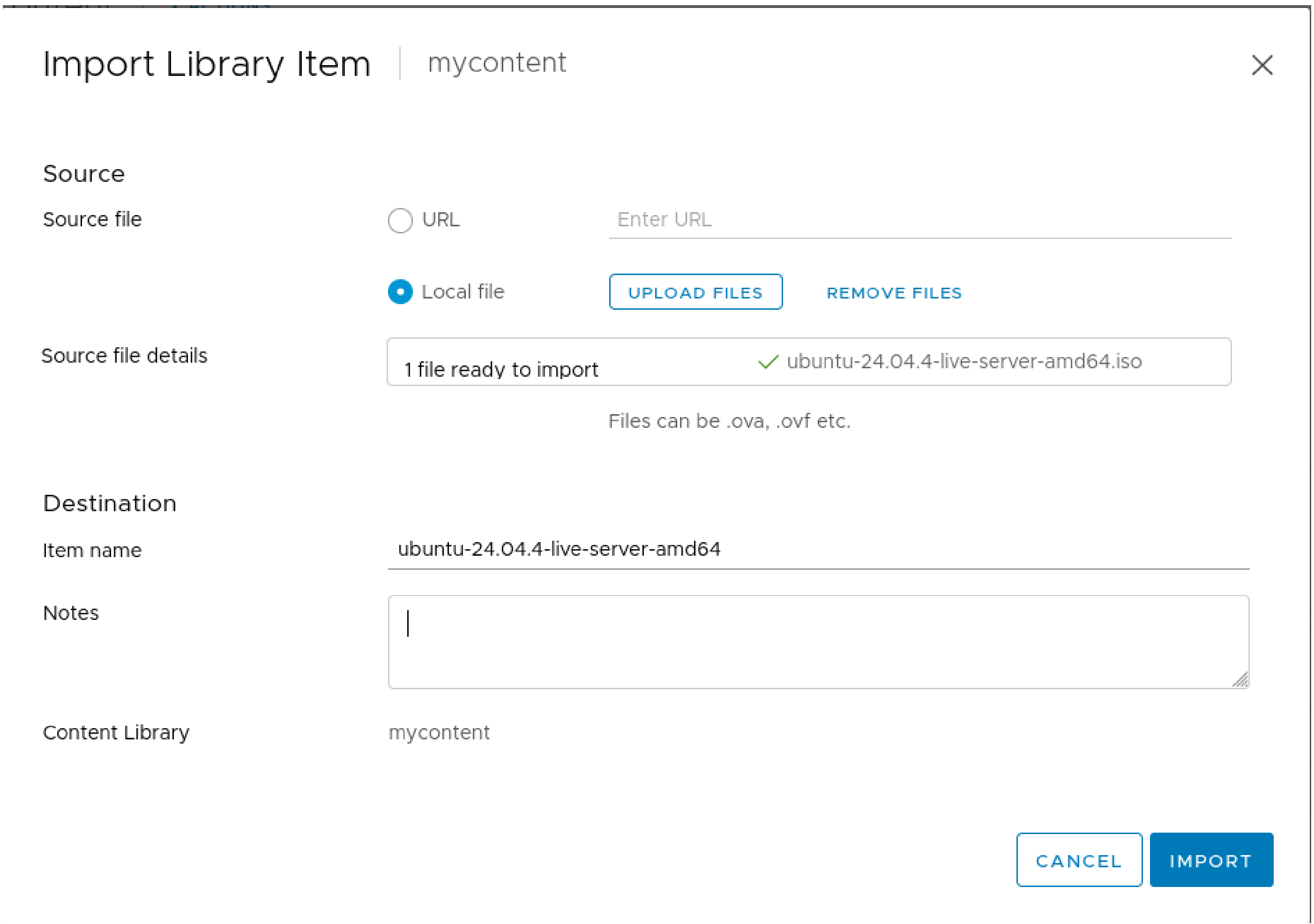


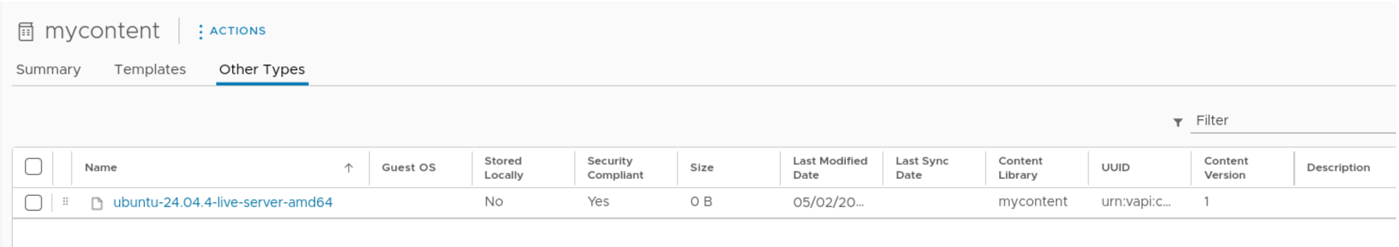

New Content Library
1 Name and location
2 Configure content library
3 Apply security policy
4 Add storage
5 Ready to complete
Add storage
Select a storage location for the library contents.
Filter
Name
Status
Type
Capacity
Free
NFS-St...
Normal
NFS 4.1
16.35 GB
13.93 GB
vcenter...
Normal
VMFS 6
599.75 GB
530.58 G
2 items
CANCEL
BACK
NEXT


New Content Library
1 Name and location
2 Configure content library
3 Apply security policy
4 Add storage
5 Ready to complete
Ready to complete
Review content library settings.
Name: mycontent
Notes:
vCenter Server: 192.168.122.200
Type: Local Content Library
Publishing: Disabled
Security policy: Not applied
Storage: NFS-Store
CANCEL
BACK
FINISH



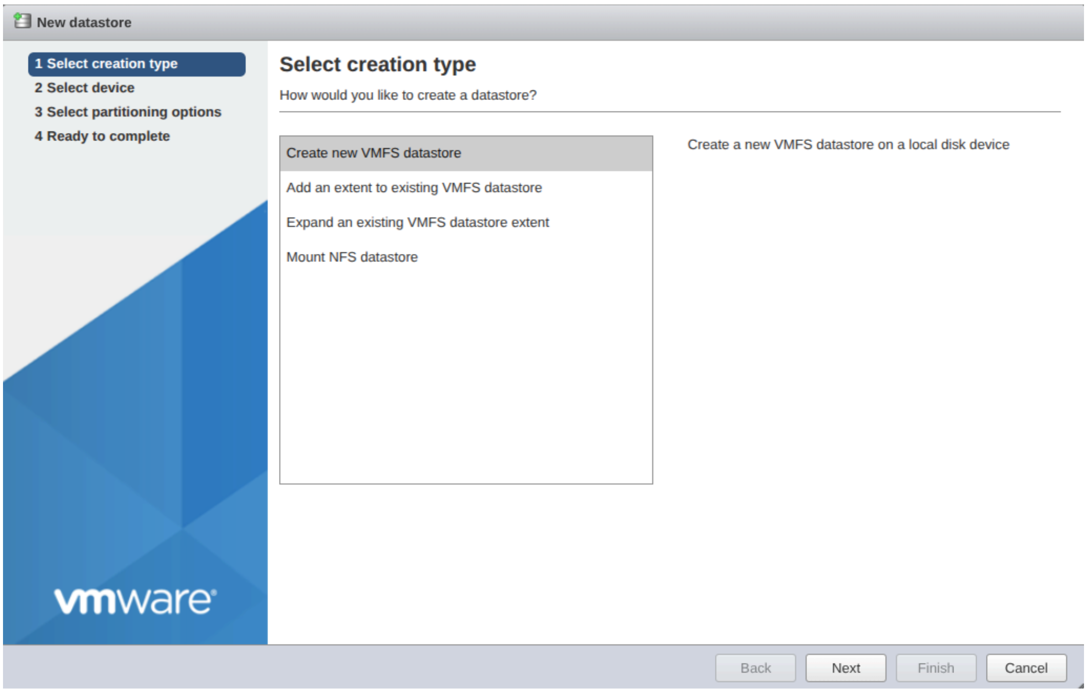
Now let upload an Ubuntu-server iso:

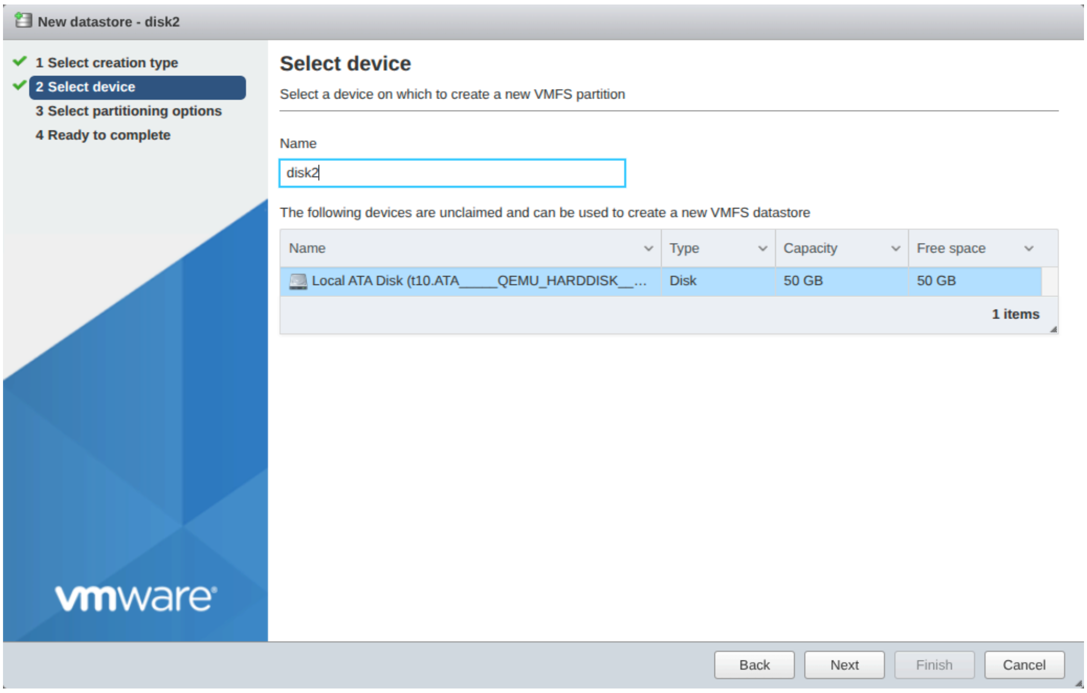


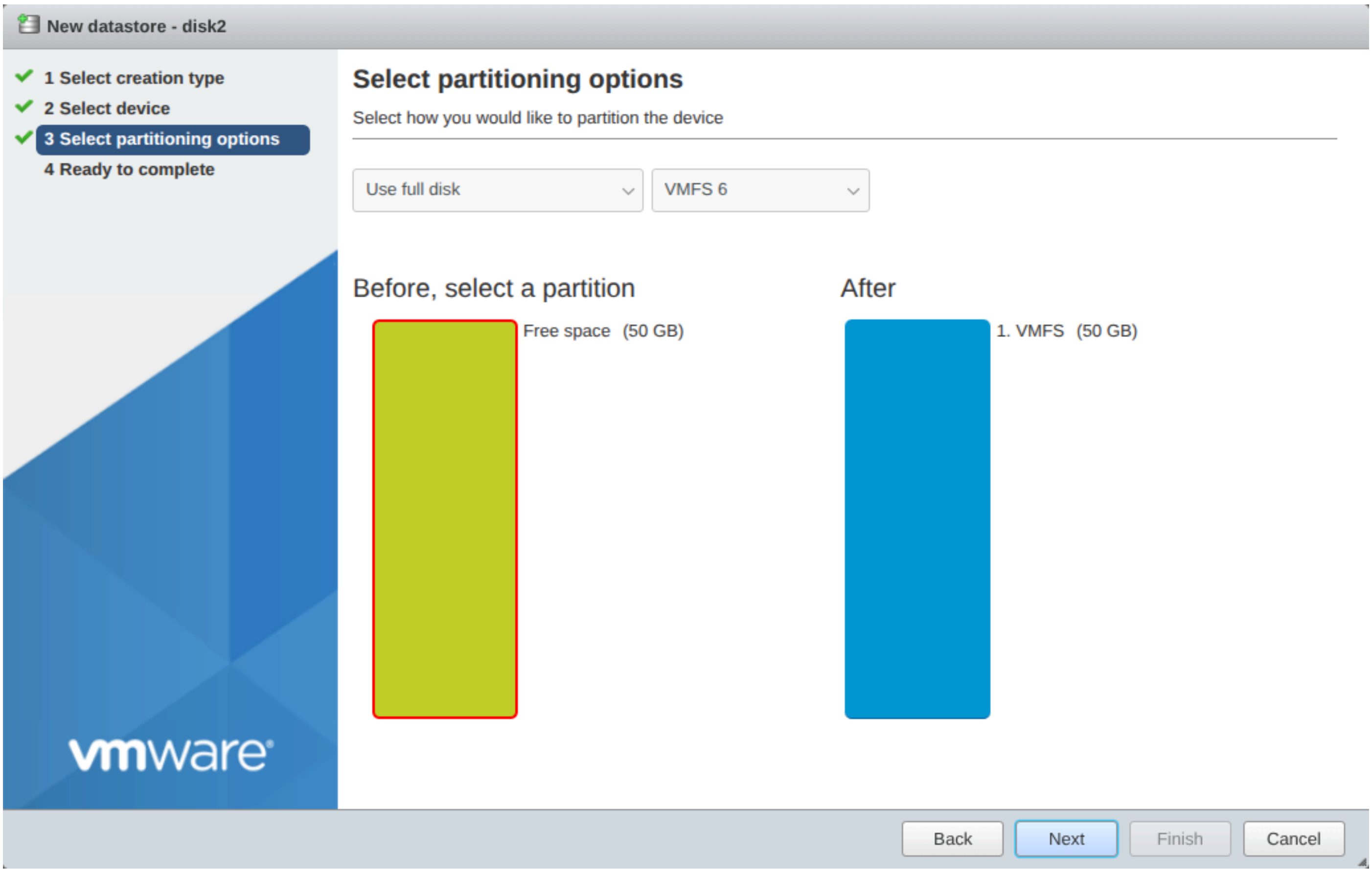


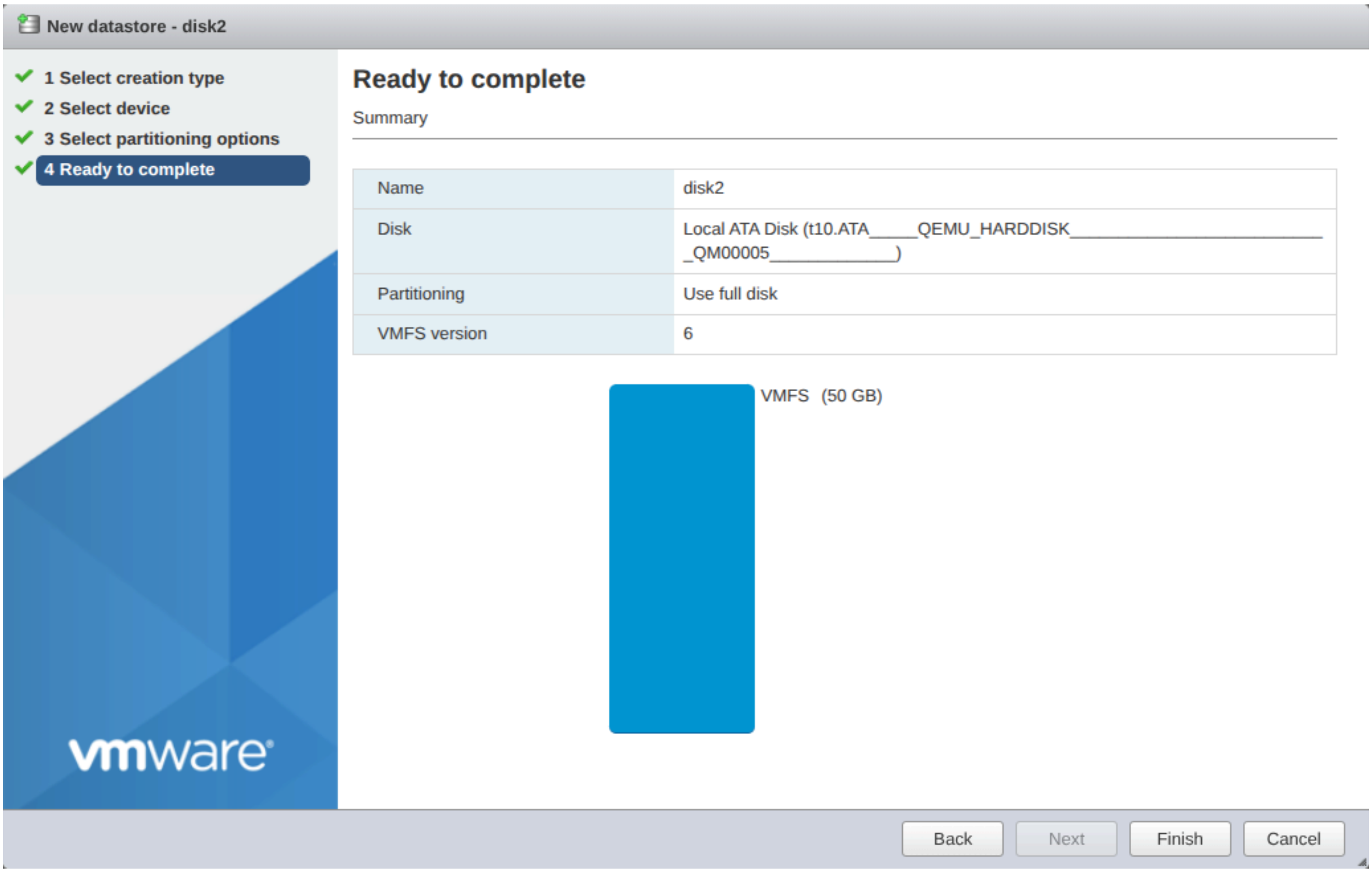
Creating a New Virtual Machine:

Before we can create a VM we need to first create a datastore from the secondary 50GB on both ESXI2 and ESXI3

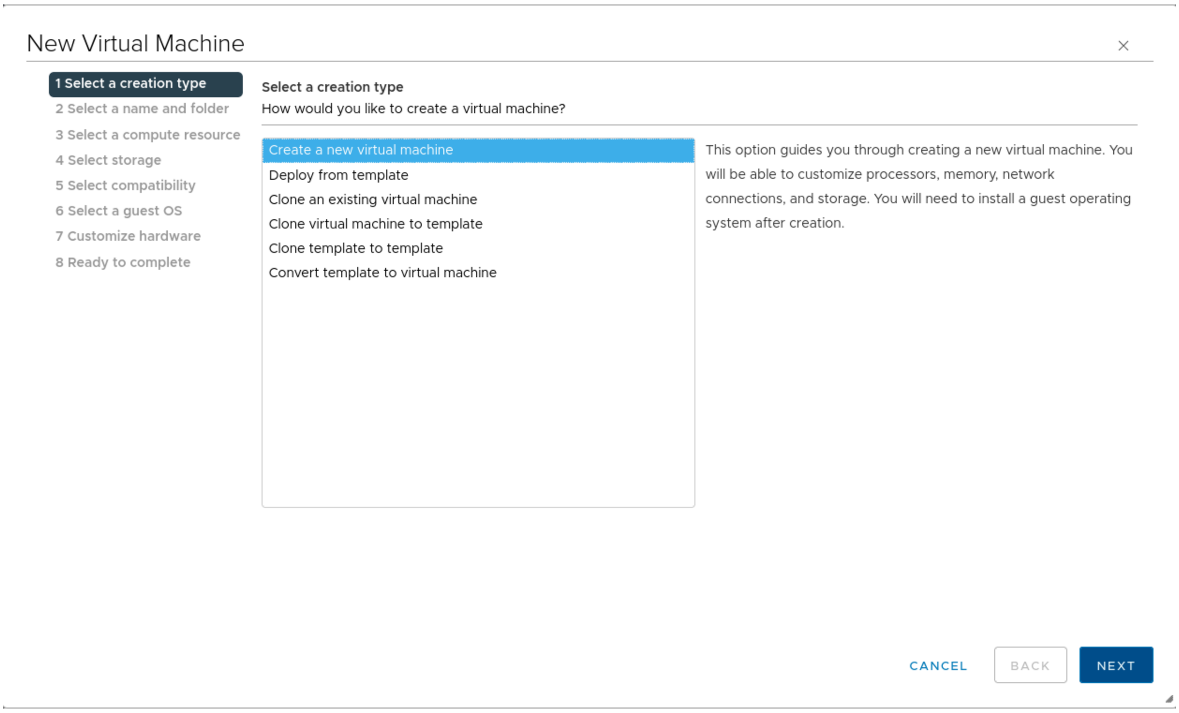


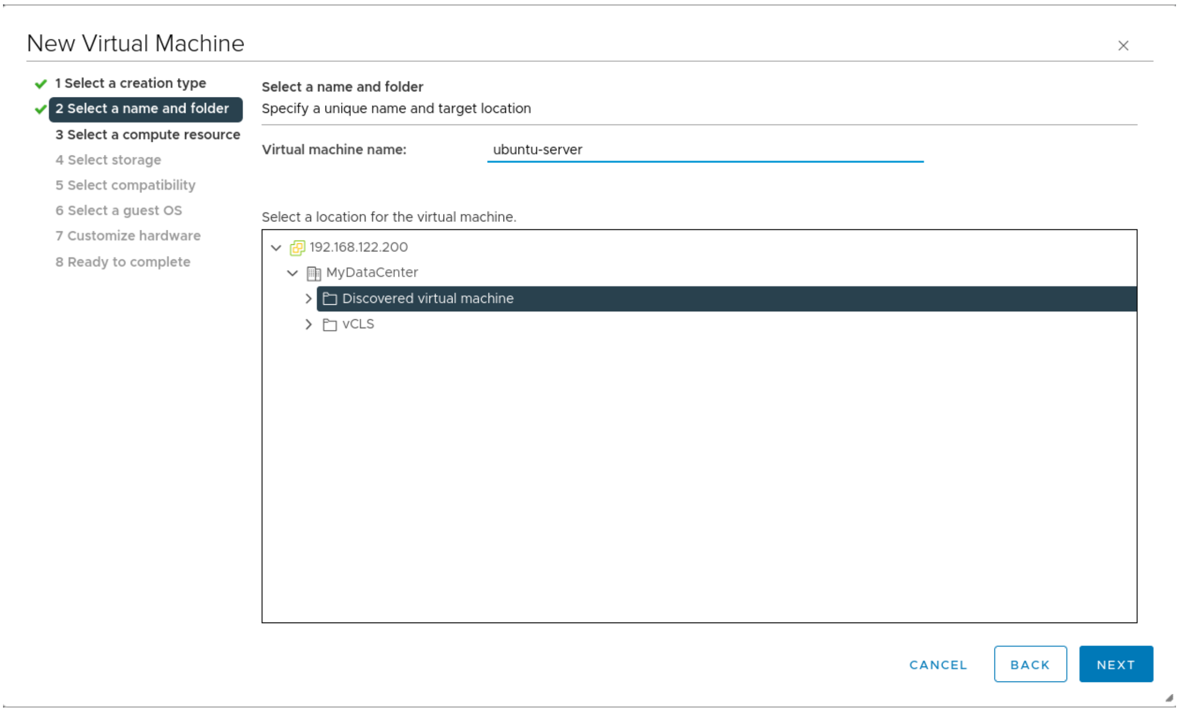


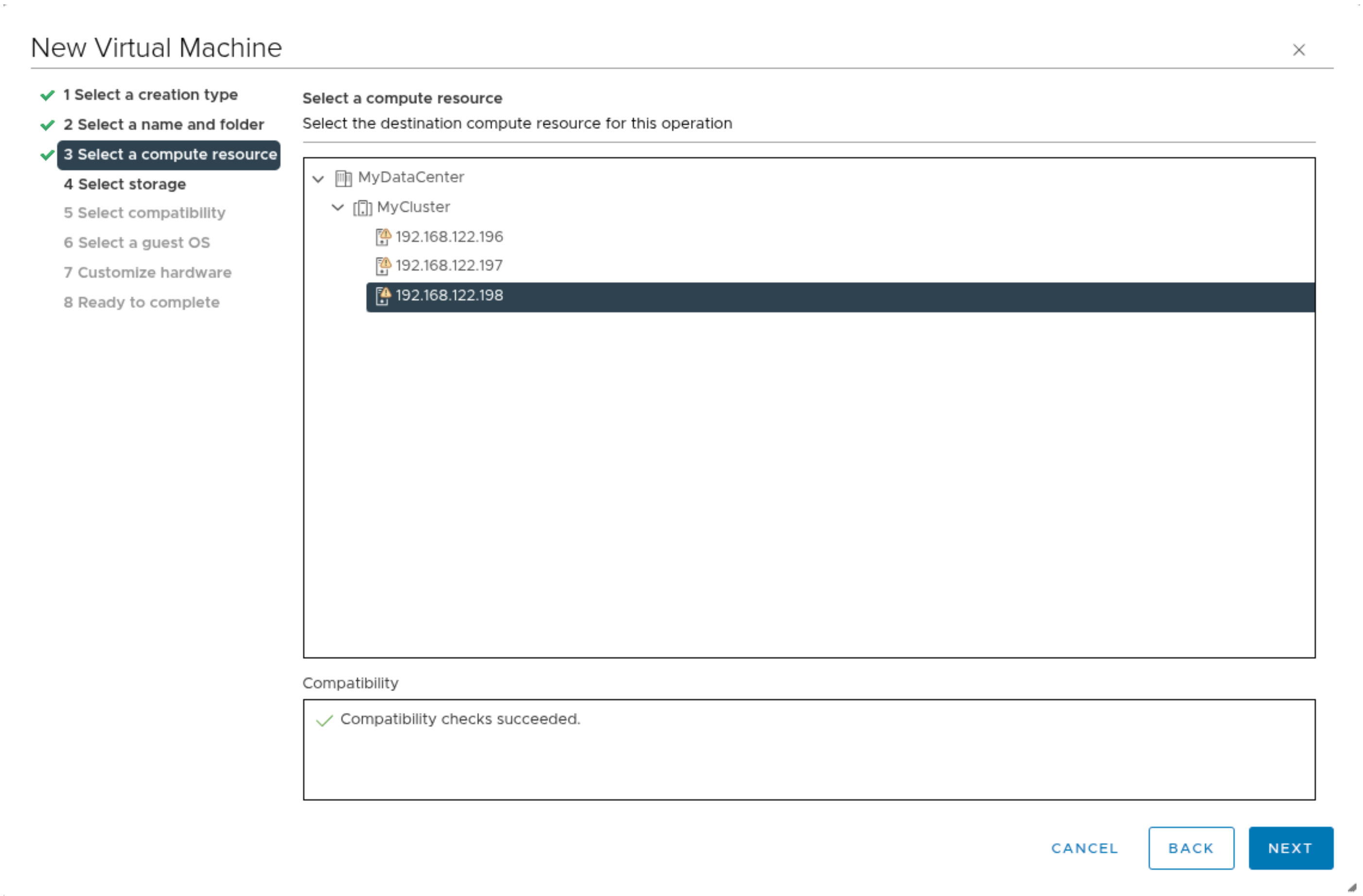

New datastore - disk2
1 Select creation type
2 Select device
3 Select partitioning options
4 Ready to complete
Select partitioning options
Select how you would like to partition the device
Use full disk
VMFS 6
Before, select a partition
Free space (50 GB)
After
1. VMFS (50 GB)
Back
Next
Finish
Cancel

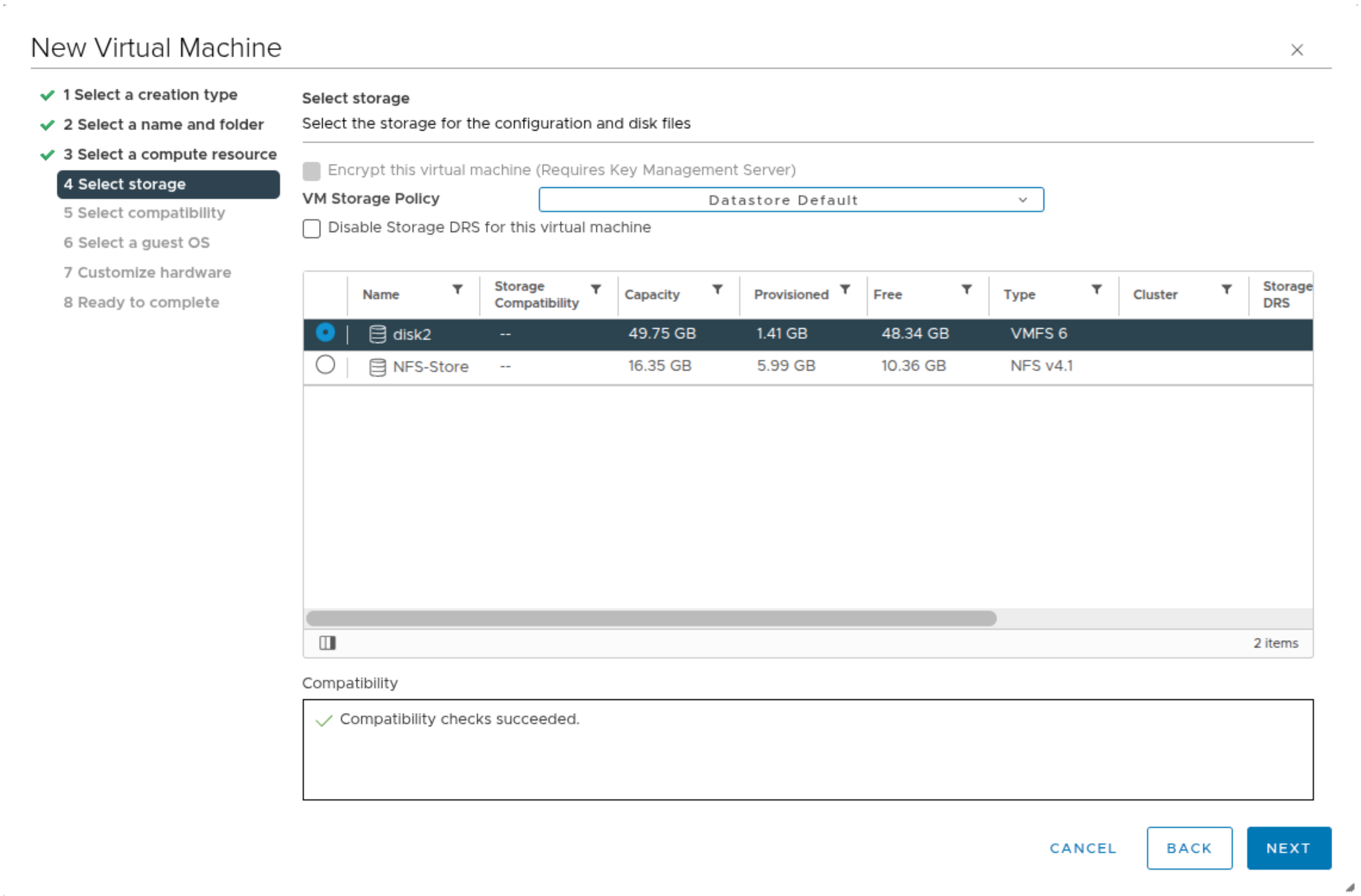

New datastore - disk2
1 Select creation type
2 Select device
3 Select partitioning options
4 Ready to complete
Ready to complete
Summary
Name
disk2
Disk
Local ATA Disk (t10.ATA_____QEMU_HARDDISK_________________________
_QM00005_____________)
Partitioning
Use full disk
VMFS version
6
VMFS (50 GB)
Back
Next
Finish
Cancel

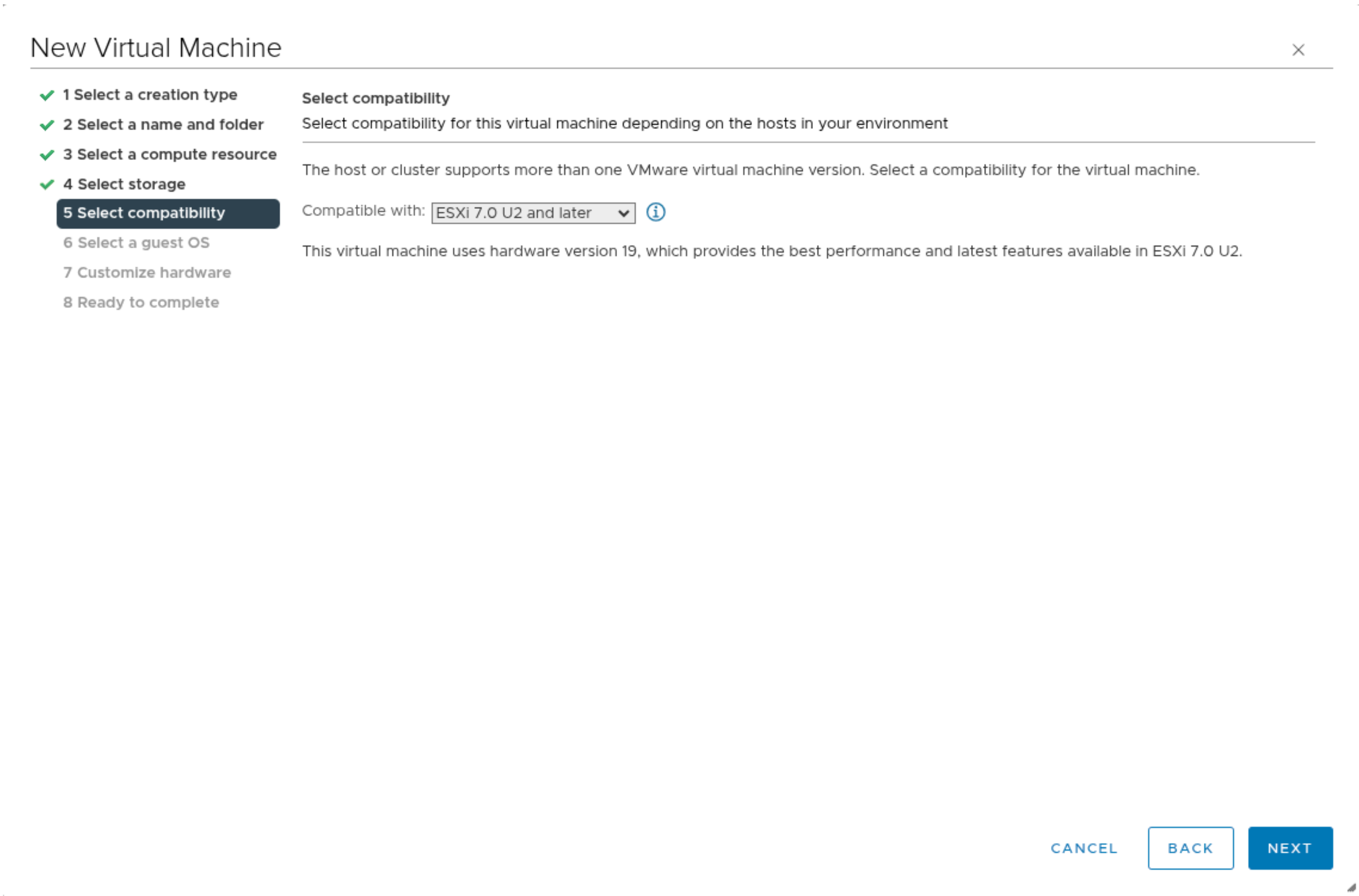
We are going to deploy this VM on ESXI3 (192.168.122.198)

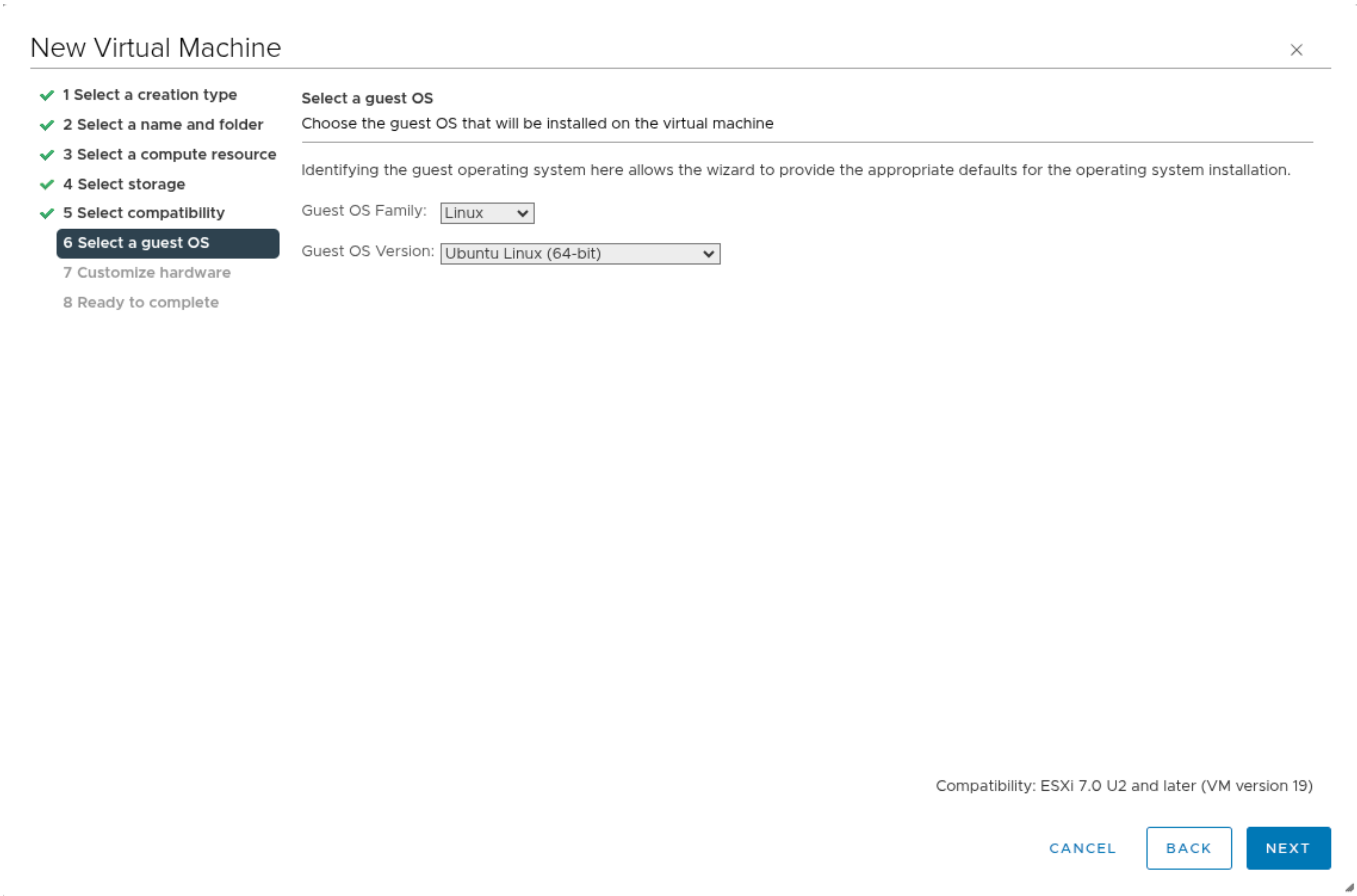


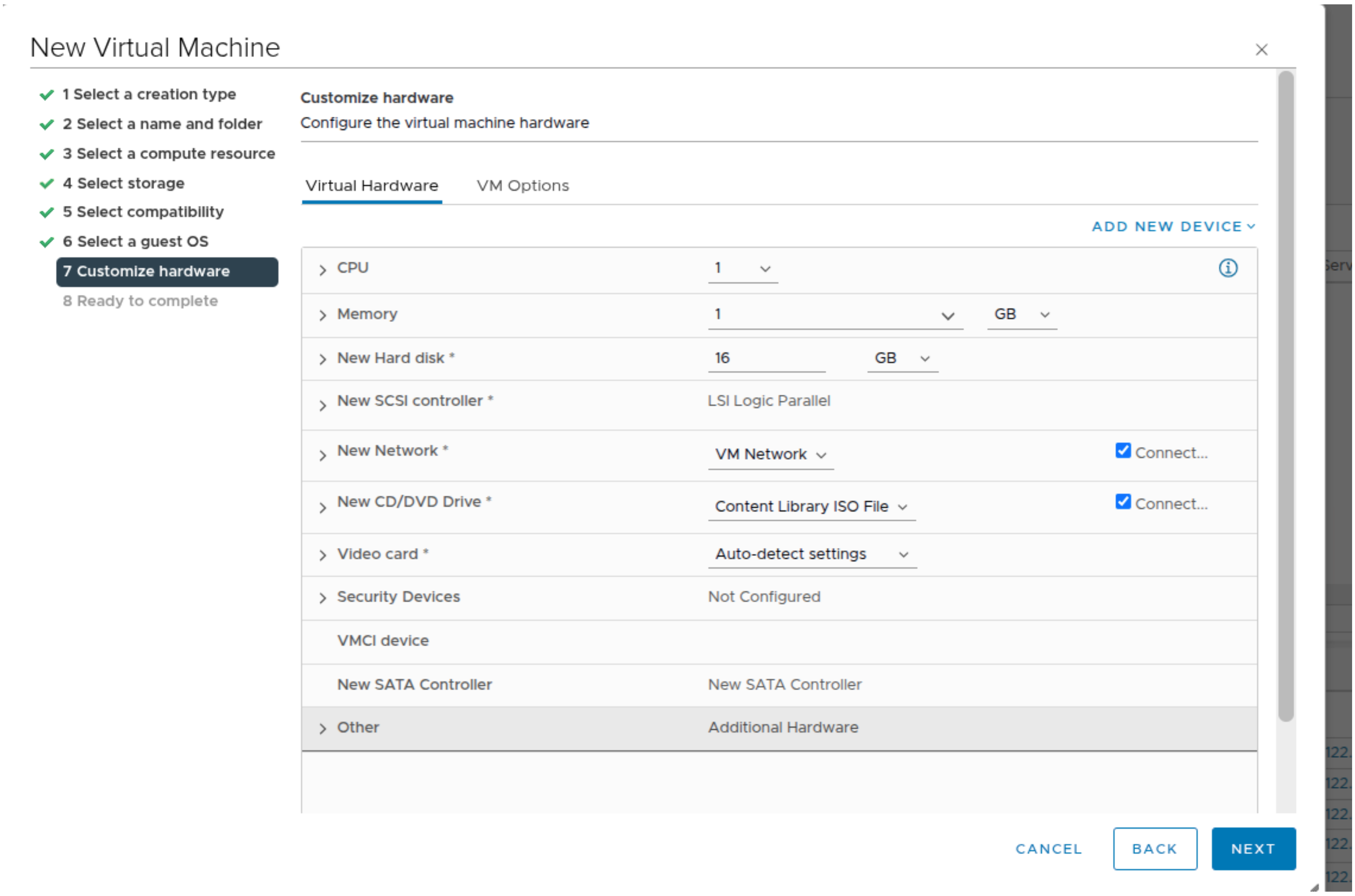


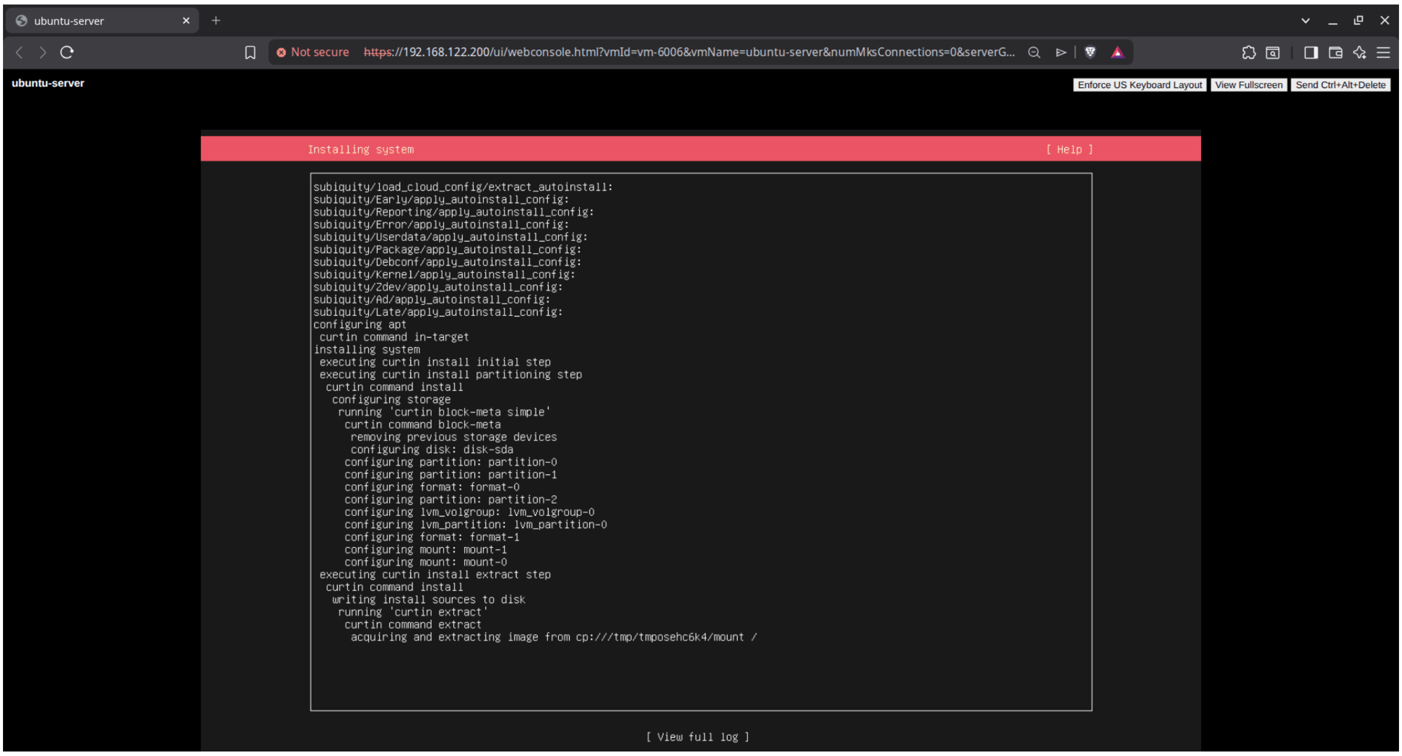

New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Select a compute resource
Select the destination compute resource for this operation
MyDataCenter
MyCluster
192.168.122.196
192.168.122.197
192.168.122.198
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT

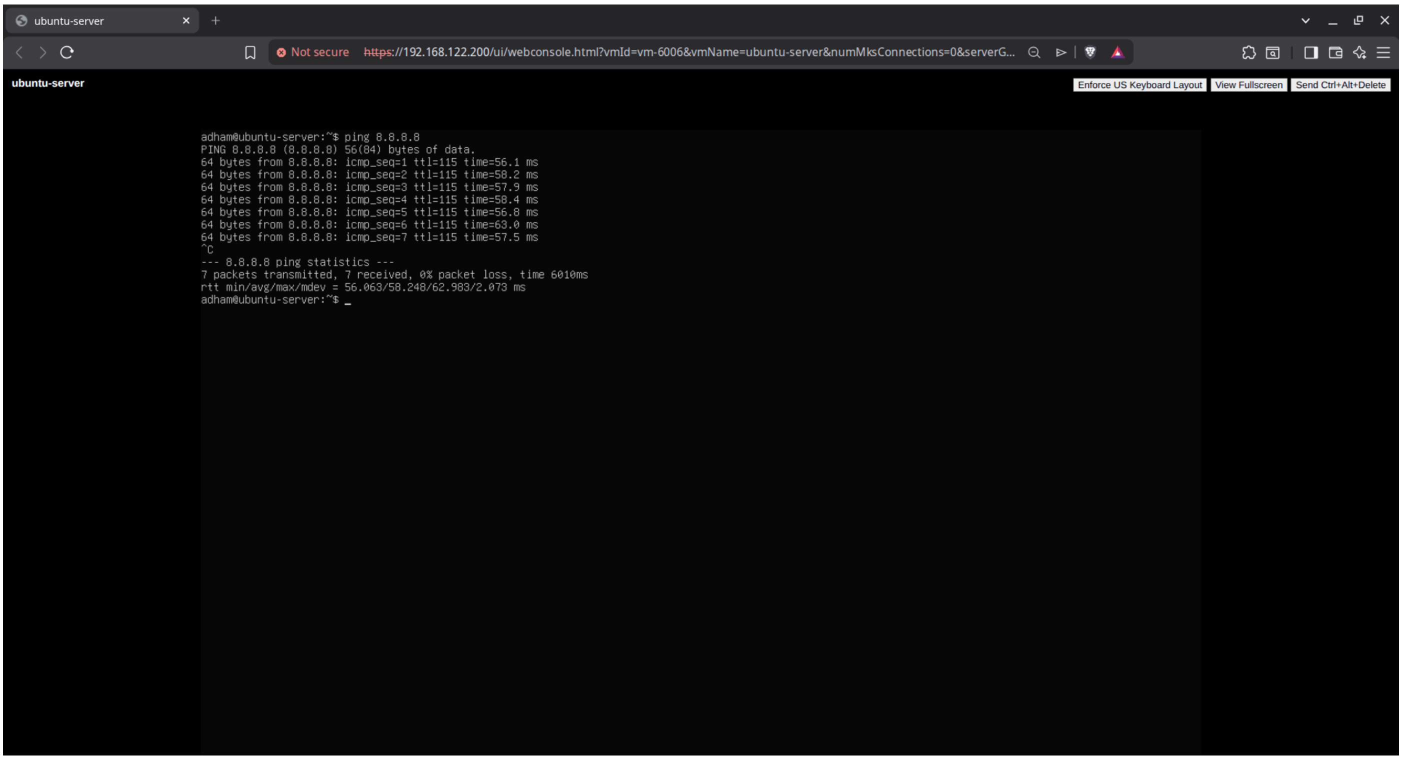

New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Select storage
Select the storage for the configuration and disk files
Encrypt this virtual machine (Requires Key Management Server)
VM Storage Policy
Datastore Default
Disable Storage DRS for this virtual machine
Name
Storage Compatibility
Capacity
Provisioned
Free
Type
Cluster
Storage DRS
disk2
--
49.75 GB
1.41 GB
48.34 GB
VMFS 6
NFS-Store
--
16.35 GB
5.99 GB
10.36 GB
NFS v4.1
2 items
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT


New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Select compatibility
Select compatibility for this virtual machine depending on the hosts in your environment
The host or cluster supports more than one VMware virtual machine version. Select a compatibility for the virtual machine.
Compatible with: ESXi 7.0 U2 and later
This virtual machine uses hardware version 19, which provides the best performance and latest features available in ESXi 7.0 U2.
CANCEL
BACK
NEXT

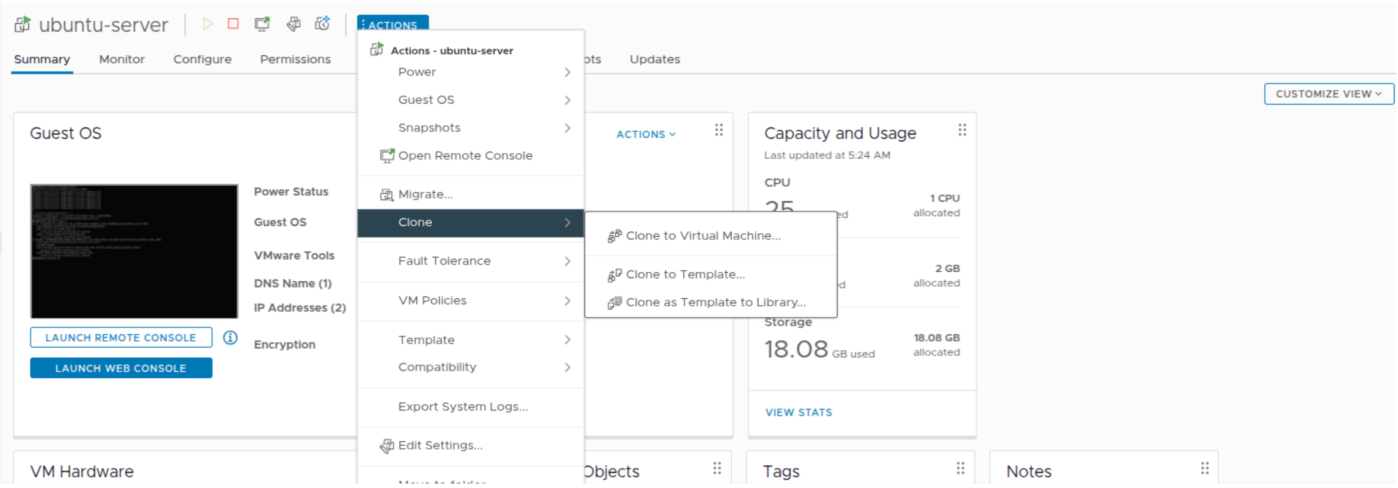

New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Select a guest OS
Choose the guest OS that will be installed on the virtual machine
Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.
Guest OS Family: Linux
Guest OS Version: Ubuntu Linux (64-bit)
Compatibility: ESXi 7.0 U2 and later (VM version 19)
CANCEL
BACK
NEXT

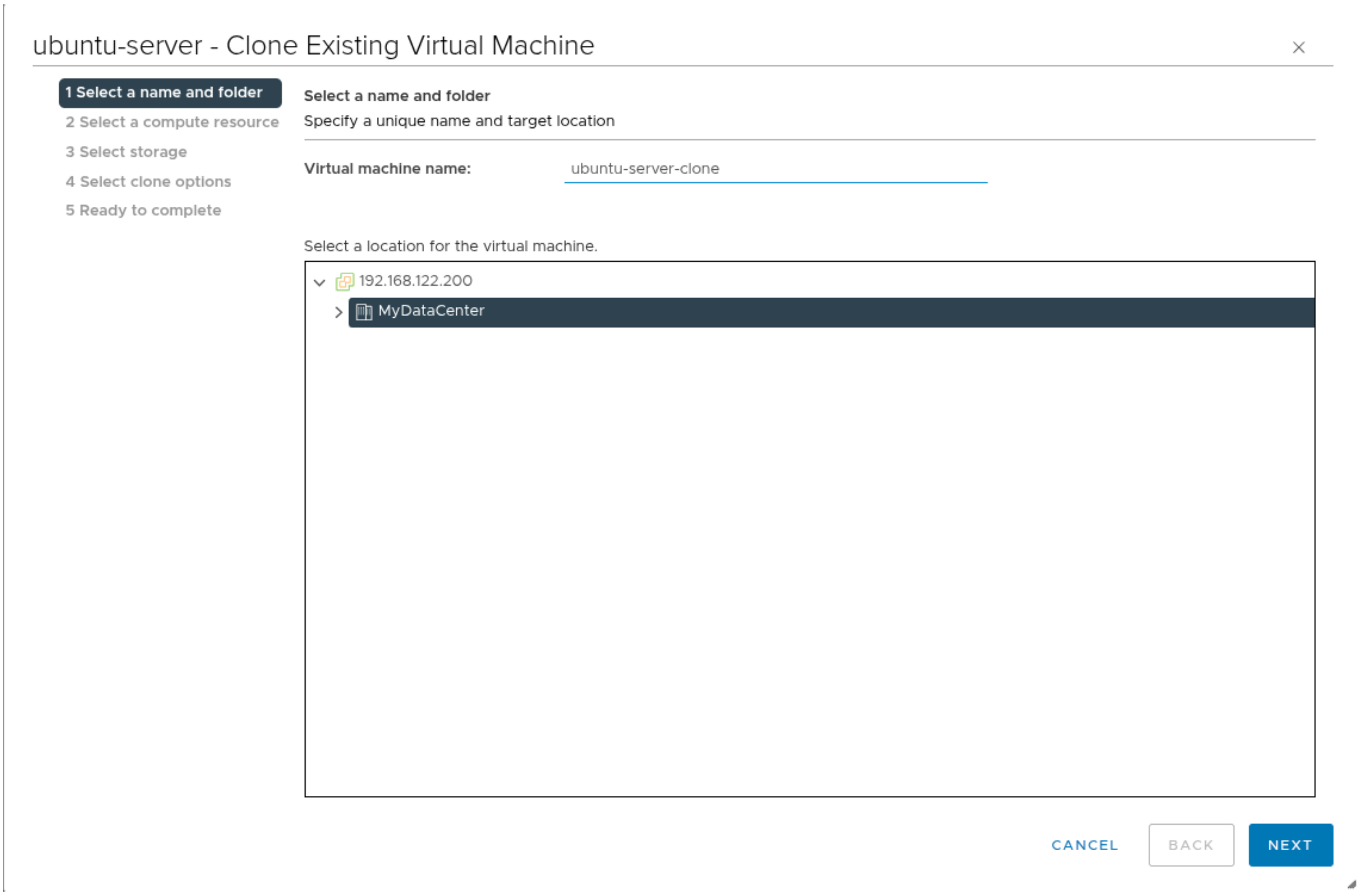

New Virtual Machine
1 Select a creation type
2 Select a name and folder
3 Select a compute resource
4 Select storage
5 Select compatibility
6 Select a guest OS
7 Customize hardware
8 Ready to complete
Customize hardware
Configure the virtual machine hardware
Virtual Hardware
VM Options
ADD NEW DEVICE
CPU 1
Memory 1 GB
New Hard disk * 16 GB
New SCSI controller * LSI Logic Parallel
New Network * VM Network Connect...
New CD/DVD Drive * Content Library ISO File Connect...
Video card * Auto-detect settings
Security Devices Not Configured
VMCI device
New SATA Controller New SATA Controller
Other Additional Hardware
CANCEL
BACK
NEXT

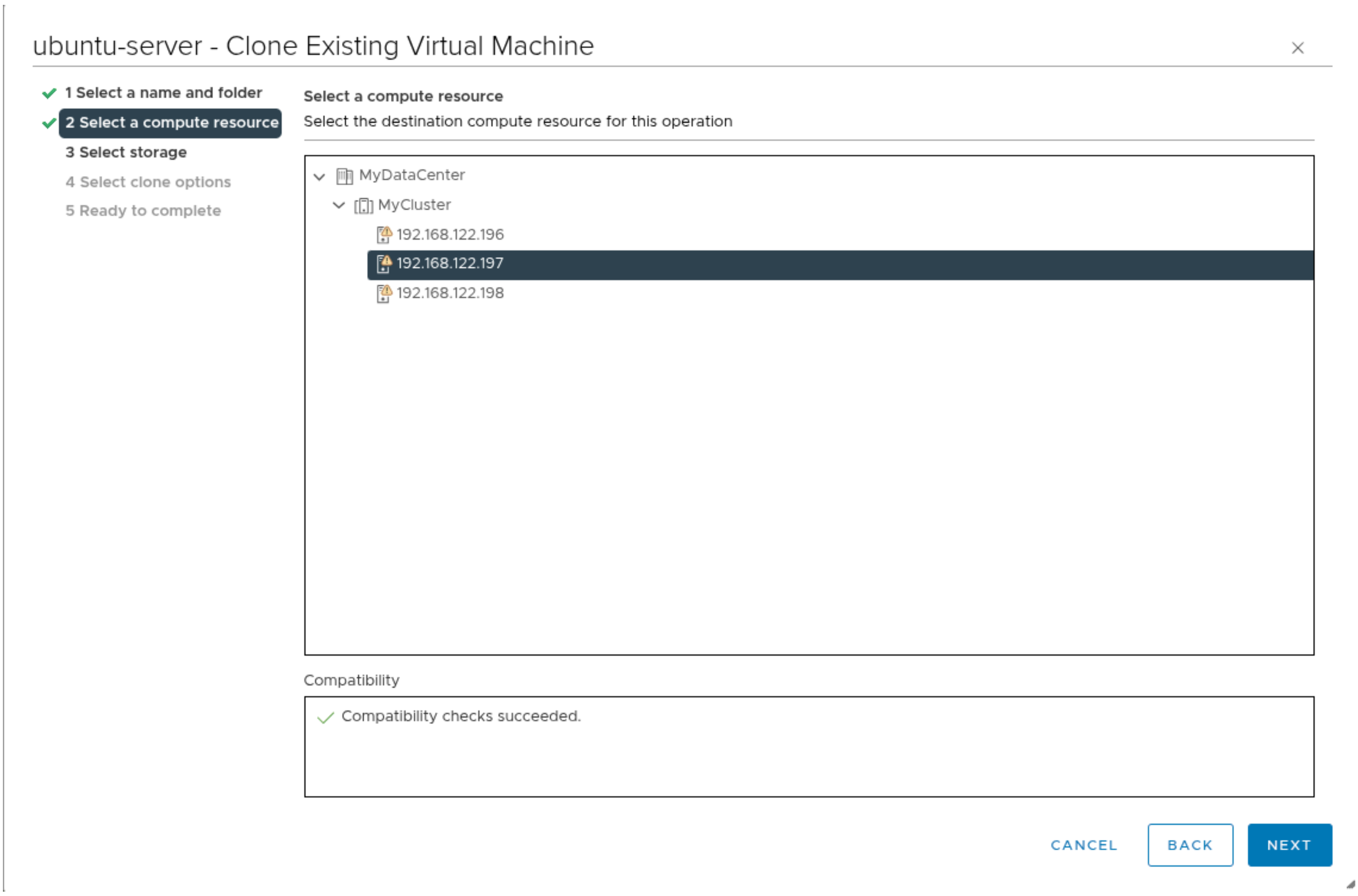

ubuntu-server
Not secure https://192.168.122.200/ui/webconsole.html?vmId=vm-6006&vmName=ubuntu-server&numMksConnections=0&serverG...
ubuntu-server
Enforce US Keyboard Layout
View Fullscreen
Send Ctrl+Alt+Delete
Installing system
[Help]
subiquity/load_cloud_config/extract_autoinstall:
subiquity/Early/apply_autoinstall_config:
subiquity/Reporting/apply_autoinstall_config:
subiquity/Error/apply_autoinstall_config:
subiquity/Userdata/apply_autoinstall_config:
subiquity/Package/apply_autoinstall_config:
subiquity/Debconf/apply_autoinstall_config:
subiquity/Kernel/apply_autoinstall_config:
subiquity/Zdev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring format: format-0
configuring partition: partition-2
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmposehc6k4/mount /
[View full log]

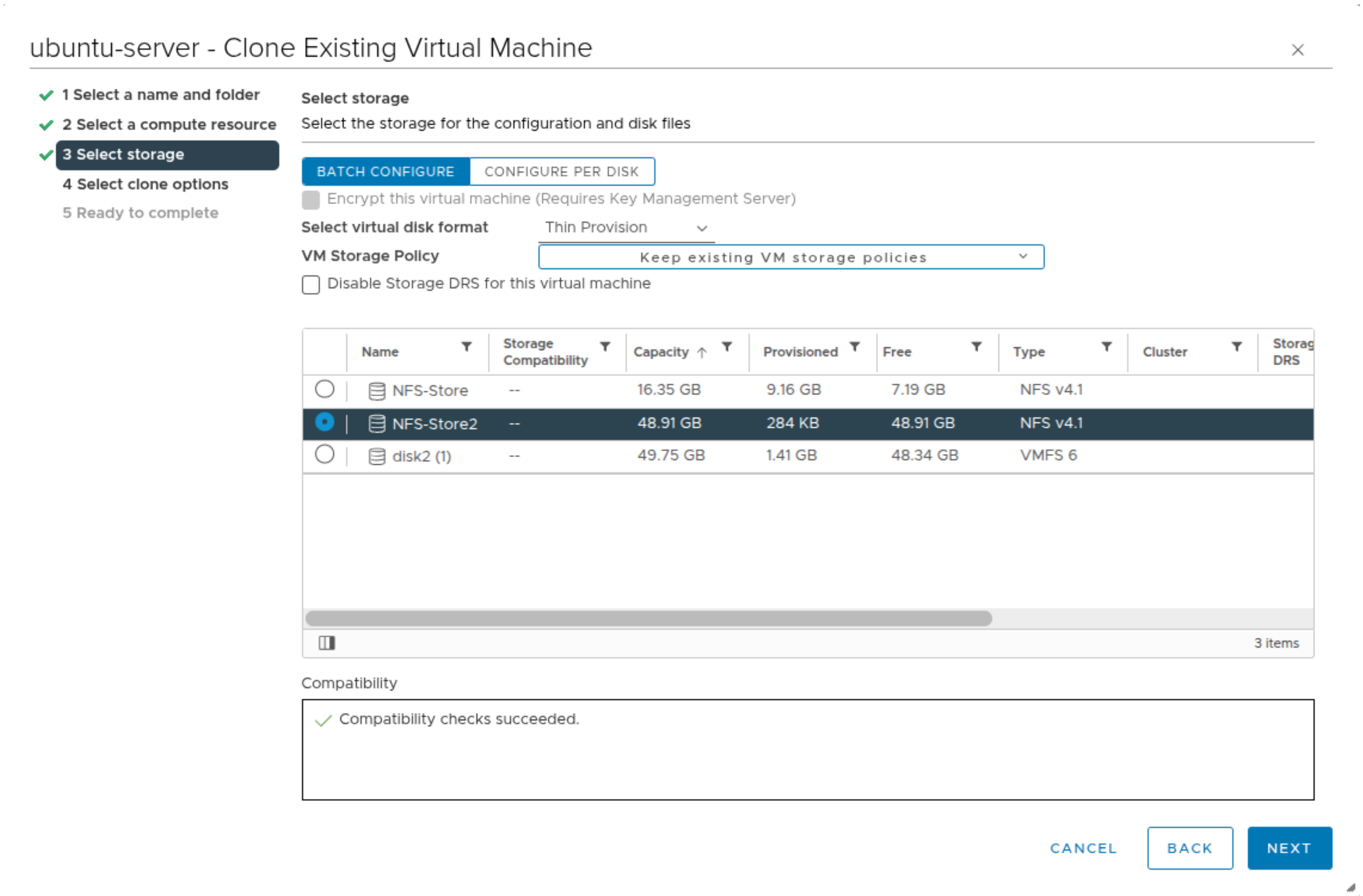


Clone an Existing Virtual Machine:

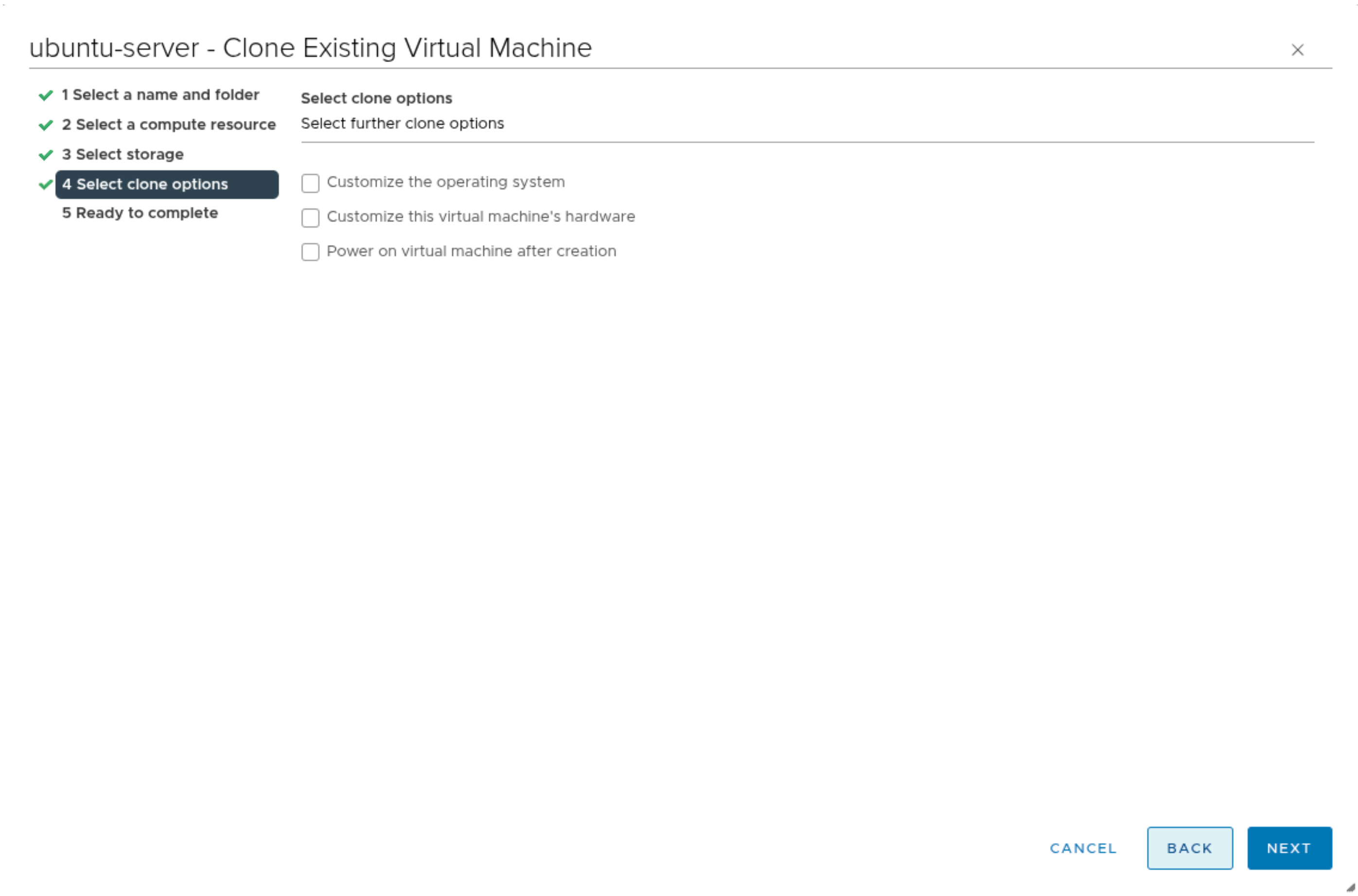


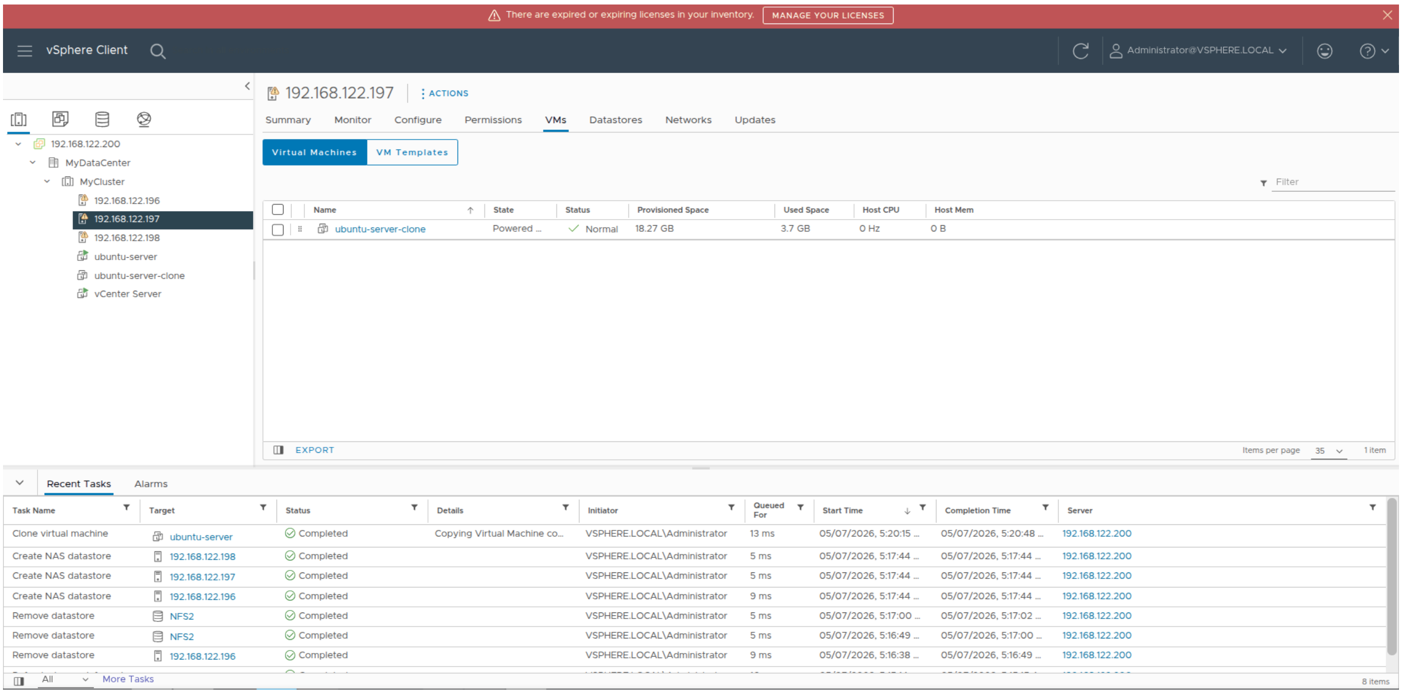

ubuntu-server - Clone Existing Virtual Machine
1 Select a name and folder
2 Select a compute resource
3 Select storage
4 Select clone options
5 Ready to complete
Select a name and folder
Specify a unique name and target location
Virtual machine name:
ubuntu-server-clone
Select a location for the virtual machine.
192.168.122.200
MyDataCenter
CANCEL
BACK
NEXT


ubuntu-server - Clone Existing Virtual Machine
1 Select a name and folder
2 Select a compute resource
3 Select storage
4 Select clone options
5 Ready to complete
Select a compute resource
Select the destination compute resource for this operation
MyDataCenter
MyCluster
192.168.122.196
192.168.122.197
192.168.122.198
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT



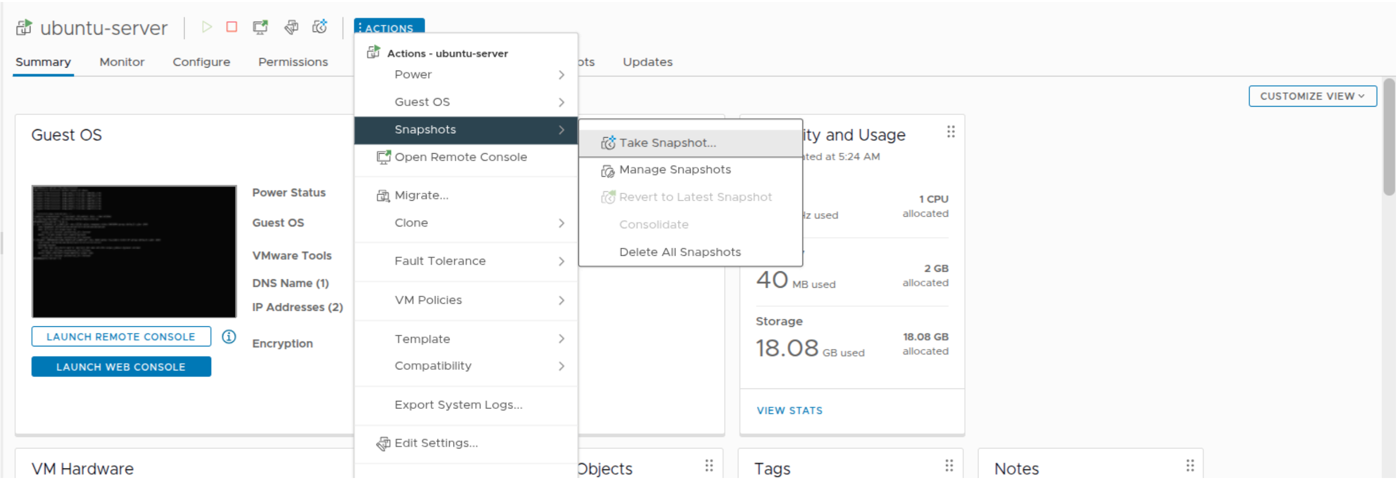
ubuntu-server - Clone Existing Virtual Machine
1 Select a name and folder
2 Select a compute resource
3 Select storage
4 Select clone options
5 Ready to complete
Select storage
Select the storage for the configuration and disk files
BATCH CONFIGURE
CONFIGURE PER DISK
Encrypt this virtual machine (Requires Key Management Server)
Select virtual disk format
Thin Provision
VM Storage Policy
Keep existing VM storage policies
Disable Storage DRS for this virtual machine
Name
Storage Compatibility
Capacity
Provisioned
Free
Type
Cluster
Storag DRS
NFS-Store -- 16.35 GB 9.16 GB 7.19 GB NFS v4.1
NFS-Store2 -- 48.91 GB 284 KB 48.91 GB NFS v4.1
disk2 (1) -- 49.75 GB 1.41 GB 48.34 GB VMFS 6
3 items
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT

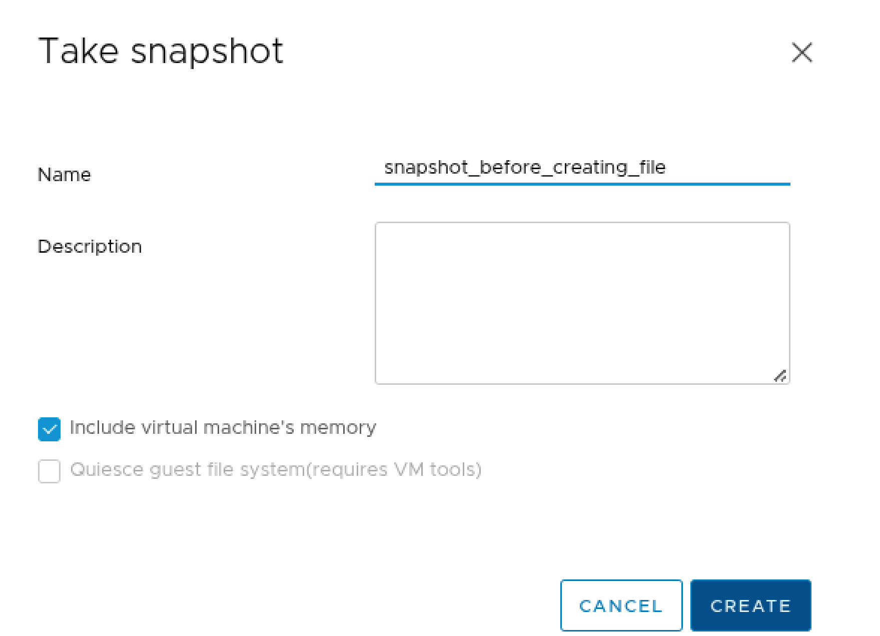

ubuntu-server - Clone Existing Virtual Machine
1 Select a name and folder
2 Select a compute resource
3 Select storage
4 Select clone options
5 Ready to complete
Select clone options
Select further clone options
Customize the operating system
Customize this virtual machine's hardware
Power on virtual machine after creation
CANCEL
BACK
NEXT


There are expired or expiring licenses in your inventory.
MANAGE YOUR LICENSES
vSphere Client
Administrator@VSPHERE.LOCAL
192.168.122.200
MyDataCenter
MyCluster
192.168.122.196
192.168.122.197
192.168.122.198
ubuntu-server
ubuntu-server-clone
vCenter Server
192.168.122.197
ACTIONS
Summary
Monitor
Configure
Permissions
VMs
Datastores
Networks
Updates
Virtual Machines
VM Templates
Name
State
Status
Provisioned Space
Used Space
Host CPU
Host Mem
ubuntu-server-clone
Powered ...
Normal
18.27 GB
3.7 GB
0 Hz
0 B
EXPORT
Recent Tasks
Alarms
Task Name
Target
Status
Details
Initiator
Start Time
Completion Time
Server
Clone virtual machine
Create NAS datastore
Remove datastore
Completed
Copying Virtual Machine co...
VSPHERE.LOCAL\Administrator
More Tasks

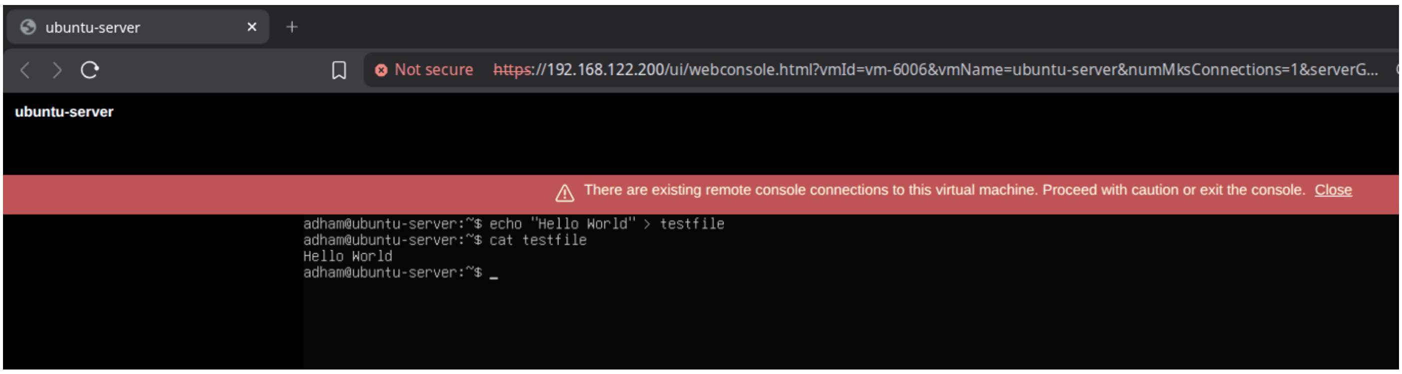
Create a snapshot of VM:

We first take a snapshot of the VM before creating a text file.

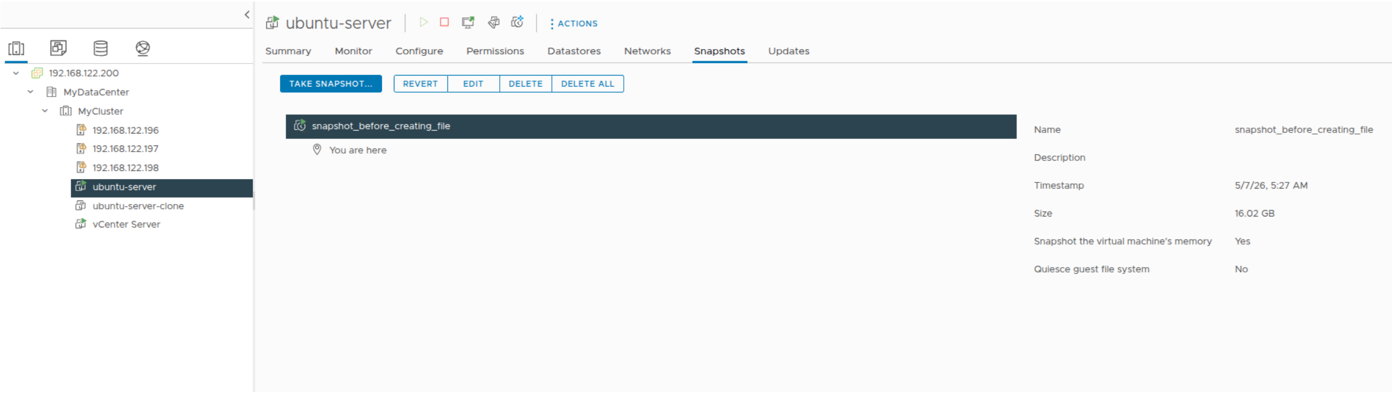


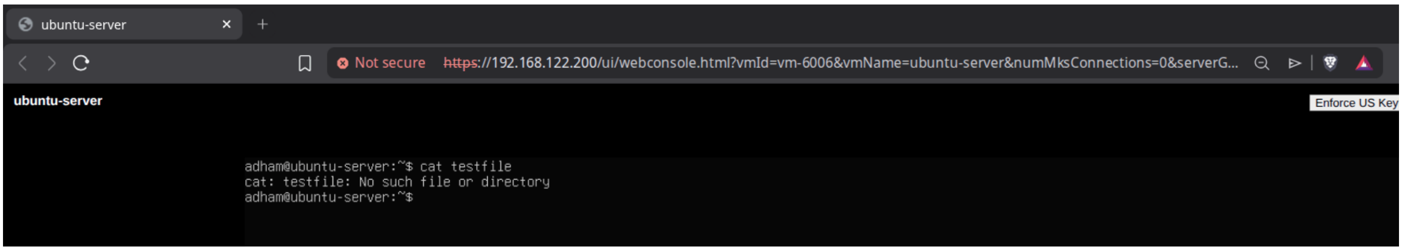


Then we create the test file.

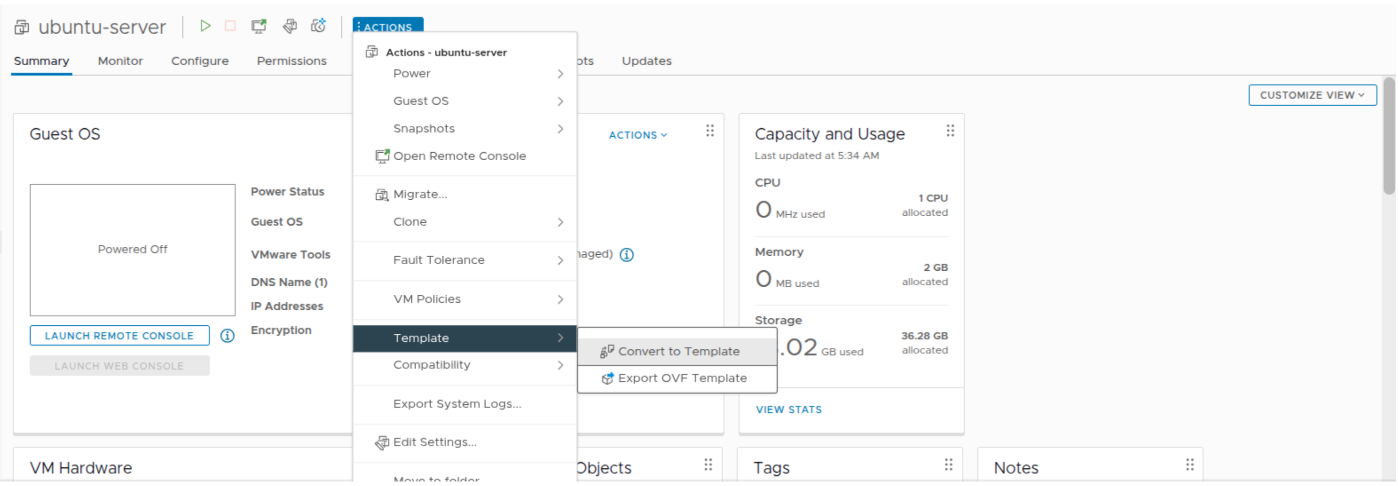


Then we revert back to the snapshot and test file should not exist.

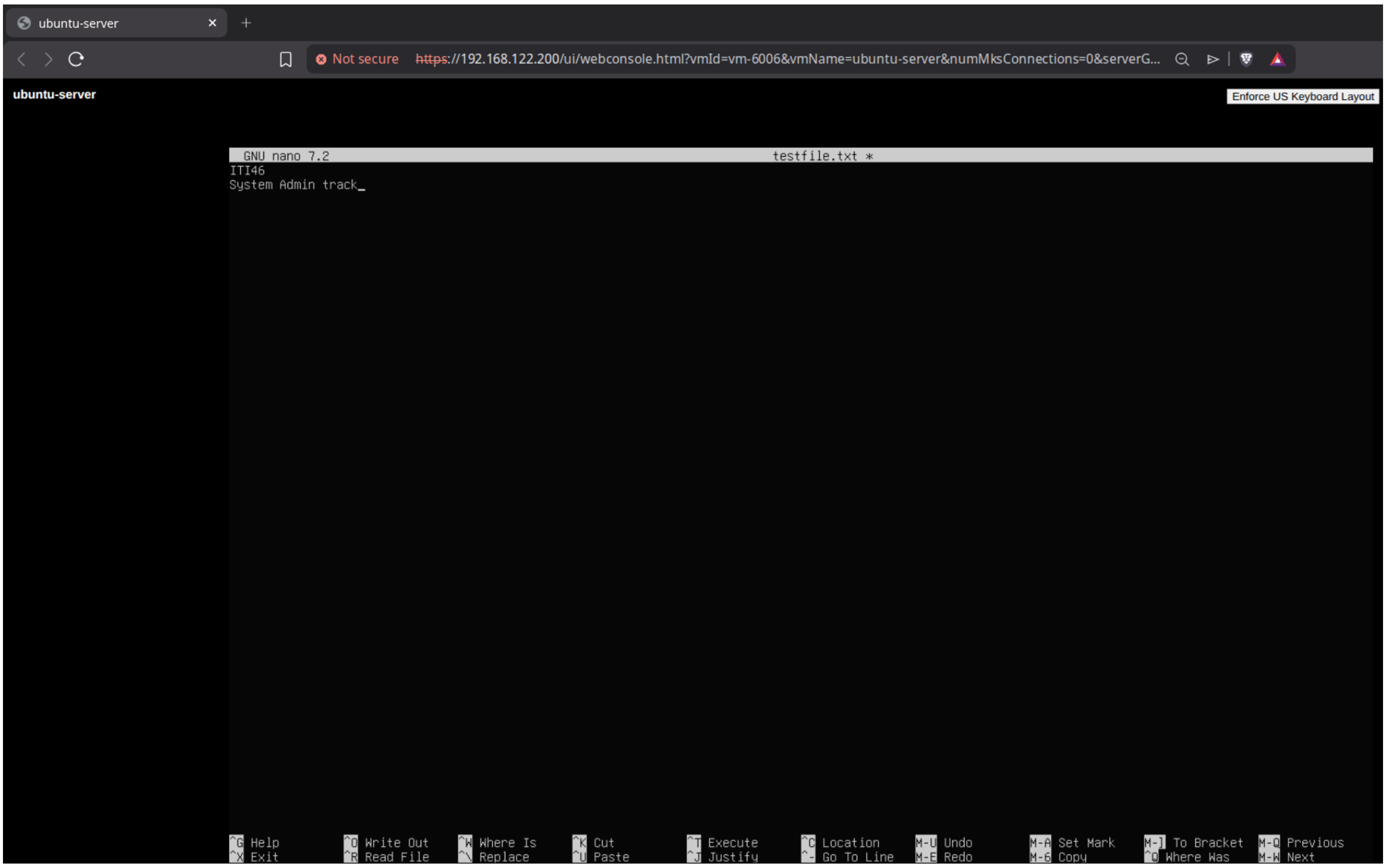


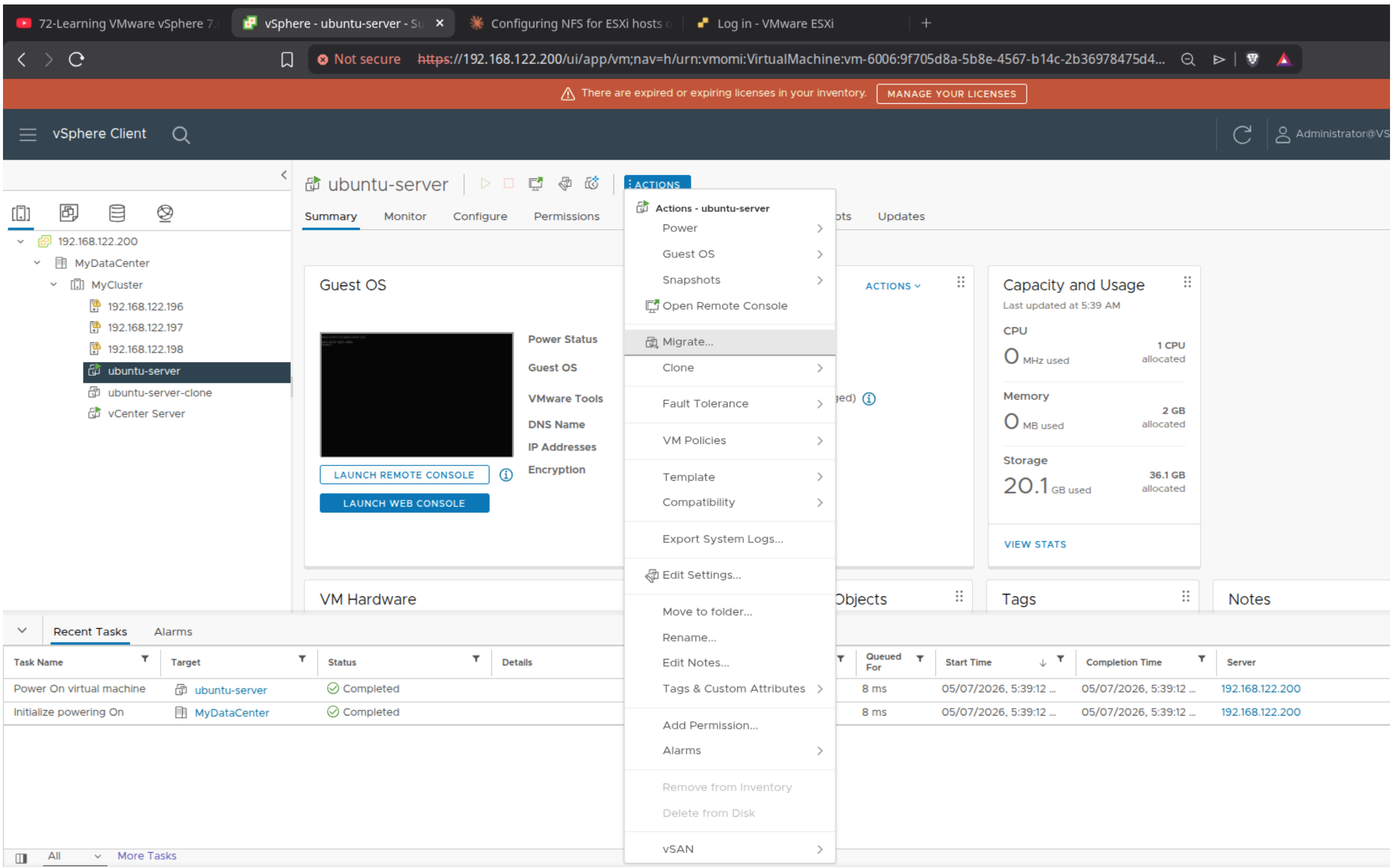


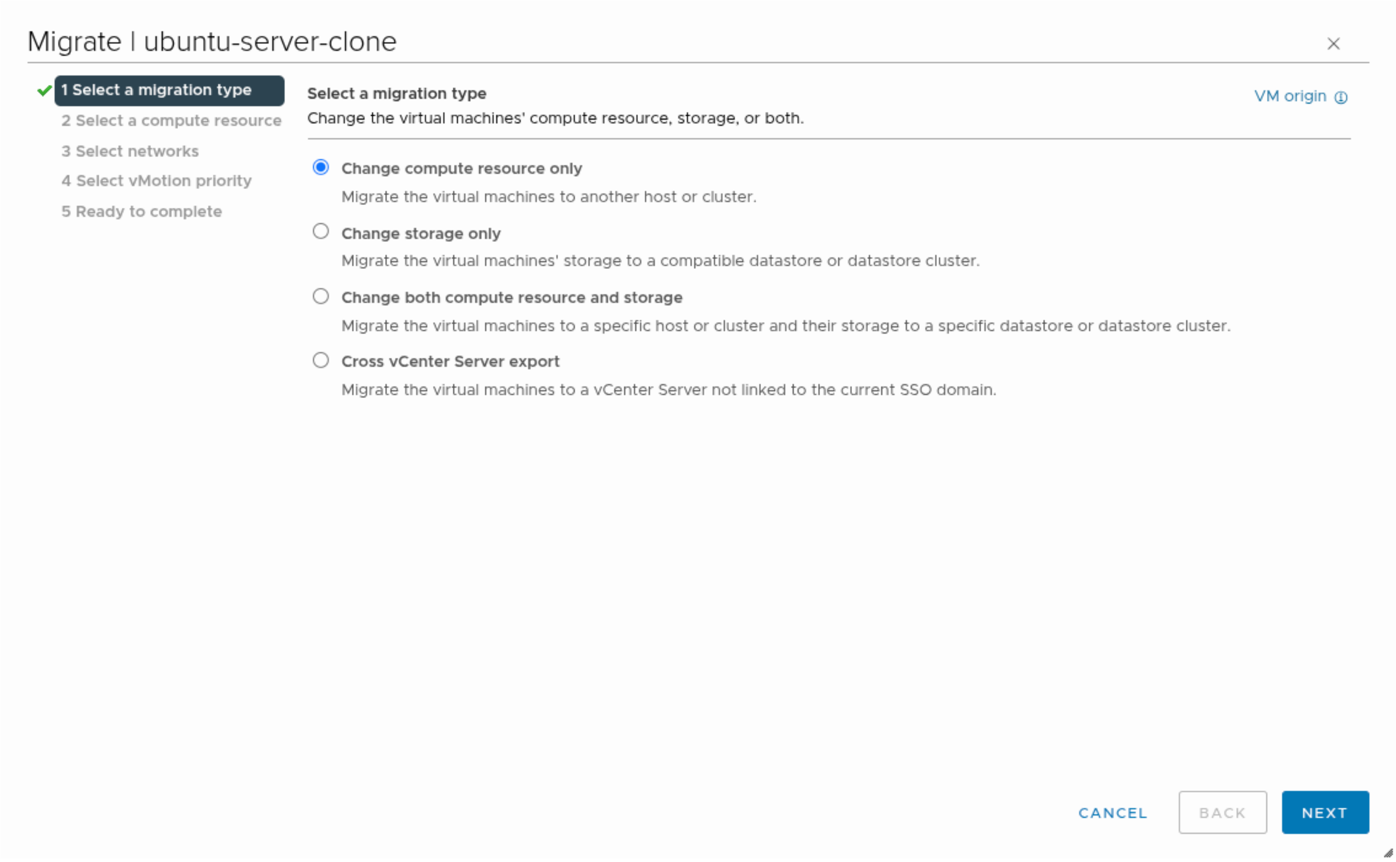
Convert a VM to a template:

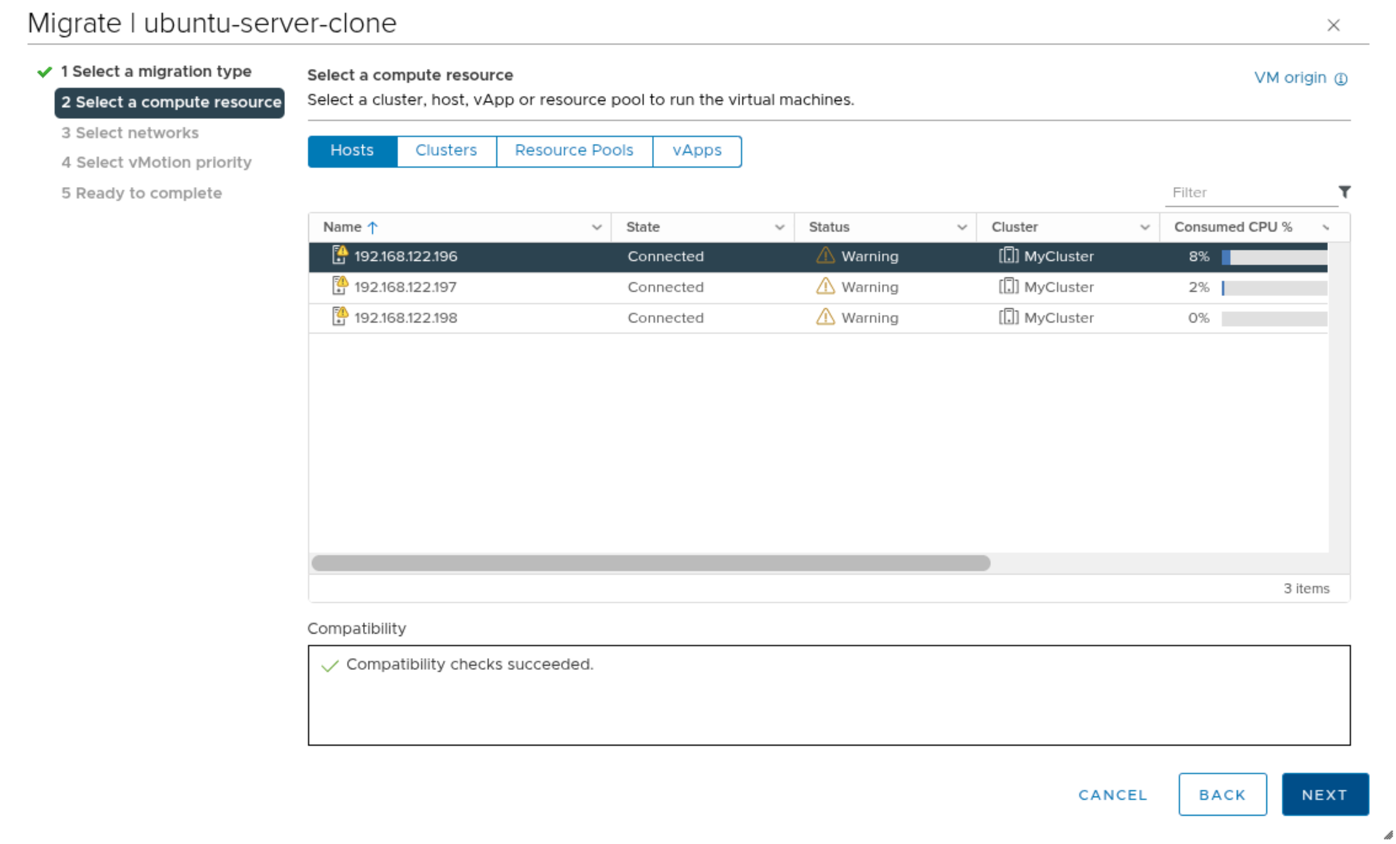


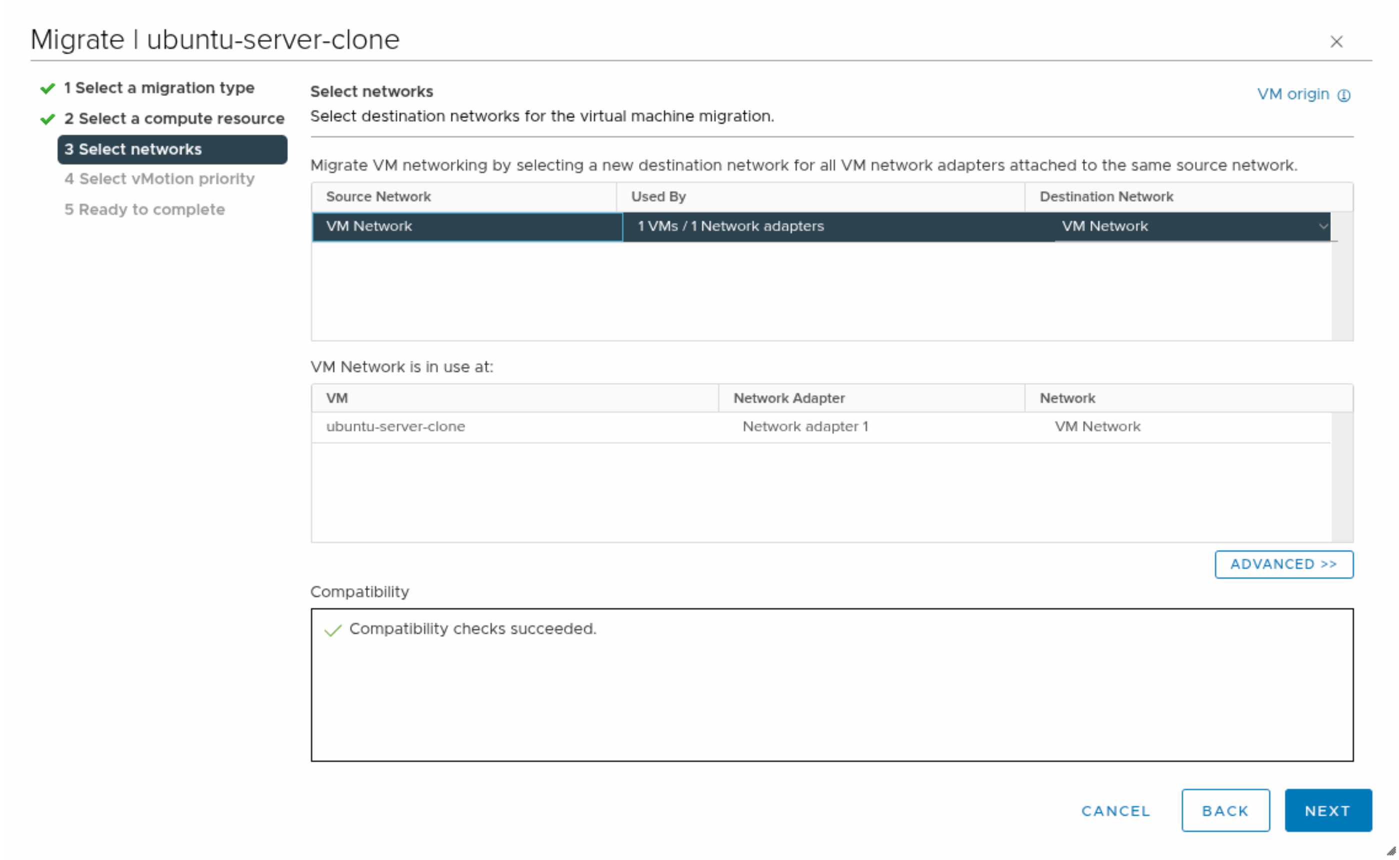
Testing Migration:

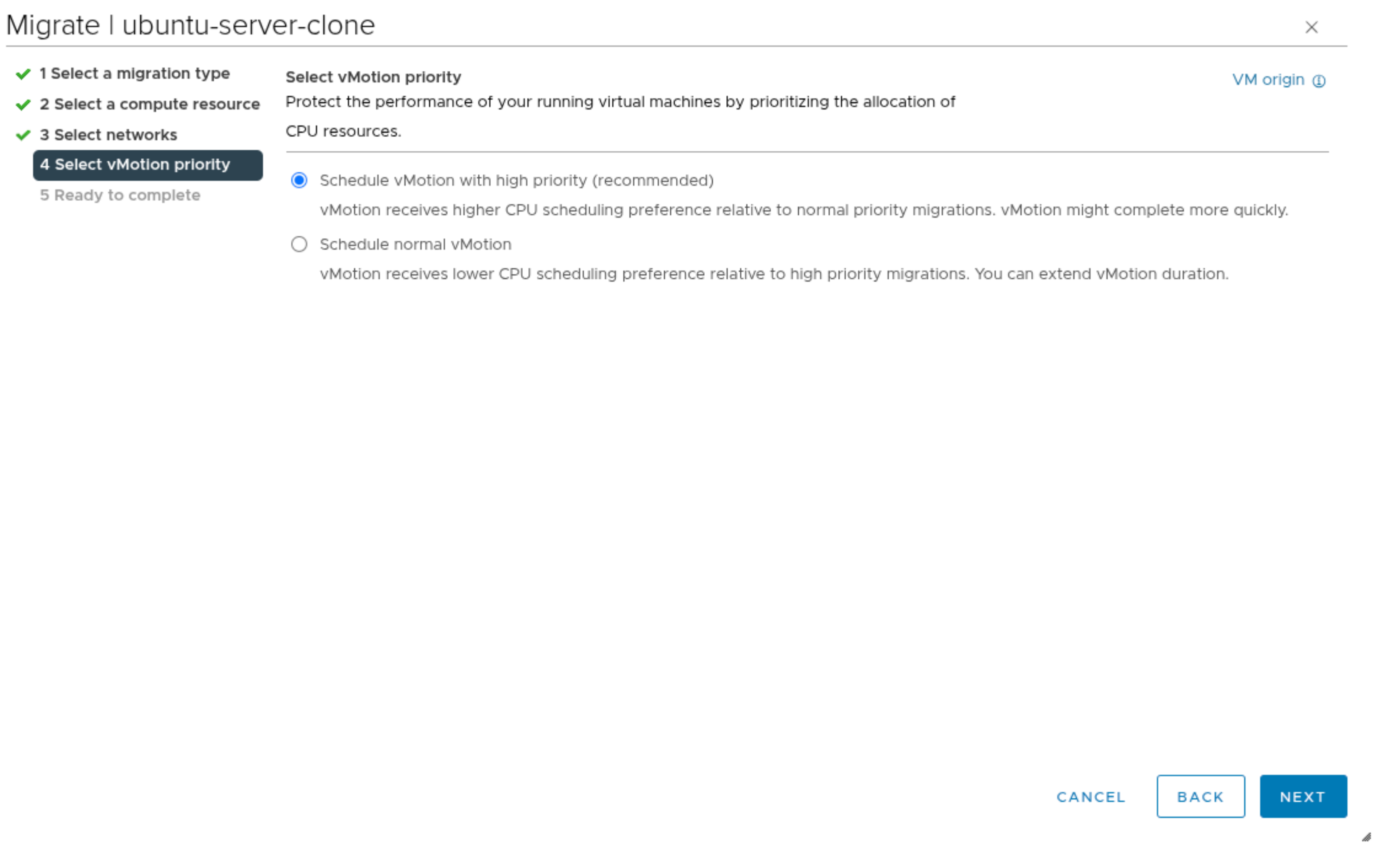


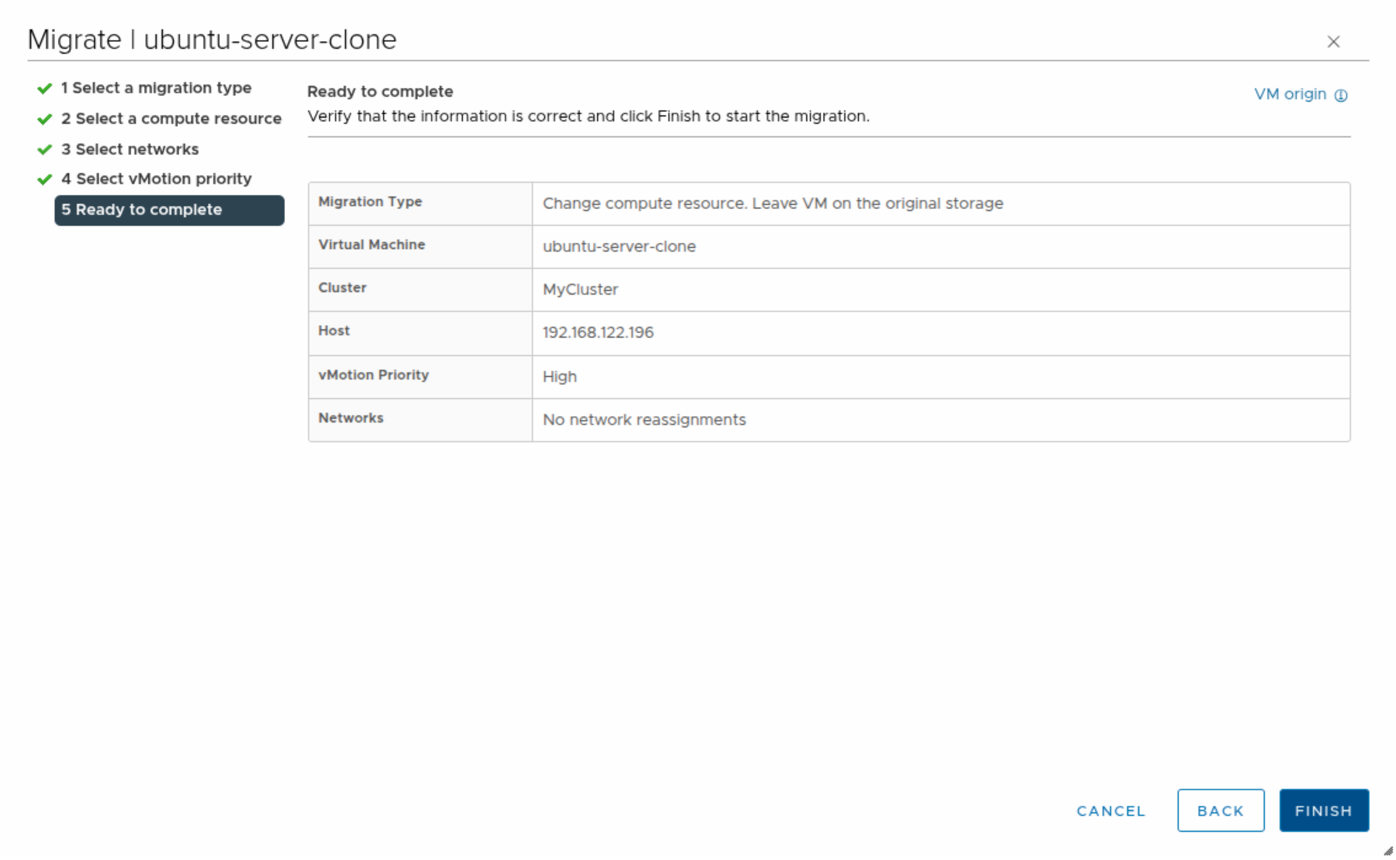


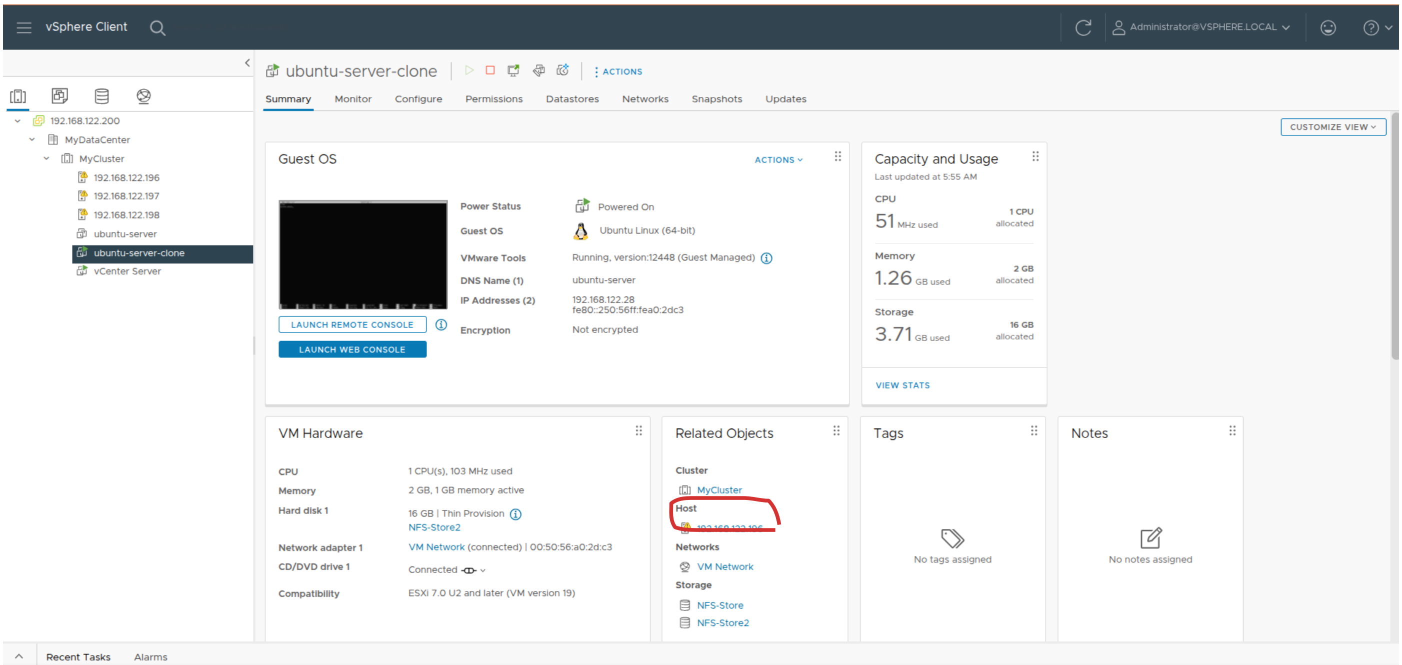

Migrate | ubuntu-server-clone
1 Select a migration type
2 Select a compute resource
3 Select networks
4 Select vMotion priority
5 Ready to complete
Select a migration type
Change the virtual machines' compute resource, storage, or both.
VM origin
Change compute resource only
Migrate the virtual machines to another host or cluster.
Change storage only
Migrate the virtual machines' storage to a compatible datastore or datastore cluster.
Change both compute resource and storage
Migrate the virtual machines to a specific host or cluster and their storage to a specific datastore or datastore cluster.
Cross vCenter Server export
Migrate the virtual machines to a vCenter Server not linked to the current SSO domain.
CANCEL
BACK
NEXT

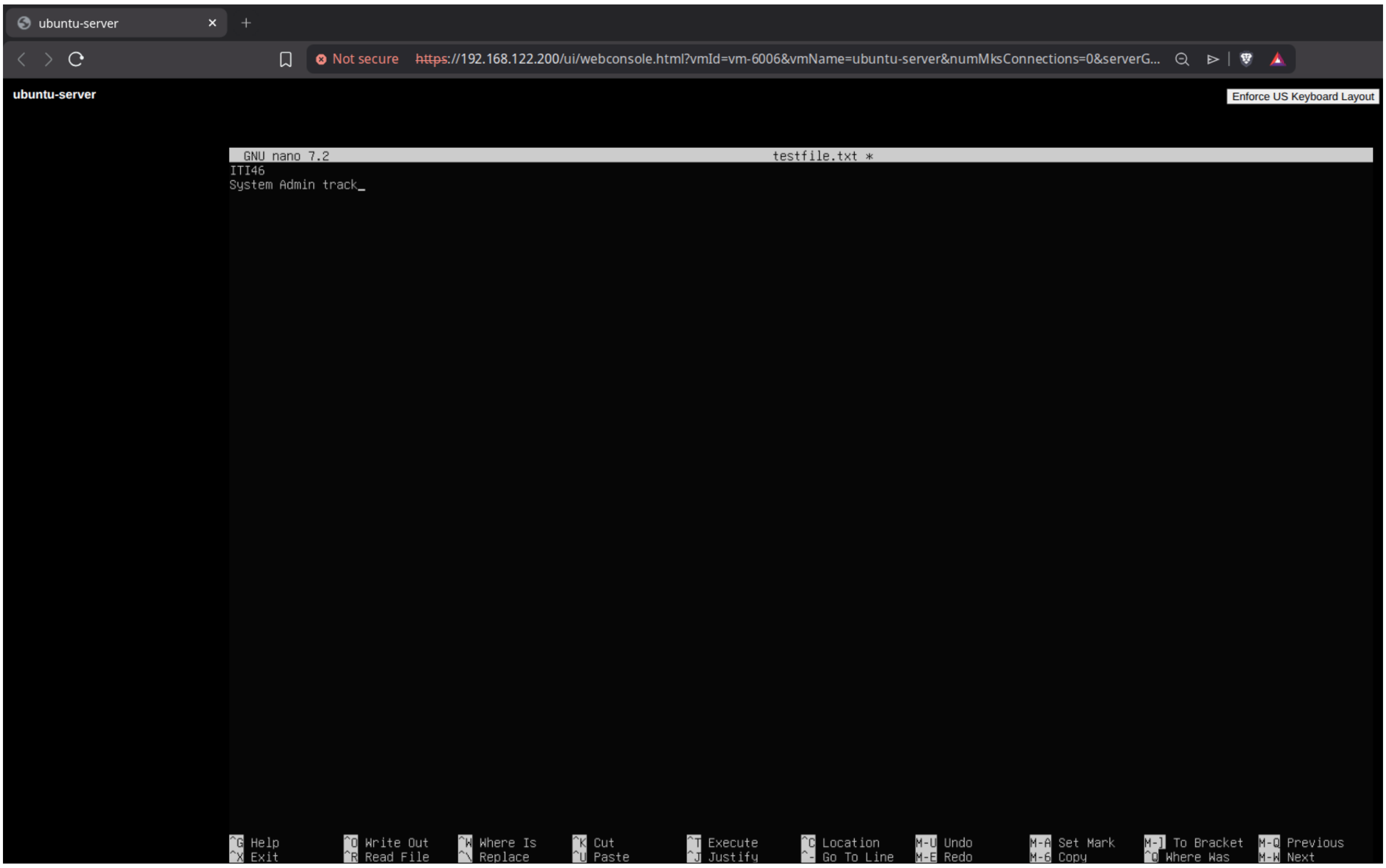

Migrate | ubuntu-server-clone
1 Select a migration type
2 Select a compute resource
3 Select networks
4 Select vMotion priority
5 Ready to complete
Select a compute resource
Select a cluster, host, vApp or resource pool to run the virtual machines.
VM origin
Hosts
Clusters
Resource Pools
vApps
Filter
Name
State
Status
Cluster
Consumed CPU %
192.168.122.196
Connected
Warning
MyCluster
8%
192.168.122.197
Connected
Warning
MyCluster
2%
192.168.122.198
Connected
Warning
MyCluster
0%
3 items
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT


Migrate | ubuntu-server-clone
1 Select a migration type
2 Select a compute resource
3 Select networks
4 Select vMotion priority
5 Ready to complete
Select networks
Select destination networks for the virtual machine migration.
VM origin
Migrate VM networking by selecting a new destination network for all VM network adapters attached to the same source network.
Source Network
Used By
Destination Network
VM Network
1 VMs / 1 Network adapters
VM Network
VM Network is in use at:
VM
Network Adapter
Network
ubuntu-server-clone
Network adapter 1
VM Network
ADVANCED >>
Compatibility
Compatibility checks succeeded.
CANCEL
BACK
NEXT


Migrate | ubuntu-server-clone
1 Select a migration type
2 Select a compute resource
3 Select networks
4 Select vMotion priority
5 Ready to complete
Select vMotion priority
Protect the performance of your running virtual machines by prioritizing the allocation of CPU resources.
VM origin
Schedule vMotion with high priority (recommended)
vMotion receives higher CPU scheduling preference relative to normal priority migrations. vMotion might complete more quickly.
Schedule normal vMotion
vMotion receives lower CPU scheduling preference relative to high priority migrations. You can extend vMotion duration.
CANCEL
BACK
NEXT

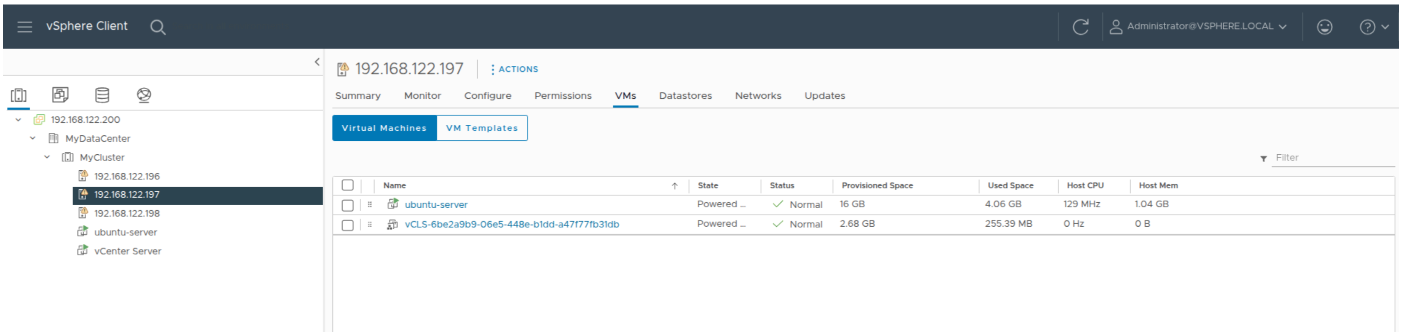

Migrate | ubuntu-server-clone
1 Select a migration type
2 Select a compute resource
3 Select networks
4 Select vMotion priority
5 Ready to complete
Ready to complete
Verify that the information is correct and click Finish to start the migration.
VM origin
Migration Type
Change compute resource. Leave VM on the original storage
Virtual Machine
ubuntu-server-clone
Cluster
MyCluster
Host
192.168.122.196
vMotion Priority
High
Networks
No network reassignments
CANCEL
BACK
FINISH


vSphere Client
Administrator@VSPHERE.LOCAL
192.168.122.200
MyDataCenter
MyCluster
192.168.122.196
192.168.122.197
192.168.122.198
ubuntu-server
ubuntu-server-clone
vCenter Server
ubuntu-server-clone
ACTIONS
Summary
Monitor
Configure
Permissions
Datastores
Networks
Snapshots
Updates
CUSTOMIZE VIEW
Guest OS
Power Status
Powered On
Guest OS
Ubuntu Linux (64-bit)
VMware Tools
Running, version:12448 (Guest Managed)
DNS Name (1)
ubuntu-server
IP Addresses (2)
192.168.122.28
fe80::250:56ff:fea0:2dc3
Encryption
Not encrypted
LAUNCH REMOTE CONSOLE
LAUNCH WEB CONSOLE
Capacity and Usage
Last updated at 5:55 AM
CPU
51 MHz used
1 CPU allocated
Memory
1.26 GB used
2 GB allocated
Storage
3.71 GB used
16 GB allocated
VIEW STATS
VM Hardware
CPU
1 CPU(s), 103 MHz used
Memory
2 GB, 1 GB memory active
Hard disk 1
16 GB | Thin Provision
NFS-Store2
Network adapter 1
VM Network (connected) | 00:50:56:a0:2d:c3
CD/DVD drive 1
Connected
Compatibility
ESXi 7.0 U2 and later (VM version 19)
Related Objects
Cluster
MyCluster
Host
Networks
VM Network
Storage
NFS-Store
NFS-Store2
Tags
No tags assigned
Notes
No notes assigned
Recent Tasks
Alarms

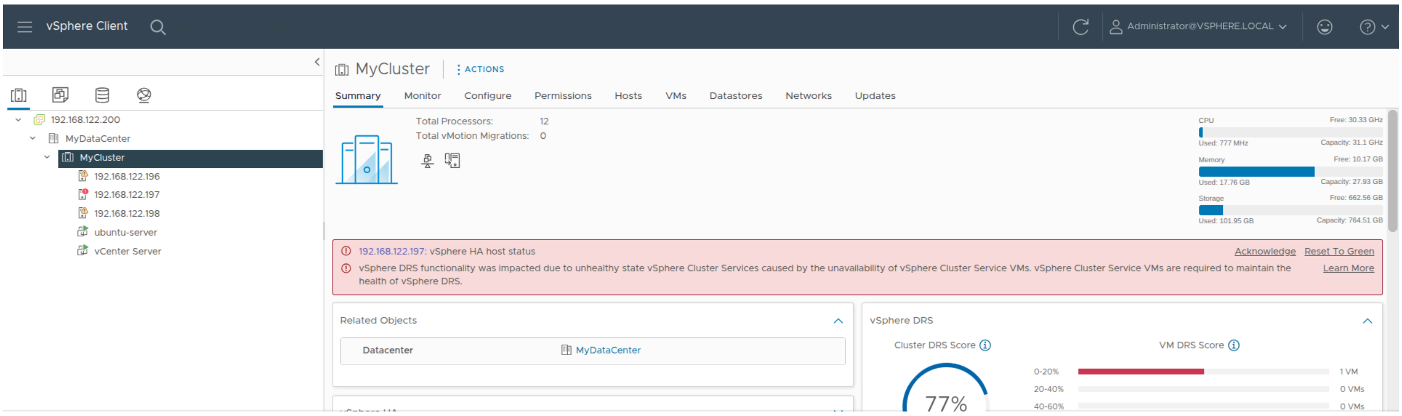


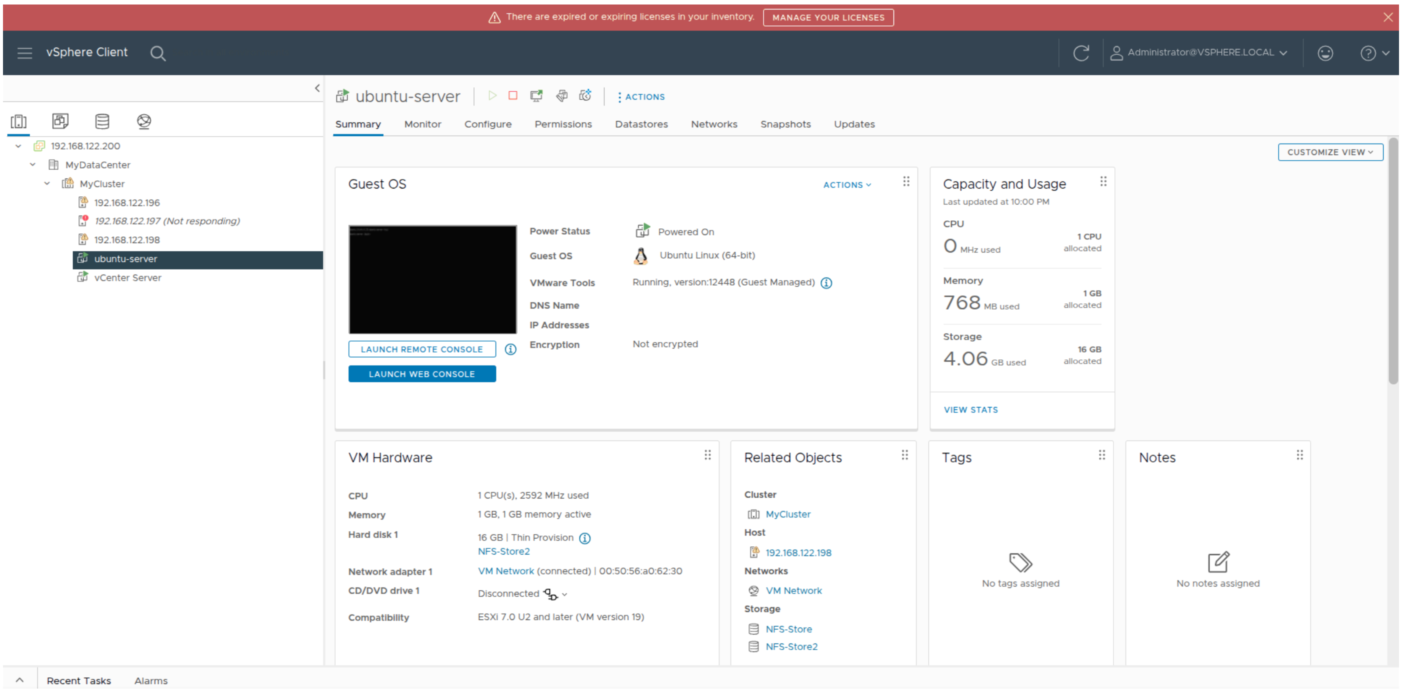
Testing HA:

We start with ubuntu-server-clone VM running on ESXI2 (192.168.122.197) and then we shutdown ESXI2 to induce a host failure so that the VM will restart on another ESXI (in this case ESXI3).



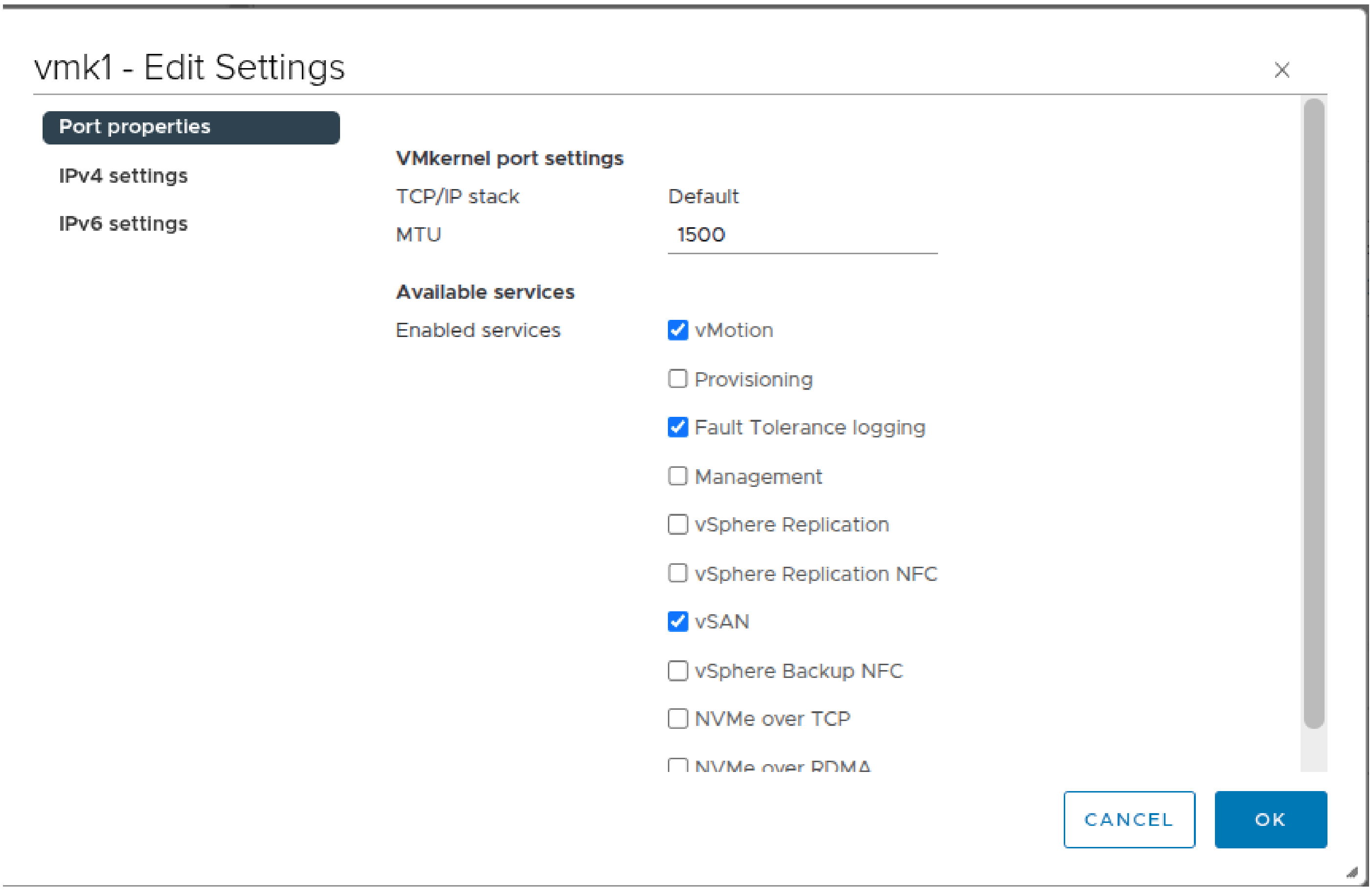
We get an HA alarm that restarts the VM on ESXI3 (192.168.122.198)



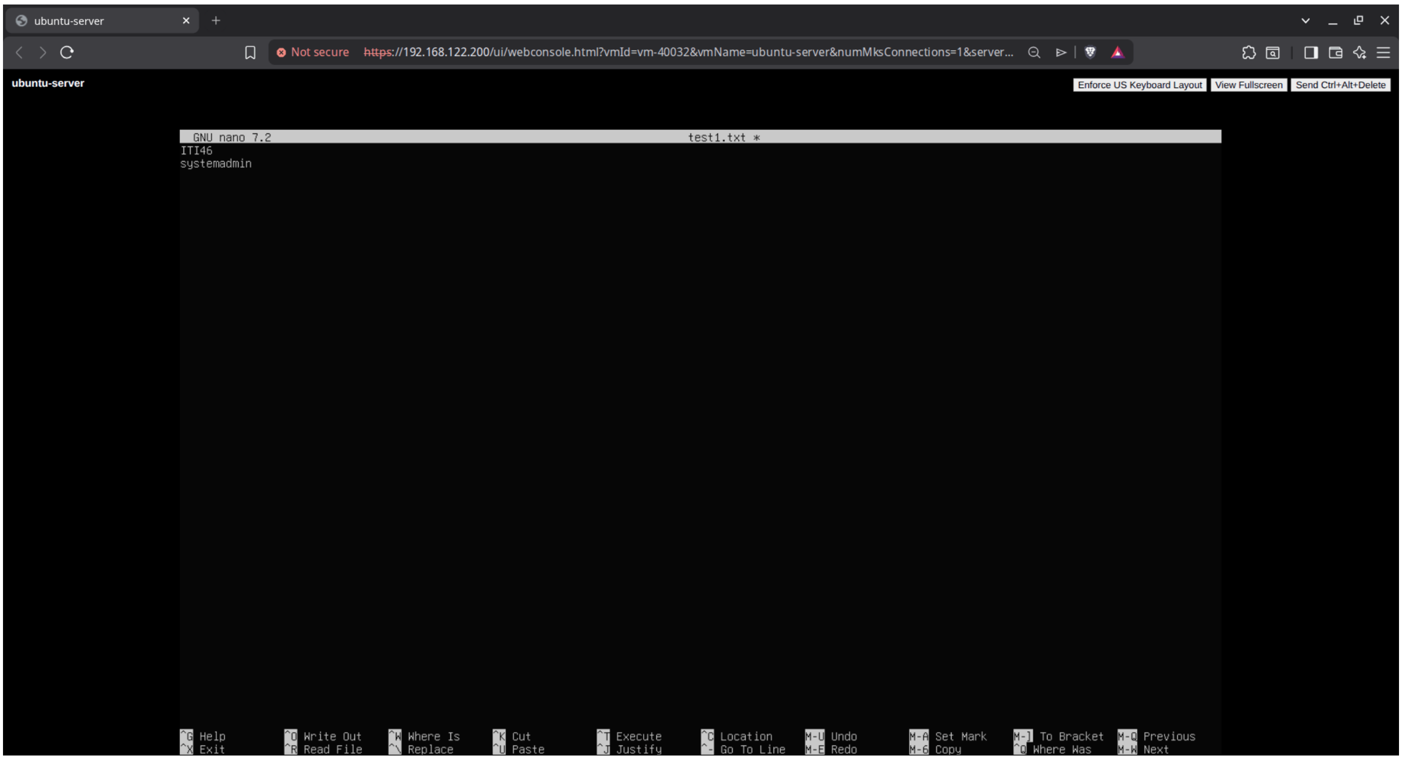


Testing Fault Tolerance:

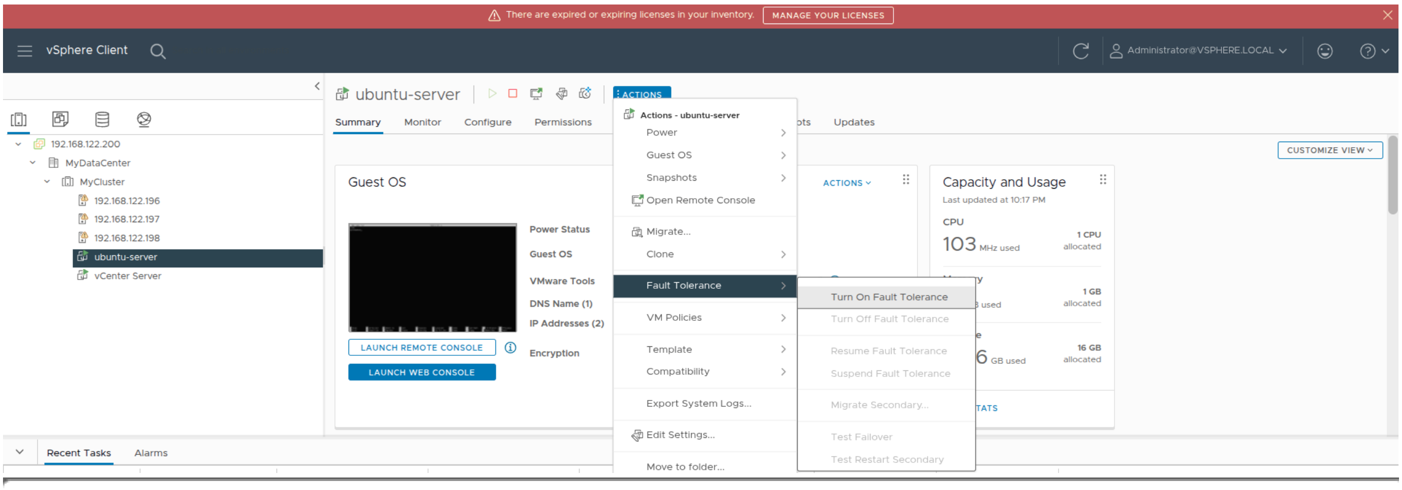
We first edit the vmkernel created so that it passes Fault tolerance logging traffic.



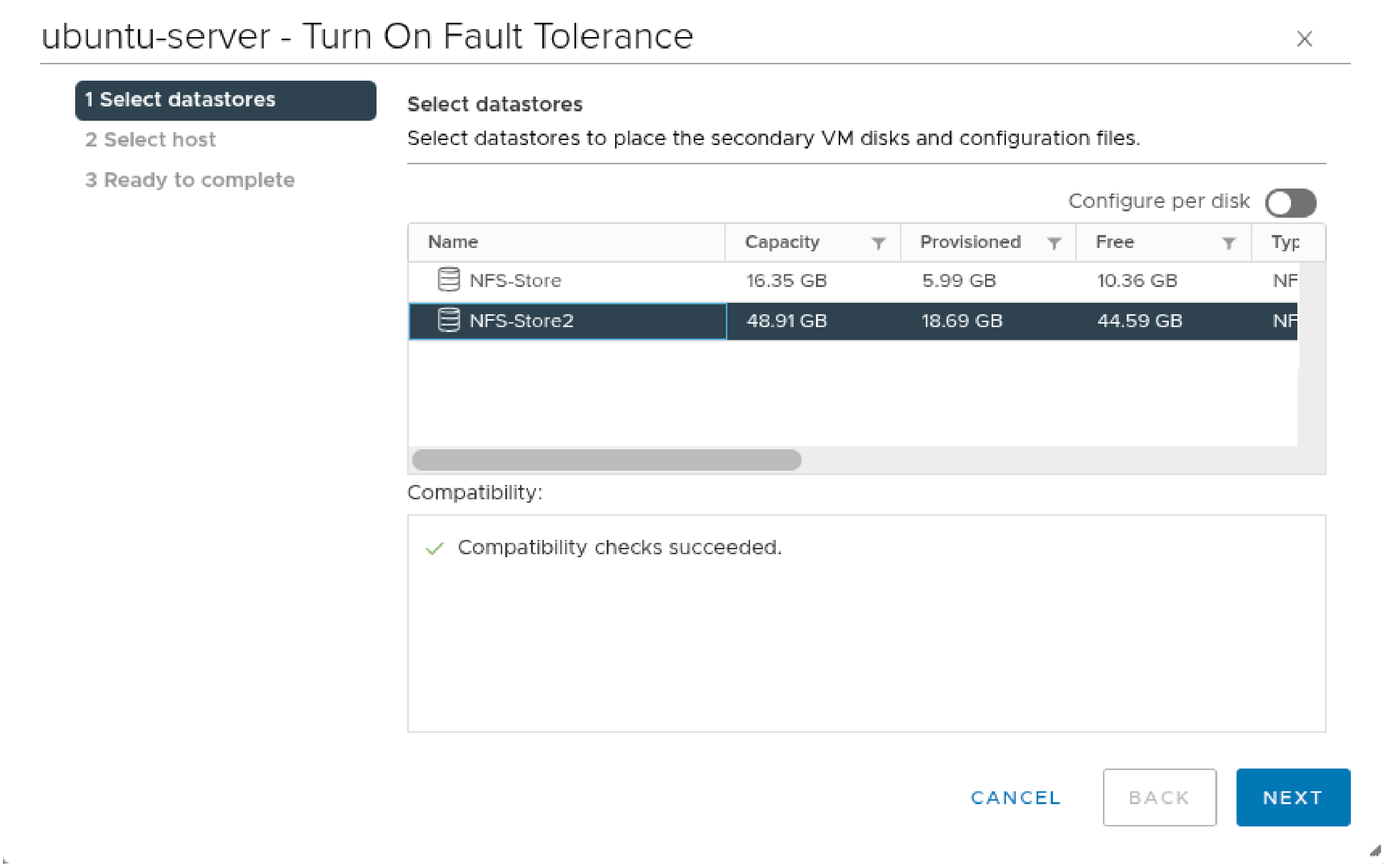
Then we start the ubuntu server on ESXI3 and run nano to write a text file and leave it in this state.



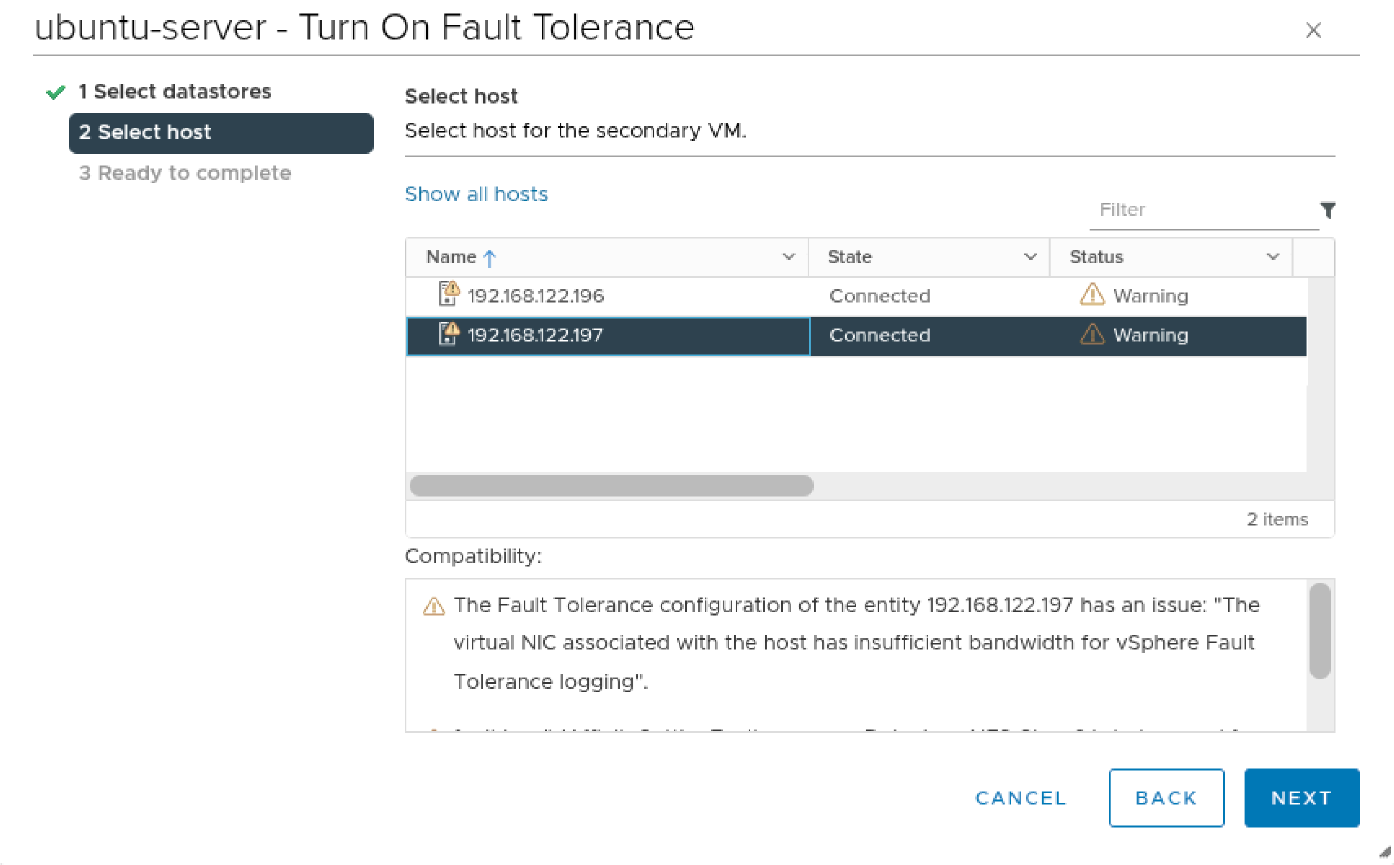
Then we create a secondary standby of this VM on ESXI2

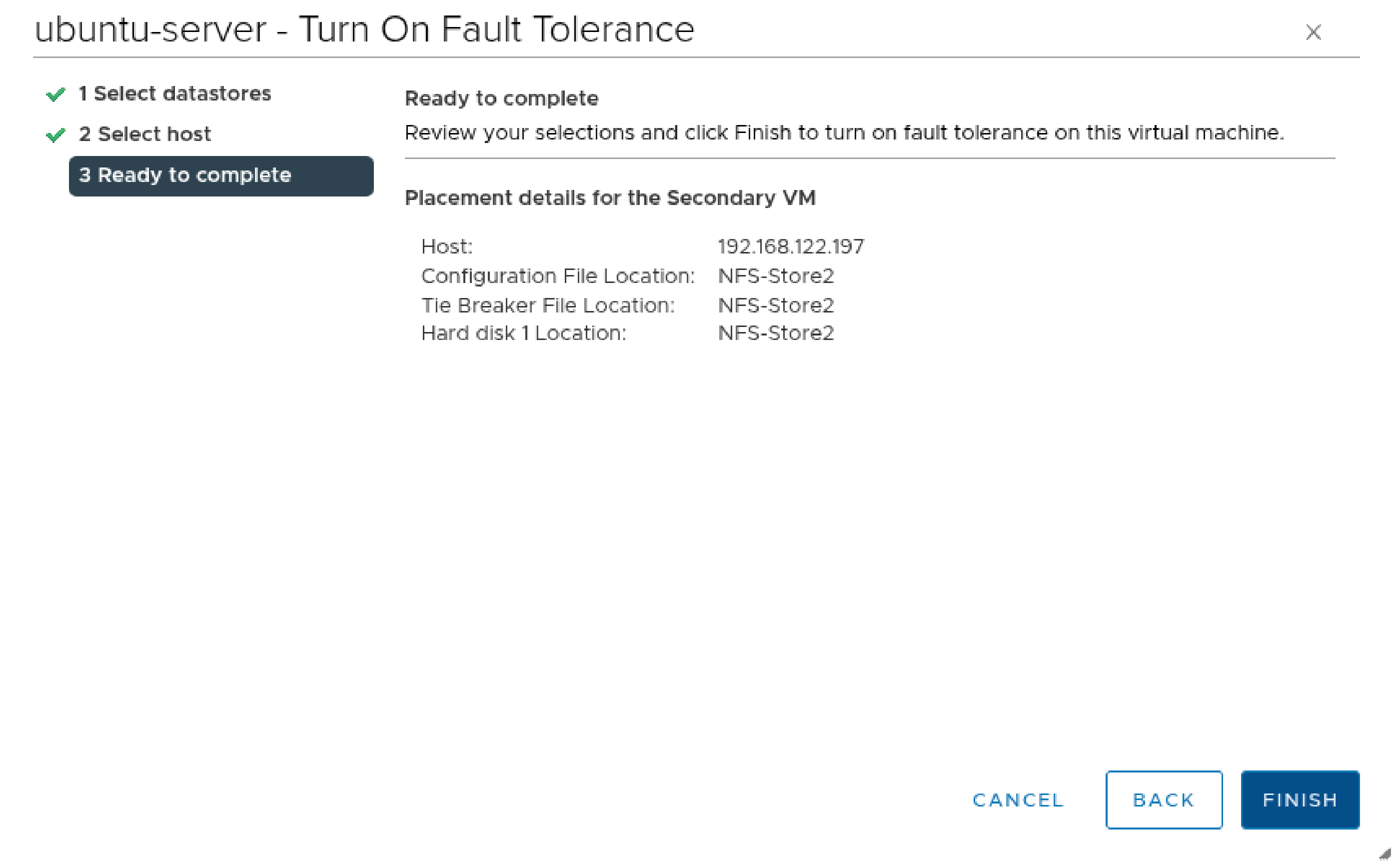


There are expired or expiring licenses in your inventory.
MANAGE YOUR LICENSES
vSphere Client
Administrator@VSPHERE.LOCAL
192.168.122.200
MyDataCenter
MyCluster
192.168.122.196
192.168.122.197
192.168.122.198
ubuntu-server
vCenter Server
ubuntu-server
ACTIONS
Summary
Monitor
Configure
Permissions
Updates
CUSTOMIZE VIEW
Guest OS
Power Status
Guest OS
VMware Tools
DNS Name (1)
IP Addresses (2)
Encryption
LAUNCH REMOTE CONSOLE
LAUNCH WEB CONSOLE
Actions - ubuntu-server
Power
Guest OS
Snapshots
Open Remote Console
Migrate...
Clone
Fault Tolerance
VM Policies
Template
Compatibility
Export System Logs...
Edit Settings...
Move to folder...
Turn On Fault Tolerance
Turn Off Fault Tolerance
Resume Fault Tolerance
Suspend Fault Tolerance
Migrate Secondary...
Test Failover
Test Restart Secondary
Capacity and Usage
Last updated at 10:17 PM
CPU
103 MHz used
1 CPU allocated
1 GB allocated
16 GB allocated
Recent Tasks
Alarms



ubuntu-server - Turn On Fault Tolerance
1 Select datastores
2 Select host
3 Ready to complete
Select datastores
Select datastores to place the secondary VM disks and configuration files.
Configure per disk
Name
Capacity
Provisioned
Free
Typ
NFS-Store
16.35 GB
5.99 GB
10.36 GB
NF
NFS-Store2
48.91 GB
18.69 GB
44.59 GB
NF
Compatibility:
Compatibility checks succeeded.
CANCEL
BACK
NEXT


ubuntu-server - Turn On Fault Tolerance
1 Select datastores
2 Select host
3 Ready to complete
Select host
Select host for the secondary VM.
Show all hosts
Filter
Name
State
Status
192.168.122.196
Connected
Warning
192.168.122.197
Connected
Warning
2 items
Compatibility:
The Fault Tolerance configuration of the entity 192.168.122.197 has an issue: "The virtual NIC associated with the host has insufficient bandwidth for vSphere Fault Tolerance logging".
CANCEL
BACK
NEXT


ubuntu-server - Turn On Fault Tolerance
1 Select datastores
2 Select host
3 Ready to complete
Ready to complete
Review your selections and click Finish to turn on fault tolerance on this virtual machine.
Placement details for the Secondary VM
Host: 192.168.122.197
Configuration File Location: NFS-Store2
Tie Breaker File Location: NFS-Store2
Hard disk 1 Location: NFS-Store2
CANCEL
BACK
FINISH

Now we shutdown ESXI3 and it should make the standby VM run on ESXI2 with zero downtime.